Simon's Multidimensional L^2 Conjecture

Operator-valued reciprocal Riesz contraction and the radial short-time problem

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Abstract

We study reciprocal transfer recursions of the form

$$\Phi_z(H) = (a\mathrm{I} + zbH)^{-1}(aH + \bar{z}b\mathrm{I}), \qquad a^2 - b^2 = 1,$$

which arise in finite-channel approximations to multidimensional Schrödinger scattering. A proposed one-cell fractional contraction with a dimension-independent gap is false: root-of-unity phases make the gap vanish as the dimension grows. We prove the correct replacement. In every fixed dimension there is strict contraction, and a finite product of fixed-height scalar cells collapses to one effective Poisson barrier. Positive drift of its hyperbolic height and Haar uniformity of its centre yield uniform fractional contraction on compact families of singular operator-valued spectral measures. Deterministic lacunary phases, compact-output stability, and a dovetailed no-escape argument remove the apparent circular dependence between group length and Galerkin dimension.

We then examine physical realization by scalar angular multipliers on $\mathbb{S}^2$. A shrinking-window carrier construction produces drift-decorated population swaps with full-operator control and arbitrarily small quadratic cost on each prescribed nonempty window. Several natural globalizations are ruled out: no fixed grouped length is uniform over all prior reciprocal histories; adjacent-band multiband tilings have a positive Zakharov–Shabat energy floor; direct scaled slabs and every bounded-depth multiscale cascade freeze spatial densities; and uniformly smooth logarithmic-shell controls converge to phase multipliers. A surviving logarithmic construction must be nonperturbative, have angular degree comparable to the shell radius, and control a growing resonant cluster collectively.

The operator-valued contraction theorem is unconditional. Its insertion into the full radial Schrödinger problem is reduced to one explicit short-angular-time radialization problem. Until that theorem is proved, the global physical contraction and the resulting counterexample to Simon's multidimensional L^2 conjecture remain conditional; no resolution of the conjecture is claimed.

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Scope of the claims. The operator-valued contraction, fixed-dimensional realizations, local shrinking-window carrier theorem, and stated no-go results are presented as unconditional. Problem 10.78 is the remaining conditional input. Consequently the global physical contraction and any counterexample to Simon's conjecture are not claimed.

1 The correction in one line

Fix $\rho > 0$ and put

$$a = \cosh\rho, \qquad b = \sinh\rho, \qquad t = \frac{b}{a} \in (0, 1). \tag{1}$$

For $H \in \mathrm{U}(n)$ and $z \in \mathbb{T}$, the ideal reciprocal factor is

$$F_z(H) = (a\mathrm{I} + zbH)^{-1}. \tag{2}$$

The originally requested estimate had one $q_\beta < 1$ for all n. The correct statement is

$$\int_{\mathbb{T}} \operatorname{tr}\Big[(F_z(H)^* A F_z(H))^\beta\Big] \,\mathrm{d}m(z) \le \kappa_{\beta,n} \operatorname{tr}(A^\beta), \qquad \kappa_{\beta,n} < 1, \tag{3}$$

where $0 < \beta < 1$, $A \ge 0$, and the constant is uniform in H and A but necessarily depends on n.

This dependence is harmless at the scale-ledger level. At generation q, with $n_q = \dim \mathcal{D}_q$, choose a repetition count r_q so large that

$$\kappa_{\beta,n_q}^{r_q} \le 2^{-4q}. \tag{4}$$

Then choose the concentration depth and physical accuracy of each of the r_q cells so that their total X_3 cost and total factorisation error are at most 2^{-q}. The radii may be enormous; existence only requires that the choices are finite at each generation.

2 Finite-dimensional strict contraction

Let m denote normalized Haar measure on $\mathbb{T}$.

Lemma 2.1 (Hardy normalization)**.** *For every finite-dimensional unitary H,*

$$\int_{\mathbb{T}} F_z(H)^* F_z(H) \,\mathrm{d}m(z) = \mathrm{I}. \tag{5}$$

The same identity holds with $F_z F_z^*$.

Proof. Write $C = tH$. Since $a^{-2} = 1 - t^2$,

$$F_z = a^{-1}(\mathrm{I} + zC)^{-1}.$$

Expanding in nonnegative powers of z, applying Parseval, and telescoping gives

$$\int_{\mathbb{T}} F_z^* F_z \,\mathrm{d}m(z) = (1 - t^2) \sum_{j \ge 0} t^{2j} \mathrm{I} = \mathrm{I}.$$

Because F_z is a function of the normal operator H, it is normal, so the second identity follows as well. ☐

Theorem 2.2 (Uniform strict contraction in every fixed dimension)**.** *Let $0 < \beta < 1$ and $n < \infty$. There is a number $\kappa_{\beta,n} = \kappa_{\beta,n}(\rho) < 1$ such that (3) holds for every $H \in \mathrm{U}(n)$ and every positive $n \times n$ matrix A.*

Proof. For a unit vector u, set

$$s_{H,u}(z) = \|F_z(H)^* u\|^2. \tag{6}$$

Lemma 2.1 gives $\int s_{H,u} \,\mathrm{d}m = 1$. We first prove the strict, uniform rank-one estimate

$$\sup_{H \in \mathrm{U}(n),\ \|u\|=1} \int_{\mathbb{T}} s_{H,u}(z)^\beta \,\mathrm{d}m(z) < 1. \tag{7}$$

Let $\nu_{H,u}$ be the spectral measure of H at u. The Poisson-kernel identity

$$|a + b\zeta|^{-2} = \frac{1 - t^2}{|1 + t\zeta|^2} = 1 + 2 \sum_{k \ge 1} (-t)^k \operatorname{Re}(\zeta^k) \tag{8}$$

shows that

$$s_{H,u}(z) = 1 + 2\sum_{k\ge 1}(-t)^k \operatorname{Re}\Big(z^k\langle u, H^k u\rangle\Big). \tag{9}$$

If $s_{H,u}$ were constant, then $\langle u, H^k u\rangle = 0$ for every $k \ge 1$. But $\nu_{H,u}$ is a probability measure supported on at most n points. If its distinct support points are $\lambda_1, \dots, \lambda_r$ with positive weights $p_1, \dots, p_r$, then the equations

$$\sum_{j=1}^{r} p_j \lambda_j^k = 0, \qquad k = 1, \dots, r,$$

form an invertible Vandermonde system and would force every p_j to vanish. This is impossible. Hence $s_{H,u}$ is nonconstant. Strict concavity of x^β, together with $\int s_{H,u} = 1$, gives

$$\int_{\mathbb{T}} s_{H,u}^\beta \, \mathrm{d}m < 1.$$

The integrand is uniformly bounded because $a - b \le |a + zb\lambda| \le a + b$. The displayed functional is continuous on the compact space $\mathrm{U}(n) \times \{u : \|u\| = 1\}$. Its supremum is therefore strictly below one. Denote that supremum by $\kappa_{\beta,n}$.

Now diagonalize $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$. Rotfel'd's trace subadditivity for $0 < \beta < 1$ gives

$$\begin{aligned}\operatorname{tr}[(F_z^* A F_z)^\beta] &\le \sum_j \operatorname{tr}\Big[\big(\lambda_j |F_z^* u_j\rangle\langle F_z^* u_j|\big)^\beta\Big] \\ &= \sum_j \lambda_j^\beta s_{H,u_j}(z)^\beta.\end{aligned}$$

Integrating and applying the rank-one bound proves (3). □

Remark 2.3 (Why this is the right matrix theorem). The proof does not assume that the spectrum of H lies in a short arc and does not assume that A commutes with H. It therefore handles precisely the phase spreading and rotation into moving angular directions that defeat the fixed-probe entropy argument.

3 A fixed contraction independent of dimension is impossible

Proposition 3.1 (Root-of-unity obstruction)**.** *For fixed $0 < \beta < 1$ and $\rho > 0$,*

$$\lim_{n\to\infty} \kappa_{\beta,n} = 1. \tag{10}$$

In particular, the proposed version of Lemma B with one fixed $q_\beta < 1$ for all growing targeted blocks is FALSE AS STATED.

Proof. Let

$$H_n = \operatorname{diag}(1, e^{2\pi i/n}, \dots, e^{2\pi i(n-1)/n}), \qquad u_n = n^{-1/2}(1, \dots, 1)^T,$$

and let $A_n = |u_n\rangle\langle u_n|$. Then

$$\begin{aligned}s_n(e^{i\phi}) &= \|F_{e^{i\phi}}(H_n)^* u_n\|^2 \\ &= \frac{1}{n}\sum_{j=0}^{n-1} |a + be^{i(\phi + 2\pi j/n)}|^{-2} \\ &= 1 + 2\sum_{r\ge 1}(-t)^{rn}\cos(rn\phi). \end{aligned} \tag{11}$$

Consequently

$$\|s_n - 1\|_{L^\infty(\mathbb{T})} \le \frac{2t^n}{1-t^n} \longrightarrow 0. \tag{12}$$

Since $\operatorname{tr}(A_n^\beta) = 1$,

$$\int_{\mathbb{T}} \operatorname{tr}[(F_z(H_n)^* A_n F_z(H_n))^\beta]\, \mathrm{d}m(z) = \int_{\mathbb{T}} s_n(z)^\beta\, \mathrm{d}m(z) \longrightarrow 1.$$

Theorem 2.2 gives the reverse inequality $\kappa_{\beta,n} < 1$ for each n, proving (10). ☐

Remark 3.2 (Interpretation). The obstruction is coherent phase averaging. A vector spread uniformly over many reciprocal eigenphases sees an almost constant average of the Poisson kernel, so one random circle phase produces almost no fractional gain. Repetition, rather than a nonexistent dimension-free gap, is the correct repair.

4 Strong barriers for singular phases

The preceding dimension obstruction fixes ρ. Physical reciprocal phases have additional structure, and allowing the hyperbolic angle to grow gives a stronger route.

Theorem 4.1 (Poisson contraction for a singular reciprocal phase)**.** *Let H be unitary on a separable Hilbert space and let u be a unit vector whose spectral measure $\nu_{H,u}$ is singular with respect to Haar measure on $\mathbb{T}$. For $0 < \beta < 1$,*

$$\lim_{\rho\to\infty} \int_{\mathbb{T}} \left\|(\cosh\rho\,\mathrm{I} + z\sinh\rho\, H)^{-*}u\right\|^{2\beta}\, \mathrm{d}m(z) = 0. \tag{13}$$

If $\mathfrak{C}$ is a compact family of pairs (H,u) in operator norm times vector norm, and every associated spectral measure is singular, then the convergence is uniform on $\mathfrak{C}$.

Proof. Put $t = \tanh\rho$. Up to the harmless rotation $z \mapsto -z$, the squared norm in (13) is the Poisson integral

$$s_{t,\nu}(z) = (P_t * \nu)(z).$$

The radial boundary theorem gives $s_{t,\nu}(z) \to 0$ for Haar-almost every z when ν is singular. Moreover, for every measurable $E \subset \mathbb{T}$,

$$\int_E s_{t,\nu}^\beta\, \mathrm{d}m \le m(E)^{1-\beta}\left(\int_E s_{t,\nu}\, \mathrm{d}m\right)^\beta \le m(E)^{1-\beta}.$$

Thus the fractional powers are uniformly integrable, and (13) follows.

For uniformity, define $Q_t(\nu) = \int (P_t * \nu)^\beta\, \mathrm{d}m$. If $0 < t_1 < t_2 < 1$, then $P_{t_1} = P_{t_1/t_2} * P_{t_2}$, and concavity gives $Q_{t_1}(\nu) \ge Q_{t_2}(\nu)$. For fixed $t < 1$, Q_t is continuous in the weak topology of probability measures. The spectral-measure map is continuous on $\mathfrak{C}$, so its image is compact. Pointwise monotone convergence to the continuous limit zero is uniform by Dini's theorem. ☐

Corollary 4.2 (Finite probe bundles need no dimension-free gap)**.** *Let $\mathfrak{C}$ be a compact unit-vector bundle containing the ranges of the finite-rank positive matrices $A(k)$ targeted in one generation. If every reciprocal phase spectral measure on $\mathfrak{C}$ is singular, then, for any $\delta > 0$, one may choose a finite ρ such that*

$$\int_{\mathbb{T}} \operatorname{tr}[(F_z(H(k))^* A(k) F_z(H(k)))^\beta]\, \mathrm{d}m(z) \le \delta\, \operatorname{tr}(A(k)^\beta) \tag{14}$$

uniformly in k.

Proof. Use the spectral decomposition of $A(k)$, Rotfel'd subadditivity, and the uniform part of Theorem 4.1. $\square$

Proposition 4.3 (Compact partial potentials have singular boundary phase)**.** *Let V be real, smooth, and compactly supported in the exterior radial problem, and let*

$$\Theta_V(k) = J_V(k)^{-1}\overline{J_V(k)}, \qquad \Theta_0(k) = J_0(k)^{-1}\overline{J_0(k)}.$$

For every $k > 0$,

$$\Theta_V(k) - \Theta_0(k) \in \mathfrak{S}_1(L^2(\mathbb{S}^2)). \tag{15}$$

Consequently $\Theta_V(k)$ has no absolutely continuous spectral subspace, and every one of its vector spectral measures is singular with respect to Haar measure on $\mathbb{T}$. On compact positive energy intervals the boundary phases are norm-continuous, so Corollary 4.2 applies uniformly to every compact finite-probe bundle.

Proof. Use the free spectral representation for the exterior Dirichlet problem. The scattering matrix of the pair $(H_0 + V, H_0)$ is, up to the fixed free incoming/outgoing diagonal factors dictated by the Jost normalization,

$$\mathsf{S}_V(k) = \Theta_V(k)\Theta_0(k)^*.$$

The stationary representation writes $\mathsf{S}_V(k) - \mathrm{I}$ as an integral operator on $\mathbb{S}^2$ whose kernel is the on-shell scattering amplitude. Compact support permits differentiation of the generalized eigenfunctions in both angular variables to every order; the kernel is therefore smooth. A smoothing operator on the compact two-dimensional sphere is trace class. Multiplication on the right by the unitary $\Theta_0(k)$ proves (15).

The free phase $\Theta_0(k)$ is diagonal in spherical harmonics and hence has pure point spectrum. The unitary Kato–Rosenblum theorem now implies that the absolutely continuous parts of $\Theta_V(k)$ and $\Theta_0(k)$ are unitarily equivalent, so the former vanishes. Standard stationary dependence away from zero gives norm continuity in k on compact intervals. $\square$

5 Exact operator-valued Riesz singularity on the phase torus

The next theorem shows that the corrected finite-dimensional contraction is already sufficient in the ideal independent-phase model.

Theorem 5.1 (Operator-valued reciprocal Riesz martingale)**.** *Fix $n < \infty$, $0 < \beta < 1$, and an invertible matrix X_0. On $\Omega = \mathbb{T}^{\mathbb{N}}$ with product Haar measure and its coordinate filtration, let H_j be any predictable $\mathrm{U}(n)$-valued process and define*

$$F_j = (a\mathrm{I} + bz_jH_j)^{-1}, \qquad X_j = F_jX_{j-1}, \qquad W_j = X_j^*X_j. \tag{16}$$

Then

$$\mathbb{E}(W_j \mid \mathcal{F}_{j-1}) = W_{j-1}, \tag{17}$$

$$\mathbb{E}\,\mathrm{tr}(W_j^\beta) \le \kappa_{\beta,n}^j\,\mathrm{tr}(W_0^\beta). \tag{18}$$

The consistent positive matrix measures with cylinder densities W_j have a nonzero weak limit $\boldsymbol{\mu}$ which is purely singular with respect to product Haar measure. Its trace measure, and hence every scalarization $u^*\boldsymbol{\mu}u$, is supported on one common Haar-null set.

Proof. Conditioning on the first $j-1$ coordinates and using Lemma 2.1 yields

$$\mathbb{E}(W_j \mid \mathcal{F}_{j-1}) = X_{j-1}^* \mathbb{E}(F_j^* F_j) X_{j-1} = W_{j-1}.$$

The nonzero eigenvalues of $X_j^* X_j$ and $X_j X_j^*$ agree. Hence, with $A_{j-1} = X_{j-1} X_{j-1}^*$,

$$\operatorname{tr}(W_j^\beta) = \operatorname{tr}[(F_j A_{j-1} F_j^*)^\beta].$$

Theorem 2.2, applied after replacing z by $\bar{z}$ and H by H^*, gives the conditional contraction. Iteration proves (18).

The matrix martingale converges entrywise almost everywhere. In finite dimension this implies convergence of its eigenvalues, and Fatou gives

$$\mathbb{E}\operatorname{tr}(W_\infty^\beta) \le \liminf_{j\to\infty} \mathbb{E}\operatorname{tr}(W_j^\beta) = 0.$$

Thus $W_\infty = 0$ almost everywhere. The martingale consistency defines a positive matrix measure on Ω whose absolutely continuous Radon–Nikodym derivative is W_∞; therefore its absolutely continuous part vanishes. Its total matrix mass equals that of W_0 and is nonzero. Finally, positivity gives $0 \le u^* \boldsymbol{\mu}(B) u \le \operatorname{tr} \boldsymbol{\mu}(B)$, so a null support for the trace measure is a common null support for every scalarization. □

6 Deterministic lacunary phases and tail locking

The torus theorem is probabilistic only in its notation. Successively large physical collar frequencies can realize the conditional phase averages on an energy interval. To avoid the moving-spike defect, one must preserve not only fractional moments but also the mass of concentration witnesses chosen at earlier stages.

Lemma 6.1 (One-step deterministic phase placement)**.** *Let K be a compact interval, let X, H be measurable matrix functions on K, with $H(k) \in \mathrm{U}(n)$, and put*

$$W = X^* X, \qquad F_z(k) = (a\mathrm{I} + zbH(k))^{-1}, \qquad W_z = (F_z X)^*(F_z X).$$

Given finitely many bounded measurable scalar test functions $\psi_1, \dots, \psi_N$ and $\epsilon > 0$, there are arbitrarily large real frequencies ν such that, with $z(k) = e^{i\nu k}$,

$$\int_K \operatorname{tr}(W_{z(k)}^\beta)\, \mathrm{d}k \le \kappa_{\beta,n} \int_K \operatorname{tr}(W^\beta)\, \mathrm{d}k + \epsilon, \tag{19}$$

$$\left| \int_K \psi_\ell(k) \big(W_{z(k)}(k) - W(k)\big)_{rs}\, \mathrm{d}k \right| < \epsilon \tag{20}$$

for every $\ell \le N$ and every matrix entry (r, s).

Proof. All denominators are bounded below by $a - b > 0$. Hence the integrands are in $L^1(K; C(\mathbb{T}))$. Approximate them by finite Fourier sums in z. Every nonzero Fourier coefficient tends to zero after substituting $z = e^{i\nu k}$, by the Riemann–Lebesgue lemma. The zero Fourier coefficient in (19) is bounded by Theorem 2.2; that in (20) equals $\psi_\ell W$ by Lemma 2.1. A finite maximum of the required frequency thresholds proves the claim. □

Theorem 6.2 (Deterministic operator-valued Riesz singularity)**.** *Fix $n < \infty$ and $0 < \beta < 1$. Starting from an integrable invertible matrix function X_0 on a compact interval K, let $H_j : K \to \mathrm{U}(n)$ be any predictable sequence. There are successively arbitrarily large collar frequencies ν_j such that the recursion*

$$X_j(k) = \big(a\mathrm{I} + be^{i\nu_j k} H_j(k)\big)^{-1} X_{j-1}(k), \qquad W_j = X_j^* X_j, \tag{21}$$

has a weak limiting positive matrix measure $\boldsymbol{\mu}$ on K which is purely singular with respect to Lebesgue measure. The choices can simultaneously satisfy any previously fixed finite lists of phase-placement inequalities.

Proof. Choose a summable error sequence. At step j, use Lemma 6.1 to obtain

$$I_j := \int_K \operatorname{tr}(W_j^{\beta})\,\mathrm{d}k \le \kappa_{\beta,n} I_{j-1} + \epsilon_j. \tag{22}$$

Choose the errors so that $I_j \to 0$. At the same step, preserve the first j members of a fixed countable dense subset of $C(K)$, against every matrix entry, with summable tail errors. The corresponding matrix moments are Cauchy. Positivity and compactness of K therefore give a weak limiting positive matrix measure $\boldsymbol{\mu}$.

It remains to prove singularity rather than only weak convergence. Select stages j_q so that I_{j_q} is as small as required below, and write $w_q = \operatorname{tr} W_{j_q}$. For a threshold T_q,

$$\left|\{w_q > T_q\}\right| \le T_q^{-1} \int_K w_q\,\mathrm{d}k, \tag{23}$$

$$\int_{\{w_q \le T_q\}} w_q\,\mathrm{d}k \le T_q^{1-\beta} I_{j_q}. \tag{24}$$

First choose T_q large and then j_q so late that there are an open set $O_q \subset K$ and a continuous function $0 \le \chi_q \le 1$, supported in O_q, for which

$$|O_q| \le 2^{-q}, \qquad \int_K (1-\chi_q) w_q\,\mathrm{d}k \le 2^{-q}. \tag{25}$$

This uses (23)–(24) and regularity of the two finite measures involved.

From stage $j_q + 1$ onward, permanently include $1 - \chi_q$ in the finite test list in Lemma 6.1. Allocate its later errors so that their sum is at most 2^{-q}. Passing to the weak limit gives

$$\operatorname{tr} \boldsymbol{\mu}(K \setminus O_q) \le 2^{1-q}. \tag{26}$$

After taking a sparse subsequence if necessary, both right-hand sides are summable. The set

$$S = \liminf_{q\to\infty} O_q$$

has Lebesgue measure zero, while the Borel–Cantelli argument applied to the trace measure and (26) gives $\operatorname{tr} \boldsymbol{\mu}(K \setminus S) = 0$. Positivity then puts every scalarization of $\boldsymbol{\mu}$ on the same null set. $\square$

Remark 6.3 (What defeats the oscillating-density counterexample). Fractional-moment decay alone is not used to pass to the weak limit. Each small concentration witness is frozen into every later collar choice. Thus future shells preserve its spectral mass up to a summable error, which is exactly the missing tail-stability input.

7 Dense probes without a growing matrix contraction

The operator-valued construction above is useful, but it is stronger than is needed to kill the absolutely continuous subspace. A countable dense family of scalar probes can be treated by a dovetailed schedule. Cells aimed at one probe need only be nonexpansive, in phase mean, for the finitely many other probes currently being protected. The exact Hardy identity supplies that nonexpansion.

Fix a triangular schedule

$$1; \quad 1,2; \quad 1,2,3; \quad \dots$$

and denote its successive entries by $\tau(j)$. Thus every positive integer occurs infinitely often and $\tau(j) \le j$.

Theorem 7.1 (Dovetailed scalar no-escape principle)**.** *Let K be a compact interval and $0 < \beta < 1$. For $p \le j$, let $w_{p,j} \ge 0$ be integrable densities on K, and put*

$$I_{p,j} = \int_K w_{p,j}^{\beta}\,\mathrm{d}k.$$

Assume that the stages can be chosen with a summable sequence $\epsilon_j > 0$ so that

$$I_{p,j} \le I_{p,j-1} + \epsilon_j, \qquad p \le j, \tag{27}$$

$$I_{\tau(j),j} \le \tfrac{1}{2} I_{\tau(j),j-1} + \epsilon_j. \tag{28}$$

Assume also that, for a fixed dense sequence (ψ_ℓ) in $C(K)$, the first j moments of the first j densities are changed by at most ϵ_j, entrywise:

$$\left| \int_K \psi_\ell (w_{p,j} - w_{p,j-1})\,\mathrm{d}k \right| \le \epsilon_j, \qquad p, \ell \le j. \tag{29}$$

Finally, whenever a continuous witness $0 \le \chi \le 1$ is declared for a probe p, assume it may be included in every later finite test list with a prescribed summable total future error.

Then the errors and witness declarations may be chosen so that every $w_{p,j}\,\mathrm{d}k$ has a weak limit μ_p, every μ_p is singular with respect to Lebesgue measure, and all μ_p are supported on one common Lebesgue-null set.

Proof. For a fixed p, inequalities (27) and (28), along the infinitely many stages at which $\tau(j) = p$, give $I_{p,j} \to 0$ after making the error tail at the qth such stage smaller than 2^{-2q}. Equation (29) makes every continuous moment Cauchy, first on the dense sequence and then on all of $C(K)$ by positivity and the uniform mass bound. Hence $w_{p,j}\,\mathrm{d}k \rightharpoonup \mu_p$.

Choose stages $j(p,q)$ with $I_{p,j(p,q)}$ sufficiently small. With $w = w_{p,j(p,q)}$, first choose a large threshold $T_{p,q}$ and then choose $j(p,q)$ so late that an open set $O_{p,q}$ and a continuous cutoff $0 \le \chi_{p,q} \le 1$, supported in $O_{p,q}$, satisfy

$$|O_{p,q}| \le 2^{-p-q}, \qquad \int_K (1 - \chi_{p,q}) w\,\mathrm{d}k \le 2^{-p-q}. \tag{30}$$

Indeed,

$$|\{w > T_{p,q}\}| \le T_{p,q}^{-1} \int_K w\,\mathrm{d}k, \qquad \int_{\{w \le T_{p,q}\}} w\,\mathrm{d}k \le T_{p,q}^{1-\beta} I_{p,j(p,q)}.$$

Permanently lock $1 - \chi_{p,q}$ with total future error at most 2^{-p-q}. Passing to the weak limit gives

$$\mu_p(K \setminus O_{p,q}) \le 2^{1-p-q}.$$

Thus μ_p is supported on $S_p = \liminf_q O_{p,q}$, and $|S_p| = 0$. The countable union $S = \bigcup_p S_p$ is still Lebesgue null and supports every μ_p. □

Corollary 7.2 (Why scalar probes suffice)**.** *Let (ϕ_p) be dense in the Hilbert space of the Schrödinger operator. If the construction in Theorem 7.1 is realized by their spectral densities on K, then*

$$P_{\mathrm{ac}}(H)\mathbf{1}_K(H) = 0.$$

Proof. Singularity of the spectral measure of ϕ_p on K says $P_{\mathrm{ac}}(H)\mathbf{1}_K(H)\phi_p = 0$. The kernel of this bounded projection is closed and contains a dense set. □

This theorem removes both the growing matrix dimension and the shrinking constant $\kappa_{\beta,n}$. At every physical stage only one current radiation vector must receive a strict contraction. All earlier probes and all locked witnesses form a finite preservation list.

8 The exact compact-output stability estimate

There is a subtle orientation issue in passing from a cell estimate to an inverse Jost column. It can be isolated completely.

Lemma 8.1 (Residual control on the ideal inverse output)**.** *Let $D_0 = a\mathrm{I} + zbH$, where H is unitary and $z \in \mathbb{T}$, and let D be invertible. For a vector x, put*

$$y_0 = D_0^{-1}x, \qquad y = D^{-1}x, \qquad M = \|D^{-1}\|.$$

If

$$\|(D - D_0)y_0\| \le \eta, \tag{31}$$

then

$$\|y - y_0\| \le M\eta, \tag{32}$$

$$\big|\|y\|^2 - \|y_0\|^2\big| \le M\eta(2e^\rho\|x\| + M\eta). \tag{33}$$

Consequently the error in the β-fractional density is bounded by the βth power of the right side of (33)*.*

Proof. Since $D_0y_0 = x = Dy$,

$$y - y_0 = -D^{-1}(D - D_0)y_0.$$

This proves (32). Also $\|D_0^{-1}\| = (a - b)^{-1} = e^\rho$, and

$$\big|\|y\|^2 - \|y_0\|^2\big| \le (\|y\| + \|y_0\|)\|y - y_0\|.$$

Use $\|y\| \le \|y_0\| + M\eta$ and $|s^\beta - t^\beta| \le |s - t|^\beta$ for $s, t \ge 0$. □

For continuous $H(k)$ and a nonvanishing continuous radiation vector $x(k)$, the family

$$\mathfrak{Y} = \Big\{(a\mathrm{I} + zbH(k))^{-1}x(k)/\|x(k)\| : k \in K,\ z \in \mathbb{T}\Big\} \tag{34}$$

is compact. Thus the Talbot cell need not be scalar on the whole angular Hilbert space. It must control its residual uniformly on $\mathfrak{Y}$, together with a finite-dimensional safety space large enough to absorb the residual. This is the correct noncircular target.

Lemma 8.2 (Reciprocal reflection gap controls the inverse)**.** *Let A, B satisfy $A^*A - B^*B = \mathrm{I}$, assume A is invertible, and put $C = BA^{-1}$. If $\|C\| \le t < 1$, then, for every unitary H and every $z \in \mathbb{T}$,*

$$\|(A + zHB)^{-1}\| \le \frac{1}{1 - t}. \tag{35}$$

For the localized hyperbolic barrier

$$A = \mathrm{I} + (\cosh\rho - 1)P, \qquad B = (\sinh\rho)P,$$

one has $C = (\tanh\rho)P$. Reflectionless unitary converters on either side do not change $\|C\|$.

Proof. Since $A^*A \ge \mathrm{I}$, $\|A^{-1}\| \le 1$. The factorization

$$A + zHB = (\mathrm{I} + zHC)A$$

and the Neumann series give (35). The localized formula is immediate, and unitary pre- and post-factors preserve singular values of the reflection contraction. □

Lemma 8.3 (A moving two-real-direction squeeze)**.** *Let H be a symmetric unitary on a complex Hilbert space, let $Q = H\mathcal{C}$, where $\mathcal{C}x = \bar{x}$, and fix a unit vector u. Put*

$$S_u = |u\rangle\langle u| + |Qu\rangle\langle Qu|. \tag{36}$$

For $\rho > 0$ define

$$A_u = \cosh(\rho S_u), \qquad B_u = H^* \sinh(\rho S_u). \tag{37}$$

Then (A_u, B_u) is reciprocal:

$$A_u^* A_u - B_u^* B_u = \mathrm{I}, \qquad A_u^* \overline{B_u} = B_u^* \overline{A_u}. \tag{38}$$

Moreover

$$A_u + zHB_u = \cosh(\rho S_u) + z \sinh(\rho S_u), \qquad z \in \mathbb{T}. \tag{39}$$

For every $0 < \beta < 1$ there is a number $q^{(2)}_{\beta,\rho} < 1$, depending only on β and ρ, such that

$$\int_{\mathbb{T}} \left\| (A_u + zHB_u)^{-1} u \right\|^{2\beta} \mathrm{d}m(z) \le q^{(2)}_{\beta,\rho}. \tag{40}$$

The assertion is dimension free. The generator S_u has rank at most two and depends norm-continuously on (H, u).

Proof. Since H is symmetric and unitary, Q is a conjugation. The operator S_u is positive, has rank at most two, and satisfies $QS_uQ = S_u$. Equivalently,

$$H\overline{S_u} = S_u H. \tag{41}$$

Functional calculus and (41) give (38); the first identity is simply $\cosh^2(\rho S_u) - \sinh^2(\rho S_u) = \mathrm{I}$. They also give (39).

Let $r = |\langle u, Qu\rangle| \in [0,1]$ and set

$$g_{\beta,\rho}(s) = \int_{\mathbb{T}} |\cosh(\rho s) + z\sinh(\rho s)|^{-2\beta}\, \mathrm{d}m(z), \qquad 0 \le s \le 2. \tag{42}$$

The spectral measure of S_u at u has atoms at $1 + r$ and $1 - r$ with weights $w_+ = (1 + r)/2$ and $w_- = (1 - r)/2$, respectively. Put $t_\pm = \tanh(\rho(1 \pm r))$ and $P_t(z) = (1 - t^2)/|1 - tz|^2$. The squared norm inside (40) is

$$s_r(z) = w_+ P_{-t_+}(z) + w_- P_{-t_-}(z). \tag{43}$$

Consequently the left side of (40) is

$$G_{\beta,\rho}(r) = \int_{\mathbb{T}} s_r(z)^\beta\, \mathrm{d}m(z). \tag{44}$$

The scalar Hardy identity gives $\int_{\mathbb{T}} s_r\, \mathrm{d}m = 1$. The first nonconstant Fourier coefficient of (43) is $-(w_+t_+ + w_-t_-)$, which is strictly negative for every $r \in [0,1]$. Hence s_r is nonconstant and strict concavity gives $G_{\beta,\rho}(r) < 1$. The integrand is jointly continuous in (r, z), including at $r = 1$, so $G_{\beta,\rho}$ is continuous on $[0,1]$. Thus

$$q^{(2)}_{\beta,\rho} := \max_{0 \le r \le 1} G_{\beta,\rho}(r) < 1.$$

Finally (36) proves the asserted norm continuity. $\square$

Remark 8.4 (Why this bypasses the root-of-unity obstruction). For a physical Jost matrix, $H = J^{-1}\overline{J}$ satisfies $H\overline{H} = \mathrm{I}$; together with unitarity this gives $H^T = H$. Lemma 8.3 therefore applies to every current scalar radiation vector. It does not squeeze a preassigned growing angular block. It moves a reciprocal rank-two squeeze with the target vector, and so the root-of-unity example in Proposition 3.1 is irrelevant to this scalar dense-probe mechanism.

At the distinguished phase $z = 1$, the sharper projection version is also useful. If P projects onto the Q-invariant space $\mathrm{span}_{\mathbb{C}}\{u, Qu\}$ and

$$A = \mathrm{I} + (\cosh\rho - 1)P, \qquad B = (\sinh\rho)H^*P,$$

then $A + HB = \mathrm{I} + (e^{\rho} - 1)P$. Thus the target is multiplied by $e^{-\rho}$ and the reciprocal phase is unchanged:

$$(A + HB)^{-1}H\,\overline{(A + HB)} = H.$$

This exact repeatability is hidden by a scalar full-block ansatz.

Lemma 8.5 (Physical reciprocal Poisson identity)**.** *Let $A, B \in \mathbb{C}^{n\times n}$ satisfy*

$$A^*A - B^*B = \mathrm{I}, \tag{45}$$

let $H \in \mathrm{U}(n)$, and put

$$F_z = (A + zHB)^{-1}, \qquad z \in \mathbb{T}.$$

Then A is invertible, F_z is invertible for every $z \in \mathbb{T}$, and

$$\int_{\mathbb{T}} F_z^*F_z \,\mathrm{d}m(z) = \mathrm{I}. \tag{46}$$

Consequently, for every matrix X and every $0 < \beta < 1$,

$$\int_{\mathbb{T}} X^*F_z^*F_zX \,\mathrm{d}m(z) = X^*X, \tag{47}$$

$$\int_{\mathbb{T}} \mathrm{tr}\Big[(X^*F_z^*F_zX)^{\beta}\Big] \,\mathrm{d}m(z) \le \mathrm{tr}\Big[(X^*X)^{\beta}\Big]. \tag{48}$$

The same conclusion for a scalar probe is

$$\int_{\mathbb{T}} \|F_zx\|^{2\beta}\,\mathrm{d}m(z) \le \|x\|^{2\beta}.$$

Proof. Equation (45) gives $A^*A \succeq \mathrm{I}$, so A is invertible. Set $C = BA^{-1}$ and $T = -HC$. Then

$$C^*C = A^{-*}B^*BA^{-1} = \mathrm{I} - A^{-*}A^{-1}, \qquad \|C\| < 1, \tag{49}$$

and

$$F_z = A^{-1}(\mathrm{I} - zT)^{-1}.$$

Therefore

$$F_z^*F_z = (\mathrm{I} - \bar{z}T^*)^{-1}(\mathrm{I} - T^*T)(\mathrm{I} - zT)^{-1}.$$

Termwise integration of the norm-convergent Neumann series gives

$$\begin{aligned}\int_{\mathbb{T}} F_z^*F_z\,\mathrm{d}m(z) &= \sum_{q\ge 0} T^{*q}(\mathrm{I} - T^*T)T^q \\ &= \mathrm{I} - \lim_{q\to\infty} T^{*(q+1)}T^{q+1} = \mathrm{I}.\end{aligned}$$

This proves (46) and (47). Operator concavity of $Y \mapsto Y^{\beta}$, followed by the trace, proves (48). $\square$

Remark 8.6 (Why this is stronger than the ideal Hardy identity). No scalarity or commutativity of A, B, and H is used. Thus an outer physical cell may behave arbitrarily away from its tuning window: after a rapid scalar collar phase is averaged, every protected quadratic density is preserved and every protected fractional moment is nonexpansive. In a finite reachable block, deterministic large collar frequencies realize these phase means by the same Fourier approximation and Riemann–Lebesgue argument as Lemma 6.1. Proposition 8.9 then transfers the assertion to the specified finite list of full outgoing probe densities, with a prescribed summable error. This is the preservation input used in Proposition 10.7; by itself it does not supply a strict contraction away from the cell's tuning window.

Thus a cell with scattering-norm error $\varepsilon < 1 - \tanh\rho$ has an inverse bound depending on ρ and ε, but not on its radial location or on an evanescent transfer condition number. Combining this with Lemma 8.1 shows that it is enough to make the direct residual $o(1 - \tanh\rho)$ on the finite safety block.

Remark 8.7 (Why a global inverse bound may not be assumed). At a fixed radial shell, sufficiently high angular momenta are evanescent. Their inward/outward transfer coefficients can be exponentially ill conditioned even though the physical scattering matrix is unitary. Hence a formal use of Lemma 8.1 with a full-space constant M is invalid. One must first eliminate the evanescent tail by an outgoing Schur complement and apply the lemma on a finite reachable block. This is the precise place where the earlier scale ledger was incomplete.

The required elimination is stable at the resolvent level. This avoids every raw evanescent transfer condition number.

Lemma 8.8 (Schur removal measured from the full outgoing problem). *Let*

$$\mathcal{A} = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix}$$

be an invertible closed block operator, with C invertible, and put $\Sigma = B^*C^{-1}B$ and $S = A - \Sigma$. If $\|\mathcal{A}^{-1}\| \le M_0$ and $M_0\|\Sigma\| < 1$, then A is invertible and

$$\|A^{-1} - S^{-1}\| \le \frac{M_0^2\|\Sigma\|}{1 - M_0\|\Sigma\|}. \tag{50}$$

In particular, the right side is at most $2M_0^2\|\Sigma\|$ when $M_0\|\Sigma\| \le 1/2$.

Proof. The upper-left block of $\mathcal{A}^{-1}$ is S^{-1}, so $\|S^{-1}\| \le M_0$. Since $A = S + \Sigma$,

$$A^{-1} = (\mathrm{I} + S^{-1}\Sigma)^{-1}S^{-1}.$$

The Neumann bound and the resolvent identity give (50). □

Proposition 8.9 (Explicit outgoing buffer stability). *Let U be a fixed bounded real potential supported in $1 \le r \le R$, let $K = [k_0, k_1] \Subset (0, \infty)$, and let $\mathcal{A}_U(k)$ be the Dirichlet problem on $(1, R)$ with the exact free outgoing Dirichlet-to-Neumann condition at R. Then*

$$M_{U,K} := \sup_{k\in K} \|\mathcal{A}_U(k)^{-1}\| < \infty. \tag{51}$$

If P_J contains the angular degrees at most J and

$$g_{J,R} = \frac{(J+1)(J+2)}{R^2} - \|U\|_\infty - k_1^2 > 0,$$

then, once $M_{U,K}\|U\|_\infty^2/g_{J,R} \le 1/2$, the full and P_J-truncated outgoing resolvents, compressed to P_J, differ by at most

$$\sup_{k\in K} \|P_J\mathcal{A}_U(k)^{-1}P_J - \mathcal{A}_{U,J}(k)^{-1}\| \le 2M_{U,K}^2\frac{\|U\|_\infty^2}{g_{J,R}}. \tag{52}$$

The same bound, multiplied by $\|f\|^2/\pi$, controls the difference of the spectral densities of every source f lying in the retained block.

Proof. The outgoing boundary problem is elliptic and Fredholm. A real-axis homogeneous solution has zero outgoing flux by Green's identity; positivity of the channel flux forces its far-field amplitude to vanish, and unique continuation gives the zero solution. Thus the problem is invertible. Continuous dependence on $k \in K$ proves (51).

On the omitted sector, centrifugal coercivity gives $\|C^{-1}\| \le g_{J,R}^{-1}$, while $\|B\| \le \|U\|_\infty$. Hence $\|\Sigma\| \le \|U\|_\infty^2/g_{J,R}$. Apply Lemma 8.8. Finally, the density is the imaginary part of the outgoing resolvent quadratic form divided by π. □

9 Full-block Talbot concentration: what actually changes

Let

$$A_3 = -\Delta_{\mathbb{S}^2},$$

and let $h_L(\theta) = h(L\theta)$ be a fast zonal factor. For an arbitrary smooth envelope $f(\theta, \varphi)$ the invariant product identity is

$$A_3(h_L f) = (A_3 h_L) f - 2\nabla h_L \cdot \nabla f + h_L A_3 f. \tag{53}$$

The first term is exactly the term in the existing zonal Talbot proof. On a fixed finite-dimensional $\mathcal{K} \subset C^\infty(\mathbb{S}^2)$, the two new terms obey

$$\sup_{\substack{f\in\mathcal{K} \\ \|f\|_2=1}} (\|\nabla h_L \cdot \nabla f\|_2 + \|h_L A_3 f\|_2) = O_{\mathcal{K},h}(L). \tag{54}$$

After a Talbot flight of length $O(L^{-2})$, this is $O(L^{-1})$. The polar term already present in $A_3 h_L$ has the known bound $O(L\sqrt{\log L})$, hence contributes $O(\sqrt{\log L}/L)$. No new φ-singularity appears in (53).

Proposition 9.1 (Finite-block extension of the Talbot lemma)**.** *The scalar polar estimate used in the zonal Talbot proposition holds by splitting the two polar caps at scale L^{-1} and integrating $\cot^2\theta \sin\theta$ on their complement. That proposition and its reciprocal inverse therefore hold, with the same $O(\sqrt{\log L}/L)$ rate, for every fixed finite-dimensional conjugation-invariant $\mathcal{K} \subset C^\infty(\mathbb{S}^2)$, without a zonality assumption. Consequently the Riesz concentration, small-support cutoff, and reverse refocusing estimates also hold uniformly on the unit sphere of $\mathcal{K}$.*

Proof. Use (53) in the same Duhamel comparison as in the zonal proof. Equation (54) is uniform because all smooth norms are equivalent on a fixed finite-dimensional space. The leading fast term and the polar remainder are unchanged. This proves the one-factor estimate. For a product, choose the frequencies successively; after finitely many previous finite-band factors the set of possible envelopes is still finite dimensional. The uniform Riemann–Lebesgue and Chernoff steps used for the tilted Riesz density apply to the compact unit sphere of this enlarged space. Reverse the factors to obtain refocusing. □

Lemma 9.2 (A fixed height realizes arbitrarily strong reciprocal barriers)**.** *Fix $k_\star > 0$. There are a number $B < \infty$ and a smooth real elementary one-dimensional cell such that, using finitely many translates of this cell and free phase intervals, one can realize at $k_\star$ every Cartan boost*

$$\begin{pmatrix} \cosh\rho & \sinh\rho \\ \sinh\rho & \cosh\rho \end{pmatrix}, \qquad \rho > 0,$$

up to arbitrary prescribed error, while the resulting potential is bounded by B. Its length and radial L^2 cost are finite numbers ℓ_ρ, C_ρ depending on ρ, and its reflection contraction is at most $\tanh(\rho + C_0)$ for one fixed C_0.

Proof. The transfer matrix of a nonreflectionless smooth real bump belongs to $\mathrm{SU}(1,1)$ and has a Cartan decomposition $R_\alpha B_{\rho_0} R_\gamma$ with $\rho_0 > 0$. Free intervals realize the rotations R_θ. Cancel the end rotations between successive copies; n aligned copies give $B_{n\rho_0}$. A final bump whose coupling is varied from zero to its fixed value has a continuous Cartan parameter joining zero to ρ_0, and therefore supplies every remainder in $[0, \rho_0]$ after its end rotations are compensated. No copy exceeds the fixed bump height. Cartan parameters are subadditive under multiplication, so every partially switched or transition cell has parameter at most $\rho + C_0$, which proves the reflection bound. Finitely many bounded compact cells have finite length and L^2 cost. ☐

Lemma 9.3 (The physical scale inequalities are compatible)**.** *Fix a finite Riesz depth m, hyperbolic angle ρ, largest integer frequency ratio Q, and accuracy $0 < \varepsilon < 1$. Choose the finitely many weak-slab lengths equal to a common T; let Λ be the common carrier, $D = Q\Lambda$ the largest physical angular frequency, R the cell radius, J the active outgoing cutoff, and L the Galerkin cutoff. Let $\ell_\rho < \infty$ be the radial length of any fixed bounded one-dimensional realization of the desired hyperbolic reflector. The scales can be chosen successively so that*

$$T^{-1} < \varepsilon, \qquad \frac{D^2 T}{R^2} + \frac{T}{R} < \varepsilon, \qquad \frac{D^2(1+\ell_\rho)}{R^2} < \varepsilon, \tag{55}$$

$$\frac{\sqrt{\log \Lambda}}{\Lambda} < \varepsilon, \qquad \frac{R}{\Lambda^2} < \varepsilon, \qquad \frac{R}{J} + \frac{J}{L} < \varepsilon. \tag{56}$$

The weak-mask X_3 cost can simultaneously be made smaller than ε. None of these choices changes the reflector cost $C_\rho e^{-cm}$.

Proof. First choose all weak-slab lengths so large that the sum of their $T^{-1}\|w\|_2^2$ costs and their one-dimensional errors is below ε. Keeping this finite T, choose Λ so large that

$$Q\Lambda\sqrt{T/\varepsilon} + T/\varepsilon + Q\Lambda\sqrt{(1+\ell_\rho)/\varepsilon} \ll \varepsilon\Lambda^2$$

and the polar error in (56) is below ε. There is then an R strictly between the left side and $\varepsilon\Lambda^2$. This gives (55) and the middle inequality in (56). Finally choose J/R and then L/J arbitrarily large. The carrier, radii, and cutoffs do not enter the charged cost, proving the last assertion. ☐

Lemma 9.4 (Window count does not consume the X_3 budget)**.** *Fix ρ and let $A_\rho < \infty$ be any accuracy amplification required by Lemmas 8.1 and 8.2. Suppose a depth-m cell has chromatic error $Cm|k/k_\star - 1| + \varepsilon$ and reflector cost $C_\rho e^{-cm}$. For every generation budget $\zeta > 0$, one can choose m, a finite cover of K by tuning windows, and the physical errors of the cells so that their total integrated β-fractional density error and their total X_3 cost are both at most ζ.*

Proof. Partition K into windows of length at most

$$h_m = \frac{1}{CA_\rho m}\left(\frac{\zeta}{4|K|}\right)^{1/\beta}.$$

There are at most $N_m \le C_K A_\rho m \zeta^{-1/\beta}$ windows. On their union the integrated fractional error caused by the chromatic term is bounded by

$$\sum_i \int_{I_i} (A_\rho C m h_m)^\beta \, \mathrm{d}k \le \zeta/4.$$

Choose the nonchromatic residual in every target window no larger than Cmh_m; it contributes the same bound. Since $N_m C_\rho e^{-cm} \to 0$, choose m so large that the total reflector cost is at most $\zeta/2$. All errors which act on the whole of K, including phase-placement and protected-test errors, are instead allocated a summable budget $O(\zeta/N_m)$ per cell. Lemma 9.3 realizes those finite requests. Further finite preservation tests only subdivide this allocation. ☐

Proposition 9.5 (Local contraction without coverage is insufficient)**.** *Let K have positive length and let (h_j) be positive with $\sum_j h_j < |K|$. For every adaptive choice of intervals $I_j \subset K$ with $|I_j| \le h_j$, there are nonnegative densities w_j such that the jth step contracts w_{j-1}^{β} by a fixed factor on I_j, is exactly nonexpansive off I_j, and may be assigned arbitrarily small formal cost, but*

$$\inf_j \int_K w_j^{\beta}\,\mathrm{d}k > 0.$$

Consequently, one-window cells plus a summable cost ledger do not imply the global hypothesis (28)*.*

Proof. Put $w_0 = 1$ and define

$$w_j(k) = \begin{cases} 2^{-1/\beta} w_{j-1}(k), & k \in I_j, \\ w_{j-1}(k), & k \notin I_j. \end{cases}$$

Every stated local property holds. But $E = K \setminus \bigcup_j I_j$ has $|E| \ge |K| - \sum_j h_j > 0$, and $w_j = 1$ on E for every j. □

Proposition 9.6 (Singular phases have no uniform chromatic modulus)**.** *Let H_N be diagonal with the Nth roots of unity and let u_N give each eigenvalue weight $1/N$. Put $t = \tanh\rho$ and*

$$Q_{\beta,N}(t) = \int_{\mathbb{T}} \|(\cosh\rho\,\mathrm{I} + z\sinh\rho\,H_N)^{-*}u_N\|^{2\beta}\,\mathrm{d}m(z).$$

For every fixed $q < 1$ there is $c_{\beta,q} > 0$ such that

$$Q_{\beta,N}(t) \le q \quad\Longrightarrow\quad \rho \ge \tfrac{1}{2}\log N - c_{\beta,q}, \qquad \frac{1}{1-t} \ge c_{\beta,q}N. \tag{57}$$

Thus singularity of the reciprocal phase, even for a finite-rank perturbation of a pure-point unitary, supplies no dimension-free lower bound on the physical tuning-window width.

Proof. The root-of-unity average collapses the Poisson series exactly:

$$\|(\cosh\rho\,\mathrm{I} + z\sinh\rho\,H_N)^{-*}u_N\|^2 = 1 + 2\sum_{j\ge1}(-t)^{jN}\cos(jN\arg z).$$

Up to a rotation and the measure-preserving map $z \mapsto z^N$, this is the Poisson kernel P_{t^N}. Hence

$$Q_{\beta,N}(t) = \int_{\mathbb{T}} P_{t^N}(z)^{\beta}\,\mathrm{d}m(z).$$

The last integral is continuous in $r = t^N$ and equals one at $r = 0$. Therefore $Q_{\beta,N}(t) \le q < 1$ forces $t^N \ge r_{\beta,q} > 0$. It follows that $1-t \le -\log t \le -N^{-1}\log r_{\beta,q}$, proving the second assertion; $e^{2\rho} = (1+t)/(1-t)$ proves the first. The example acts on a finite subspace and can be extended by any fixed pure-point diagonal unitary on its orthogonal complement, changing that unitary by finite rank. □

10 Synchronization before contraction: the grouped Lemma B

The one-step root-of-unity obstruction does not survive a finite reciprocal burn-in. The reason is that the same scalar cells which may initially have almost no fractional-moment gap synchronize every finite reciprocal phase. After synchronization, a fixed number of further cells contracts every vector in the block. This is the missing grouping which is absent from the one-step formulation of Lemma B.

Put

$$\phi_z(h) = \frac{ah + \bar{z}b}{a + zbh}, \qquad h, z \in \mathbb{T}, \tag{58}$$

where $a = \cosh\rho$ and $b = \sinh\rho$.

Lemma 10.1 (Uniform reciprocal synchronization). *There are $0 < \alpha < 1$ and $0 < \sigma < 1$, depending only on ρ, such that*

$$\int_{\mathbb{T}} |\phi_z(h) - \phi_z(\lambda)|^\alpha \, \mathrm{d}m(z) \le \sigma |h - \lambda|^\alpha, \qquad h, \lambda \in \mathbb{T}. \tag{59}$$

Consequently, if $H_0 \in \mathrm{U}(n)$ and $H_j = \phi_{z_j}(H_{j-1})$ for independent Haar phases, then for every $0 < \delta < 2$,

$$\mathbb{P}\{\operatorname{diam} \sigma(H_M) > \delta\} \le n^2 (2/\delta)^\alpha \sigma^M. \tag{60}$$

Proof. The exact chord identity is

$$\frac{|\phi_z(h) - \phi_z(\lambda)|}{|h - \lambda|} = \frac{1}{|a + zbh| \, |a + zb\lambda|}. \tag{61}$$

Let

$$Y_z = -\log|a + zbh| - \log|a + zb\lambda|.$$

Since $a - b = e^{-\rho}$ and $a + b = e^\rho$, one has $|Y_z| \le 2\rho$. Jensen's formula gives, uniformly in $h, \lambda \in \mathbb{T}$,

$$\int_{\mathbb{T}} Y_z \, \mathrm{d}m(z) = -2 \log a < 0.$$

Hoeffding's lemma therefore yields

$$\int_{\mathbb{T}} e^{\alpha Y_z} \, \mathrm{d}m(z) \le \exp\Big(-2\alpha \log a + 2\alpha^2 \rho^2\Big). \tag{62}$$

Choose $0 < \alpha < \min\{1, (\log a)/\rho^2\}$ and call the right side $\sigma < 1$. This proves (59).

The eigenvalues of H_j are obtained by applying the same scalar map (58) to the eigenvalues of H_{j-1}. Iterate (59), apply Markov's inequality to every pair, and take the union bound over at most n^2 ordered pairs. This proves (60). $\square$

Theorem 10.2 (Dimension-free grouped contraction). *Fix $\rho > 0$. There are $0 < \beta < 1$, an integer $R \ge 1$, and a constant $q_\star < 1$, depending only on ρ, with the following property. For every $n \ge 1$ there is an integer*

$$M(n) \le C_\rho \log(n + 1) + C_\rho \tag{63}$$

such that, for every $H_0 \in \mathrm{U}(n)$ and every matrix X_0, the ideal scalar recursion

$$F_j = (a\mathrm{I} + z_j b H_{j-1})^{-1}, \qquad X_j = F_j X_{j-1}, \tag{64}$$

$$H_j = (a\mathrm{I} + z_j b H_{j-1})^{-1}(a H_{j-1} + \bar{z}_j b \mathrm{I}) \tag{65}$$

satisfies

$$\int_{\mathbb{T}^{M(n)+R}} \operatorname{tr}\Big[(X_{M(n)+R}^* X_{M(n)+R})^\beta\Big] \, \mathrm{d}m^{M(n)+R} \le q_\star \operatorname{tr}\Big[(X_0^* X_0)^\beta\Big]. \tag{66}$$

The same estimate holds, with the same $q_\star$, for $\sum_{p=1}^N \|x_{p,M(n)+R}\|^{2\beta}$ starting from any finite family of vectors and evolving each one by the same factors F_j. Thus one group *of scalar full-block cells has a fixed contraction factor even though no single cell does.*

Proof. Take α, σ from Lemma 10.1. Choose $\beta > 0$ so small that

$$\theta := \sigma e^{2\beta\rho} < 1. \tag{67}$$

For

$$q_{\beta,\rho} = \int_{\mathbb{T}} |a + zb|^{-2\beta} \, \mathrm{d}m(z) < 1,$$

choose $\delta_0 > 0$ so small that

$$q_c := (1 - be^{\rho}\delta_0)^{-2\beta} q_{\beta,\rho} < 1. \tag{68}$$

Then choose R with $q_c^R \le 1/4$ and put $\delta = \delta_0 e^{-2\rho R}$.

Let $\mathcal{G}$ be the event $\operatorname{diam}\sigma(H_M) \le \delta$. On $\mathcal{G}$, choose one eigenvalue λ of H_M. Normality gives $\|H_M - \lambda \mathrm{I}\| \le \delta$. The chord identity shows that one further Möbius step can enlarge spectral diameter by at most $e^{2\rho}$. Therefore throughout the next R steps the current phase is within δ_0 of a scalar unitary. The resolvent factorization

$$(a\mathrm{I} + zbH)^{-1} = (a + zb\lambda)^{-1}\left[\mathrm{I} + \frac{zb}{a + zb\lambda}(H - \lambda\mathrm{I})\right]^{-1} \tag{69}$$

and (68) imply, conditionally at each of these R steps,

$$\int_{\mathbb{T}} \operatorname{tr}[(X^* F_z^* F_z X)^{\beta}]\, \mathrm{d}m(z) \le q_c \operatorname{tr}[(X^* X)^{\beta}]. \tag{70}$$

During the first M steps, Lemma 8.5 gives exact phase-mean nonexpansion. Pointwise, however, $\|F_j\| \le (a - b)^{-1} = e^{\rho}$, so

$$\operatorname{tr}[(X_M^* X_M)^{\beta}] \le e^{2\beta\rho M} \operatorname{tr}[(X_0^* X_0)^{\beta}]. \tag{71}$$

Combining (60), (67), and (71) gives

$$\int_{\mathcal{G}^c} \operatorname{tr}[(X_M^* X_M)^{\beta}]\, \mathrm{d}m^M \le n^2 (2/\delta)^{\alpha} \theta^M \operatorname{tr}[(X_0^* X_0)^{\beta}]. \tag{72}$$

Choose $M = M(n)$ so that the coefficient on the right is at most $1/4$. This has the logarithmic bound (63). On $\mathcal{G}$, iterate (70); on $\mathcal{G}^c$, use phase-mean nonexpansion for the last R cells. The two contributions are at most $1/4$ each. Hence (66) holds with $q_\star = 1/2$. For a finite vector family, apply the scalar form of Lemma 8.5 during burn-in and sum the good-event resolvent bound and the bad-event pointwise bound over p. No constant depends on the number of vectors. □

Theorem 10.3 (Fixed-height grouped contraction for singular bundles)**.** *Fix $\rho > 0$ and $0 < \beta < 1$. Let $\mathfrak{M}$ be a weakly compact family of probability measures on $\mathbb{T}$, every member of which is singular with respect to Haar measure. For every $0 < q < 1$ there is an integer $M = M(\rho, \beta, \mathfrak{M}, q)$ such that the following holds.*

For $z \in \mathbb{T}$ put

$$S(z) = \begin{pmatrix} a & \bar{z}b \\ zb & a \end{pmatrix}, \qquad S(z_M)\cdots S(z_1) = \begin{pmatrix} A_M & \overline{B_M} \\ B_M & \overline{A_M} \end{pmatrix}. \tag{73}$$

Then

$$\sup_{\nu \in \mathfrak{M}} \int_{\mathbb{T}^M} \left(\int_{\mathbb{T}} |B_M h + \overline{A_M}|^{-2}\, \mathrm{d}\nu(h) \right)^{\beta} \mathrm{d}m^M \le q. \tag{74}$$

Consequently, if H is unitary and the normalized spectral measures of a finite vector family $x_1, \dots, x_N$ belong to $\mathfrak{M}$, then the ideal recursion (64)–(65) satisfies

$$\int_{\mathbb{T}^M} \sum_{p=1}^{N} \|x_{p,M}\|^{2\beta}\, \mathrm{d}m^M \le q \sum_{p=1}^{N} \|x_p\|^{2\beta}. \tag{75}$$

The height ρ is fixed; only the finite group length depends on the singular bundle.

Proof. The matrices in (73) lie in $\mathrm{SU}(1,1)$, so

$$|A_M|^2 - |B_M|^2 = 1. \tag{76}$$

Let $r_M = |B_M/A_M|$. Conjugation by $K_\vartheta = \mathrm{diag}(e^{i\vartheta/2}, e^{-i\vartheta/2})$ sends $S(z)$ to $S(ze^{-i\vartheta})$. Product Haar measure is invariant under this simultaneous rotation, while conjugation fixes A_M and rotates B_M. Hence, conditionally on r_M, the argument of $-B_M/\overline{A_M}$ is Haar-uniform. By (76), averaging first over that argument gives the exact identity

$$\mathbb{E}\left(\int_{\mathbb{T}} |B_M h + \overline{A_M}|^{-2}\, \mathrm{d}\nu(h)\right)^\beta = \mathbb{E}\, Q_{r_M}(\nu), \tag{77}$$

where $Q_r(\nu) = \int_{\mathbb{T}} (P_r * \nu)^\beta\, \mathrm{d}m$ is the quantity used in the proof of Theorem 4.1.

It remains to show that r_M approaches one. With $L_j = \log|A_j|$ and the product taken in the order in (73),

$$L_j - L_{j-1} = \log\left|a + \bar{z}_j b \frac{B_{j-1}}{A_{j-1}}\right|.$$

Jensen's formula and $|B_{j-1}/A_{j-1}| < 1$ give

$$\mathbb{E}(L_j - L_{j-1} \mid z_1, \ldots, z_{j-1}) = \log a > 0, \tag{78}$$

and every increment lies in $[-\rho, \rho]$. Azuma–Hoeffding therefore gives

$$\mathbb{P}\left\{L_M < \tfrac{1}{2} M \log a\right\} \le \exp\left[-\frac{M(\log a)^2}{8\rho^2}\right]. \tag{79}$$

On the complementary event,

$$r_M = \sqrt{1 - |A_M|^{-2}} \ge \sqrt{1 - a^{-M}}. \tag{80}$$

The proof of Theorem 4.1 shows that $Q_r(\nu) \downarrow 0$ as $r \uparrow 1$ for every singular ν, uniformly on the compact set $\mathfrak{M}$ by monotonicity, continuity, and Dini's theorem. Also $0 \le Q_r(\nu) \le 1$. Choose $r_0 < 1$ so that $\sup_{\nu \in \mathfrak{M}} Q_{r_0}(\nu) \le q/2$, and then choose M so large that (79) is at most $q/2$ and the right side of (80) is at least r_0. Equation (77) proves (74).

Finally, direct multiplication of the scalar Möbius factors gives

$$x_{p,M} = (B_M H + \overline{A_M}\mathrm{I})^{-1} x_p. \tag{81}$$

Apply (74) to each normalized vector spectral measure, multiply by $\|x_p\|^{2\beta}$, and sum over p. $\square$

Proposition 10.4 (Grouped root-of-unity lower bound and the Zeno certificate)**.** *Fix $\rho > 0$, $0 < \beta < 1$, and $0 < q < 1$. Let ν_N give mass $1/N$ to the Nth roots of unity. For the product in (73), put*

$$\mathcal{Q}_{M,N} = \int_{\mathbb{T}^M} \left(\int_{\mathbb{T}} |B_M h + \overline{A_M}|^{-2}\, \mathrm{d}\nu_N(h)\right)^\beta \mathrm{d}m^M. \tag{82}$$

There is $C_{\beta,q,\rho} < \infty$ such that

$$\mathcal{Q}_{M,N} \le q \quad \Longrightarrow \quad M \ge \frac{\log N}{2\rho} - C_{\beta,q,\rho}. \tag{83}$$

Consequently singularity alone gives no nonsummable lower bound for the windows certified by the present Riesz–Talbot estimate. Indeed, if a group of M_j depth-m_j cells is certified only through

$$\mathrm{err}_j(k) \le C M_j m_j |k/k_j - 1| + \varepsilon_j, \tag{84}$$

then the standard certified half-width at fixed error tolerance η is $h_j = \eta/(CM_jm_j)$. Taking $N_j = \lceil e^{j^2} \rceil$ in (83) gives

$$\sum_{j\geq 1} h_j < \infty \qquad (m_j \geq 1). \tag{85}$$

Thus the implication "singular phase plus one-window contraction implies global hazard divergence" is false at the level of the currently proved black-box estimates. A proof of Problem 10.8 must use additional structure of the actually generated phase bundles or a genuinely broadband cancellation; it cannot follow from singularity alone.

Proof. Write $r_M = |B_M/A_M|$. The Cartan length on $\mathrm{SU}(1,1)$ is subadditive, so

$$r_M \leq \tanh(M\rho). \tag{86}$$

As in the root-of-unity calculation in Proposition 9.6, averaging over the conditional Haar centre of the product gives

$$\mathcal{Q}_{M,N} = \mathbb{E}\, Q_\beta(r_M^N), \qquad Q_\beta(s) = \int_{\mathbb{T}} P_s(z)^\beta \,\mathrm{d}m(z). \tag{87}$$

The function Q_β decreases continuously from 1 to 0. Hence (86) implies

$$\mathcal{Q}_{M,N} \geq Q_\beta(\tanh(M\rho)^N).$$

If the left side is at most q, then $\tanh(M\rho)^N \geq s_{\beta,q} > 0$. Therefore $1 - \tanh(M\rho) \leq C_{\beta,q}/N$. Since $1 - \tanh x = 2/(e^{2x}+1)$, this is (83). Substituting $N_j = \lceil e^{j^2} \rceil$ into the stated certified width proves (85). □

Proposition 10.5 (A pure-point Möbius orbit can accumulate on Haar measure)**.** *For*

$$g = \begin{pmatrix} A & \overline{B} \\ B & \overline{A} \end{pmatrix} \in \mathrm{SU}(1,1), \qquad |A|^2 - |B|^2 = 1,$$

define the projective action on probability measures on $\mathbb{T}$ by

$$\Psi_g(h) = \frac{Ah + \overline{B}}{Bh + \overline{A}}, \tag{88}$$

$$\mathcal{T}_g\nu = \frac{(\Psi_g)_*(|Bh + \overline{A}|^{-2}\nu)}{\displaystyle\int_{\mathbb{T}} |Bh + \overline{A}|^{-2} \,\mathrm{d}\nu(h)}. \tag{89}$$

There are a pure-point probability measure ν on $\mathbb{T}$ and elements $g_n \in \mathrm{SU}(1,1)$ such that

$$\mathcal{T}_{g_n}\nu \longrightarrow m \quad \text{weakly}, \tag{90}$$

where m is Haar measure.

Consequently, for every fixed group length M and fixed cell height $\rho > 0$,

$$\sup_{g\in \mathrm{SU}(1,1)} \int_{\mathbb{T}^M} \left(\int_{\mathbb{T}} |B_M h + \overline{A_M}|^{-2} \,\mathrm{d}(\mathcal{T}_g\nu)(h) \right)^\beta \mathrm{d}m^M = 1. \tag{91}$$

Thus even a single pure-point starting phase need not have a compact singular Möbius-history bundle. The group length in Theorem 10.3 cannot be made uniform over all prior ideal scalar histories without an additional restriction.

Proof. Write
$$K_g(h) = |Bh + \overline{A}|^{-2}.$$
Up to a rotation, K_g is a Poisson kernel. A boost whose attracting endpoint tends to $p \in \mathbb{T}$ has
$$K_g(h) \longrightarrow 0 \quad \text{uniformly on compact subsets of } \mathbb{T} \setminus \{p\}. \tag{92}$$
The unnormalized transformations
$$\mathcal{L}_g \nu = (\Psi_g)_*(K_g \nu)$$
compose according to the group law. In particular, $\mathcal{T}_g$ is invertible with inverse $\mathcal{T}_{g^{-1}}$, and if $\sigma = \mathcal{T}_{g^{-1}}\mu$, then
$$\mathcal{L}_g \sigma = Z_g(\sigma)\mu, \qquad Z_g(\sigma) = \frac{1}{\displaystyle\int_{\mathbb{T}} K_{g^{-1}} \, d\mu}. \tag{93}$$

Let μ_n be normalized counting measure on the N_nth roots of unity, where $N_n \to \infty$, and let $\varepsilon_n \downarrow 0$. We construct a purely atomic mixture
$$\nu = a_0 \delta_{h_0} + \sum_{n \geq 1} a_n \sigma_n, \qquad a_n > 0, \quad \sum_{n \geq 0} a_n = 1,$$
by induction. Suppose the first $n-1$ finite clusters have been chosen. Choose two boundary points which avoid the finite support already present and the finite support of μ_n. Along boosts g with these attracting and repelling endpoints, (92) makes
$$\int K_g \, d\left(a_0 \delta_{h_0} + \sum_{j<n} a_j \sigma_j \right) \longrightarrow 0,$$
while (93), with $\sigma = \mathcal{T}_{g^{-1}}\mu_n$, makes $Z_g(\sigma)$ tend to infinity. After choosing $a_n > 0$ small enough to satisfy all previously imposed tail bounds, take the boost sufficiently long that the past contribution is at most
$$\varepsilon_n a_n Z_{g_n}(\sigma_n), \qquad \sigma_n = \mathcal{T}_{g_n^{-1}}\mu_n.$$
Once g_n is fixed, K_{g_n} is bounded. Choose all later masses so that their total K_{g_n} contribution is at most the same quantity. A standard diagonal choice, for example making each new mass smaller than one half of every outstanding tail allowance, satisfies all these inequalities and leaves a positive residual mass a_0.

By linearity of $\mathcal{L}_{g_n}$ before normalization, the nth cluster contributes exactly $a_n Z_{g_n}(\sigma_n)\mu_n$, while the combined past and future contributions have at most $2\varepsilon_n$ times its mass. Hence
$$\|\mathcal{T}_{g_n}\nu - \mu_n\|_{\mathrm{TV}} \leq 4\varepsilon_n.$$
Since $\mu_n \to m$ weakly, this proves (90).

For fixed M, the functional on the left of (91) is weakly continuous in its measure argument: every finite product kernel is continuous, and it is bounded by a constant depending only on (M, ρ, β). At Haar measure, Poisson normalization gives
$$\int_{\mathbb{T}} |B_M h + \overline{A_M}|^{-2} \, dm(h) = 1$$
for every phase history. The functional therefore converges to one along $\mathcal{T}_{g_n}\nu$. The physical Hardy identity and concavity bound it above by one, proving (91). $\square$

Lemma 10.6 (Finite spectral model for a finite probe bundle)**.** *Let H be unitary on a separable Hilbert space and let $x_1, \dots, x_N$ be vectors. Given an integer T and $\varepsilon > 0$, there are a finite-dimensional subspace $\mathcal{E}$, a unitary $\widehat{H} \in \mathrm{U}(\mathcal{E})$, and vectors $\widehat{x}_p \in \mathcal{E}$ such that every scalar reciprocal recursion of length at most T, for every phase history in $\mathbb{T}^T$, has the same vector norms and β-fractional norms as the recursion from (H, x_p) up to error ε. If $H(k)$ and $x_p(k)$ are norm-continuous, the same model works on a sufficiently small energy neighbourhood of any fixed $k_\star$.*

Proof. Partition $\mathbb{T}$ into finitely many arcs Γ_s of chord diameter at most δ. With E_H the spectral resolution of H, put

$$\mathcal{E}_0 = \operatorname{span}\{E_H(\Gamma_s)x_p : s, p\}, \qquad \widehat{H}_0|_{E_H(\Gamma_s)\mathcal{E}_0} = \lambda_s \mathrm{I},$$

where $\lambda_s \in \Gamma_s$. The arc subspaces are orthogonal, $\widehat{H}_0$ is unitary, and

$$\|(H - \widehat{H}_0)y\| \le \delta \|y\|, \qquad y \in \mathcal{E}_0. \tag{94}$$

Every denominator $a\mathrm{I} + zbH$ has inverse norm at most e^ρ. Repeated use of the resolvent identity therefore bounds the difference of all recursions of length at most T by $C_{\rho,T}\delta$, uniformly in the phase history. Choose δ accordingly. Approximate an orthonormal basis of $\mathcal{E}_0$ and its conjugates by smooth angular vectors and polar-correct the Gram matrix; one finite safety enlargement changes the preceding bound by an arbitrarily small amount. This gives $\mathcal{E}, \widehat{H}$ and $\widehat{x}_p$. Norm continuity of the data and the same resolvent estimate give the last assertion. ☐

Proposition 10.7 (One-window physical grouped contraction)**.** *Let W be a smooth real finite partial potential, let $K \Subset (0, \infty)$, fix $k_\star \in K$, and fix finitely many compact probes. Given $\zeta > 0$, finitely many protected fractional moments, finitely many continuous spectral tests, and a prescribed inner radius, there are a number $h > 0$ and a finite group of smooth real scalar reciprocal cells outside that radius such that, with $I = K \cap (k_\star - h, k_\star + h)$,*

1. *the sum of the targeted β-fractional moments on I is contracted by a factor at most $1/2$, up to error ζ;*
2. *every protected fractional moment is nonexpansive up to ζ, and every protected spectral test changes by at most ζ;*
3. *the group has X_3 cost at most ζ and a pointwise amplitude bound depending only on the fixed hyperbolic height ρ.*

The contraction factor is independent of the resolved angular dimension, but no uniform lower bound for h is asserted.

Proof. Fix the hyperbolic height ρ once and for all, and let $x_p(k)$ be the finitely many current targeted radiation vectors. Choose $\tau > 0$ so small that the total initial fractional moment on the sets where $\|x_p(k)\| < \tau$ is below the allocated error. On the closed sets in K where $\|x_p(k)\| \ge \tau$, the normalized pairs

$$\left(\Theta_W(k), \frac{x_p(k)}{\|x_p(k)\|} \right)$$

form a compact bundle. Proposition 4.3 says that every associated spectral measure is singular. Theorem 10.3, with $q = 1/2$, therefore gives one finite group length M valid uniformly on all of K, before any angular cutoff is chosen. The exact phase-mean nonexpansion in Lemma 8.5 controls the omitted small-norm part, so the ideal group contracts the full targeted moment on any energy subinterval by at most $1/2$ plus its allocated error.

Now, and only now, apply Lemma 10.6 at the prescribed $k_\star$ with the already fixed recursion length $T = M$. Norm continuity gives a neighbourhood $I = K \cap (k_\star - h, k_\star + h)$, with $h >$

0, on which that model and its residual estimates remain valid. Proposition 9.1 makes the transmission and reflection coefficients of the M cells arbitrarily close to $(a\mathrm{I}, b\mathrm{I})$ on the resulting finite model. Since the group and its phase torus are finite, all ideal inverse outputs form a compact bundle. Lemma 8.1, the reflection gap in Lemma 8.2, and one finite safety enlargement transfer (75) to the exact physical outputs. There is no dimension–length fixed point: M was selected from the infinite-dimensional singular spectral bundle before the finite model was introduced.

Choose the free collar frequencies successively. Lemma 6.1 makes each targeted probe integral on I agree, to its allocated error, with its $\mathbb{T}^M$ mean. In the same placement step include every protected fractional moment on K and every protected continuous test. Lemma 8.5 gives nonexpansion or exact martingale preservation for their torus means. Thus the group contracts the current targeted moment on I and protects every listed quantity on K, up to its assigned error.

The group length M is finite. A depth-m realization has total reflector cost at most $MC_\rho e^{-cm}$, which tends to zero. Choose m large and allocate the weak-slab, placement, and residual errors cell by cell. Lemmas 9.3 and 9.4 meet these finite requests. Finally take the active cutoff and then the outgoing Galerkin buffer as in (286). Because all of these choices follow the already fixed M, the outgoing buffer introduces no circular dependence. This proves the three assertions. □

Remaining proof obligation 10.8 (Global physical grouped Lemma B)**.** *Under the hypotheses of Proposition 10.7, prove that finitely many such groups can be assembled into one generation which contracts every targeted fractional moment on all of K by a factor at most $3/4$, preserves the finite protected list up to ζ, and has total X_3 cost at most ζ with one pointwise amplitude bound.*

Theorem 10.3 removes the angular dimension and finite-model circularities from each individual window. The remaining issue is global chromatic coverage: groups aimed at earlier windows change the reciprocal phase seen by later ones, and the resulting positive window widths are not presently known to have a finite-cover lower bound. Equivalently, it is enough to prove the hazard divergence condition (98) below, or a finite broadband scalar synthesis which covers K in one generation.

Proposition 10.9 (Adaptive centre averaging and the exact hazard)**.** *Let $K = [k_0, k_1]$, $|K| > 0$, and let $f \in L^1(K)$ be nonnegative. Fix $0 < h \le |K|/2$ and $0 < \gamma < 1$. Suppose that, for every centre $c \in K$, a phase-placed physical cell tuned on*

$$I(c) = K \cap (c - h, c + h)$$

can be chosen so that its updated fractional density f_c^+ satisfies

$$\int_K f_c^+ \,\mathrm{d}k \le \int_K f \,\mathrm{d}k - \gamma \int_{I(c)} f \,\mathrm{d}k + \varepsilon. \tag{95}$$

Then some centre $c \in K$ satisfies

$$\int_K f_c^+ \,\mathrm{d}k \le \left(1 - \frac{\gamma h}{|K|}\right) \int_K f \,\mathrm{d}k + \varepsilon. \tag{96}$$

For an adaptive sequence of such steps, with parameters $(\gamma_j, h_j, \varepsilon_j)$, one has

$$I_N \le I_0 \exp\left(-\frac{1}{|K|}\sum_{j=1}^{N} \gamma_j h_j\right) + \sum_{j=1}^{N} \varepsilon_j \exp\left(-\frac{1}{|K|}\sum_{s=j+1}^{N} \gamma_s h_s\right), \tag{97}$$

where $I_j = \int_K f_j \, \mathrm{d}k$. In particular, summable errors and

$$\sum_{j=1}^{\infty} \gamma_j h_j = \infty \tag{98}$$

force $I_j \to 0$ after the error tail is chosen in the usual diagonal way.

Proof. For every $k \in K$, the set of centres $c \in K$ for which $k \in I(c)$ has length at least h. Tonelli therefore gives

$$\frac{1}{|K|} \int_K \left(\int_{I(c)} f(k) \, \mathrm{d}k \right) \mathrm{d}c \geq \frac{h}{|K|} \int_K f(k) \, \mathrm{d}k.$$

Average (95) in c and select one centre no worse than the average. This proves (96). Iteration and $1 - x \leq e^{-x}$ give (97); the last assertion is immediate. □

Corollary 10.10 (What the physical coverage problem has become)**.** *Proposition 10.7, the physical Poisson identity, deterministic phase placement, and the compact-output residual estimate give (95) with $f = w^\beta$ and a fixed gap*

$$\gamma_0 = \tfrac{1}{2} - o(1) > 0. \tag{99}$$

Thus Proposition 10.9 reduces the global Problem 10.8 to the accumulated hazard $\sum_j \gamma_j h_j$. Theorem 10.3 removes the former angular dimension and finite-model fixed point, but it does not bound the group length uniformly over all singular spectral measures. Consequently it supplies no uniform lower bound on h_j. Proposition 10.4 proves that the currently available black-box estimates can have a summable certified hazard. The unresolved statement is precisely (98), or an equivalent finite broadband scalar synthesis. No counterexample follows from the one-window theorem alone.

Proposition 10.11 (Positive-flight recurrence cannot supply a broadband inverse)**.** *Let D be self-adjoint and suppose that it has orthonormal eigenvectors e_0, e_1 with eigenvalues 0 and $\lambda > 0$. Fix a nondegenerate compact interval $S = [s_0, s_1] \Subset (0, \infty)$ and $\tau > 0$. Then there is $c = c(S, \lambda, \tau) > 0$ such that, for every $T \geq 0$ and every function $\chi : S \to \mathbb{T}$,*

$$\sup_{s \in S} \left\| e^{-isTD} - \chi(s) e^{is\tau D} \right\|_{\mathrm{span}\{e_0, e_1\} \to \mathcal{H}} \geq c. \tag{100}$$

Thus the usual compact-group recurrence argument at one energy does not turn the physical positive Talbot drift into a signed control uniformly on an energy interval, even modulo an energy-dependent scalar phase.

Proof. Put $a = \lambda(T + \tau)$. Testing the two eigenvectors and using the triangle inequality gives

$$2 \sup_{s \in S} \left\| e^{-isTD} - \chi(s) e^{is\tau D} \right\| \geq \sup_{s \in S} |e^{-ias} - 1|. \tag{101}$$

If $a(s_1 - s_0) \geq 2\pi$, the phase interval $[as_0, as_1]$ contains an odd multiple of π, so the right side is 2. On the remaining compact range

$$\lambda \tau \leq a \leq \frac{2\pi}{s_1 - s_0},$$

the continuous function $a \mapsto \sup_{s \in S} |e^{-ias} - 1|$ has a strictly positive minimum: a zero would force $e^{-ias} = 1$ on a nondegenerate interval and hence $a = 0$. Equation (101) proves the claim. □

Proposition 10.12 (Fejér conjugation repairs the ideal two-level drift)**.** *Let $X, Y, Z \in \mathfrak{su}(2)$ be normalized so that*

$$\mathrm{Ad}_{e^{\theta X}} Z = \cos\theta\, Z + \sin\theta\, Y. \tag{102}$$

Fix a compact interval $S = [s_0, s_1] \Subset (0, \infty)$. Suppose that the available ensemble words may use signed mixer pulses

$$e^{usX}, \qquad u \in \mathbb{R}, \tag{103}$$

and only forward drift flights

$$e^{tsZ}, \qquad t \geq 0. \tag{104}$$

Then, for every $\tau > 0$ and $\varepsilon > 0$, a finite word $\mathcal{V}_s$ made only from (103) and (104) satisfies

$$\sup_{s \in S} \|\mathcal{V}_s - e^{-\tau s Z}\| < \varepsilon. \tag{105}$$

Thus an exact noncommuting signed mixer removes the positive-flight semigroup obstruction on an arbitrary compact detuning band.

Proof. Choose $\delta > 0$ so that

$$0 < \delta s_0 \leq \delta s \leq \delta s_1 < 2\pi \qquad (s \in S).$$

The Fejér kernel is

$$F_N(x) = \frac{1}{N+1}\left(\frac{\sin((N+1)x/2)}{\sin(x/2)}\right)^2 = 1 + 2\sum_{n=1}^{N}\left(1 - \frac{n}{N+1}\right)\cos(nx). \tag{106}$$

Because δS stays a positive distance from $2\pi\mathbb{Z}$, $F_N(\delta s) \to 0$ uniformly for $s \in S$.

Put $\alpha_{n,N} = 1 - n/(N+1) > 0$ and

$$V_{n,\pm}(s) = \mathrm{Ad}_{e^{\pm n\delta s X}}(sZ).$$

Equation (102) gives the exact cancellation

$$G_N(s) := \sum_{n=1}^{N} \alpha_{n,N}\big(V_{n,+}(s) + V_{n,-}(s)\big) = s\big(F_N(\delta s) - 1\big)Z. \tag{107}$$

Hence $G_N \to -sZ$ uniformly on S.

Every positive-time flow of a summand is an admissible word:

$$e^{t\alpha_{n,N}V_{n,\pm}(s)} = e^{\pm n\delta s X} e^{t\alpha_{n,N} s Z} e^{\mp n\delta s X}, \qquad t \geq 0. \tag{108}$$

For fixed N, the Lie product formula applied to the finite sum in (107) approximates $e^{\tau G_N(s)}$ uniformly in $s \in S$ by products of the admissible factors (108). First choose N so that $e^{\tau G_N(s)}$ is uniformly close to $e^{-\tau s Z}$ and then choose the Lie-product subdivision sufficiently fine. This proves (105). $\square$

Proposition 10.13 (No exact scalar Galerkin mixer)**.** *Let $\mathcal{E} \subset C^\infty(\mathbb{S}^2)$ be a nonzero finite-dimensional subspace invariant under $A_3 = -\Delta_{\mathbb{S}^2}$. If a smooth real function q also satisfies*

$$M_q \mathcal{E} \subset \mathcal{E}, \tag{109}$$

then q is constant. In particular, a nontrivial signed mixer for Proposition 10.12 cannot be obtained by replacing the physical multiplier M_q with its compression $P_{\mathcal{E}} M_q P_{\mathcal{E}}$ and silently treating $\mathcal{E}$ as an invariant physical block.

Proof. Since A_3 is self-adjoint and $\mathcal{E}$ is finite dimensional and invariant, $\mathcal{E}$ is spanned by finitely many spherical harmonics. Consequently every element of $\mathcal{E}$ is real analytic. The restriction $M_q|_{\mathcal{E}}$ is self-adjoint, so it has a nonzero eigenvector $f \in \mathcal{E}$:

$$qf = \lambda f.$$

The zero set of the nonzero real-analytic function f has empty interior. Hence $q = \lambda$ on the dense open set where $f \neq 0$, and continuity gives $q \equiv \lambda$ on $\mathbb{S}^2$. $\square$

Proposition 10.14 (Scalar conjugate averages cannot implement the Fejér repair)**.** *Let $S = [s_0, s_1] \Subset (0,\infty)$ be nondegenerate and let $\mathcal{E} \subset C^\infty(\mathbb{S}^2)$ be finite dimensional and invariant under $A_3 = -\Delta_{\mathbb{S}^2}$. Write $P = P_{\mathcal{E}}$, $d = \dim \mathcal{E}$, and*

$$A_{\mathcal{E}} = A_3|_{\mathcal{E}}, \qquad A_{\mathcal{E}}^\circ = A_{\mathcal{E}} - \frac{\operatorname{tr} A_{\mathcal{E}}}{d}\mathrm{I}.$$

Assume that $A_{\mathcal{E}}^\circ \neq 0$. For arbitrary finite families $w_j \geq 0$, $c_j \in \mathbb{R}$, and $q_j \in C^\infty(\mathbb{S}^2;\mathbb{R})$, put

$$K_s = P\sum_j w_j e^{isc_jM_{q_j}} A_3 e^{-isc_jM_{q_j}} P\Big|_{\mathcal{E}}. \tag{110}$$

Define

$$\eta_S = \inf_{a,b\in\mathbb{C}} \sup_{s\in S} |1 + as + bs^2|. \tag{111}$$

Then $\eta_S > 0$ and

$$\inf_{\lambda\in C(S;\mathbb{R})} \sup_{s\in S} \|K_s - (-A_{\mathcal{E}} + \lambda(s)\mathrm{I})\| \geq \frac{\eta_S}{2}\|A_{\mathcal{E}}^\circ\| > 0. \tag{112}$$

In particular,

$$\eta_S \geq \frac{(s_1 - s_0)^2}{s_0^2 + 6s_0s_1 + s_1^2}. \tag{113}$$

Consequently the ideal construction in Proposition 10.12 cannot be transferred by merely replacing its $\mathrm{SU}(2)$ *adjoint rotations with positive Lie–Trotter averages of scalar-phase conjugates of the forward Laplacian, even if asymmetric conjugates are allowed.*

Proof. For real q the product rule gives the exact quadratic identity

$$e^{iscM_q}A_3e^{-iscM_q} = A_3 + scB_q + s^2c^2M_{|\nabla q|^2}, \qquad B_q = i\left(2\nabla q\cdot\nabla + \Delta_{\mathbb{S}^2}q\right). \tag{114}$$

Thus, for some matrices L, R on $\mathcal{E}$ and $W = \sum_j w_j \geq 0$,

$$K_s = WA_{\mathcal{E}} + sL + s^2R. \tag{115}$$

Let Φ be a norm-one linear functional on the matrix space with $\Phi(A_{\mathcal{E}}^\circ) = \|A_{\mathcal{E}}^\circ\|$. Taking traceless parts removes every scalar phase, and applying Φ to (115) gives, with

$$c = (W+1)\|A_{\mathcal{E}}^\circ\| > 0, \qquad a = \frac{\Phi(L^\circ)}{c}, \qquad b = \frac{\Phi(R^\circ)}{c},$$

the estimate

$$\sup_{s\in S}\|(K_s + A_{\mathcal{E}})^\circ\| \geq c\sup_{s\in S}|1 + as + bs^2| \geq c\eta_S \geq \eta_S\|A_{\mathcal{E}}^\circ\|.$$

Since $\|T^\circ\| \leq 2\|T\|$, this proves (112).

It remains to quantify η_S. Put $s_m = (s_0+s_1)/2$ and let ℓ_0, ℓ_m, ℓ_1 be the Lagrange coefficients which evaluate a quadratic at zero from its values at s_0, s_m, s_1. For every $p(s) = 1 + as + bs^2$,

$$1 = p(0) = \ell_0p(s_0) + \ell_mp(s_m) + \ell_1p(s_1).$$

A direct calculation gives

$$|\ell_0| + |\ell_m| + |\ell_1| = \frac{s_0^2 + 6s_0s_1 + s_1^2}{(s_1 - s_0)^2}.$$

Therefore $\|p\|_{C(S)} \geq (s_1 - s_0)^2/(s_0^2 + 6s_0s_1 + s_1^2)$, proving (113). In particular $\eta_S > 0$. $\square$

Proposition 10.15 (Drift-decorated broadband transpositions close the inverse)**.** *Let $A \geq 0$ be self-adjoint and let*

$$\mathcal{E} = \operatorname{span}\{e_1, \dots, e_d\}, \qquad Ae_j = \lambda_j e_j, \qquad \lambda_* = \max_j \lambda_j.$$

Choose orthonormal auxiliary eigenvectors $f_1, \dots, f_d \perp \mathcal{E}$,

$$Af_j = \mu_j f_j, \qquad \Delta_j = \mu_j - \lambda_j > 0, \qquad \min_j \Delta_j > d\lambda_*, \tag{116}$$

and put $\mathcal{F} = \operatorname{span}\{e_1, \dots, e_d, f_1, \dots, f_d\}$. Let P_j be the unitary on $\mathcal{F}$ which swaps e_j and f_j and fixes the other displayed basis vectors.

Suppose that, for every $\delta > 0$ and every j, there is a physically admissible positive-flight scalar-mask word $\mathcal{S}_{j,s}$, with total free-flight time $T_j \geq 0$, and a continuous phase $\xi_j : S \to \mathbb{T}$ such that

$$\sup_{s \in S} \|(\mathcal{S}_{j,s} - \xi_j(s) e^{-isT_jA} P_j)|_{\mathcal{F}}\| < \delta. \tag{117}$$

Then for every $\tau > 0$ and $\varepsilon > 0$ a physically admissible word $\mathcal{W}_s$ and a continuous $\chi : S \to \mathbb{T}$ satisfy

$$\sup_{s \in S} \|(\mathcal{W}_s - \chi(s) e^{is\tau A})|_{\mathcal{E}}\| < \varepsilon. \tag{118}$$

The same implication holds after a prescribed finite compact-output safety space is included in an A-invariant enlargement in (117).

Proof. Set

$$a_0 = \sum_{j=1}^{d} \frac{1}{\Delta_j}, \qquad b_0 = \sum_{j=1}^{d} \frac{\lambda_j}{\Delta_j}, \qquad r = \frac{1 + b_0}{a_0}.$$

The gap condition gives

$$r \geq \frac{1}{a_0} \geq \frac{\min_j \Delta_j}{d} > \lambda_*.$$

Put $T_\Sigma = \sum_j T_j$. Choose $D > 0$ so large that all the numbers

$$x_j = \frac{D(r - \lambda_j) + \tau(b_0/a_0 - \lambda_j) - T_\Sigma/a_0}{\Delta_j} > T_j, \tag{119}$$

and put

$$d_j = x_j - T_j > 0, \qquad C = Dr + \tau \frac{b_0}{a_0} - \frac{T_\Sigma}{a_0}. \tag{120}$$

The definitions give

$$\sum_{j=1}^{d} x_j = Ca_0 - (D + \tau) b_0 = D - T_\Sigma$$

and, for every j,

$$D\lambda_j + x_j \Delta_j = C - \tau \lambda_j. \tag{121}$$

Let $\widetilde{\mathcal{S}}_{j,s} = e^{-isT_jA} P_j$. Since $P_j A P_j$ and A are diagonal in the displayed basis,

$$\widetilde{\mathcal{S}}_{j,s} e^{-isd_jA} \widetilde{\mathcal{S}}_{j,s} = \exp[-is\{T_jA + (d_j + T_j) P_j A P_j\}]\,. \tag{122}$$

All generators on the right commute. Consider the ideal positive-flight echo

$$\mathcal{V}_s = \prod_{j=1}^{d} \left(\widetilde{\mathcal{S}}_{j,s} e^{-isd_jA} \widetilde{\mathcal{S}}_{j,s} \right).$$

On e_k the accumulated eigenvalue is

$$\sum_{j\neq k}(d_j + 2T_j)\lambda_k + T_k\lambda_k + (d_k + T_k)\mu_k = D\lambda_k + x_k\Delta_k = C - \tau\lambda_k.$$

Consequently

$$\mathcal{V}_s|_{\mathcal{E}} = e^{-isC}e^{is\tau A}|_{\mathcal{E}}. \tag{123}$$

Replace both occurrences of P_j in the jth echo by $\mathcal{S}_{j,s}$. Since every factor is unitary, (117) and one insertion of the ideal $\mathcal{F}$-invariant evolution on each side give

$$\sup_{s\in S}\left\|\mathcal{S}_{j,s}e^{-isd_jA}\mathcal{S}_{j,s} - \xi_j(s)^2\widetilde{\mathcal{S}}_{j,s}e^{-isd_jA}\widetilde{\mathcal{S}}_{j,s}\right\|_{\mathcal{F}\to\mathcal{H}} \leq 2\delta.$$

A telescoping product estimate therefore bounds the error of all d echoes on $\mathcal{E}$ by $2d\delta$. Take $\delta < \varepsilon/(2d)$ and absorb $e^{-isC}\prod_j \xi_j(s)^2$ into $\chi(s)$. Concatenating the physical swap words with the flights $d_j > 0$ is again a word of the form (226). The identical telescoping argument applies after adjoining any finite safety space required by the compact-output estimate. □

Proposition 10.16 (Internal SU(2) swap phases cancel in the echo)**.** *Retain the notation and the gap hypothesis of Proposition 10.15. Its conclusion remains true if assumption (117) is replaced by the following weaker population-inversion hypothesis. For every $\delta > 0$ and every j, there are a physically admissible word $\mathcal{S}_{j,s}$ with total flight time T_j, a continuous scalar phase $\xi_j : S \to \mathbb{T}$, and unitaries $R_{j,s}$ on $\mathcal{F}$ such that*

$$\sup_{s\in S}\|(\mathcal{S}_{j,s} - \xi_j(s)e^{-isT_jA}R_{j,s})|_{\mathcal{F}}\| < \delta, \tag{124}$$

where $R_{j,s}$ fixes $\operatorname{span}\{e_j, f_j\}^{\perp}\cap\mathcal{F}$ *and, on the ordered basis (e_j, f_j), has the form*

$$R_{j,s}|_{\operatorname{span}\{e_j,f_j\}} = \begin{pmatrix} 0 & b_j(s) \\ a_j(s) & 0 \end{pmatrix}, \qquad |a_j(s)| = |b_j(s)| = 1, \qquad a_j(s)b_j(s) = -1. \tag{125}$$

Thus no uniform control of the internal population-transfer phase is needed. It is enough to produce an isolated two-level inversion whose interaction-picture block lies in SU(2) *and whose complementary safety vectors are refocused.*

Proof. Let

$$D_j = \mathrm{I}_{\mathcal{F}} - 2(|e_j\rangle\langle e_j| + |f_j\rangle\langle f_j|).$$

Equation (125) gives

$$R_{j,s}^2 = D_j, \qquad R_{j,s}AR_{j,s}^{-1} = P_jAP_j.$$

The second identity is independent of $a_j(s)$ and $b_j(s)$. With $d_j > 0$ chosen exactly as in (119)–(120), the ideal population-swap echo is therefore

$$\begin{aligned}(e^{-isT_jA}&R_{j,s})e^{-isd_jA}(e^{-isT_jA}R_{j,s}) \\ &= \exp[-is\{T_jA + (d_j + T_j)P_jAP_j\}]\, D_j.\end{aligned} \tag{126}$$

All the D_j commute with A and with every P_kAP_k. Moreover,

$$\left.\prod_{j=1}^{d} D_j\right|_{\mathcal{E}} = -\mathrm{I}_{\mathcal{E}},$$

because precisely one factor changes the sign of each e_k. Consequently the product of the ideal echoes equals $-e^{-isC}e^{is\tau A}$ on $\mathcal{E}$. The minus sign and the factors $\xi_j(s)^2$ are common scalar phases. Replacing the ideal factors by (124) and using the same unitary telescoping estimate as in Proposition 10.15 completes the proof. □

Proposition 10.17 (Point-to-point symmetrization gives a selective universal rotation)**.** *Let*

$$I_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad I_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad I_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and let V_ν be the endpoint of the two-quadrature Bloch pulse

$$\dot{V}_\nu(t) = -i\{\nu I_z + x(t)I_x + y(t)I_y\}V_\nu(t), \qquad V_\nu(0) = \mathrm{I}, \tag{127}$$

on $0 \le t \le T$. Define its time-and-phase reversal by

$$x^{\mathrm{tr}}(t) = x(T-t), \qquad y^{\mathrm{tr}}(t) = -y(T-t),$$

and denote the corresponding endpoint by V_ν^{tr}. Then

$$V_\nu^{\mathrm{tr}} = e^{-i\pi I_y} V_\nu^{-1} e^{i\pi I_y}. \tag{128}$$

Consequently, if α is a continuous real function on a compact offset set Ω and

$$V_\nu I_y V_\nu^{-1} = I_y \cos\frac{\alpha(\nu)}{2} + I_z \sin\frac{\alpha(\nu)}{2}, \qquad \nu \in \Omega, \tag{129}$$

then the positive-time concatenation of the reversed pulse followed by the original pulse satisfies

$$V_\nu V_\nu^{\mathrm{tr}} = e^{-i\alpha(\nu)I_x}, \qquad \nu \in \Omega. \tag{130}$$

If the left side of (129) differs from its right side by at most ε in operator norm, then the error in (130) is at most $\pi\varepsilon$.

In particular, on two disjoint compact offset sets one may prescribe $\alpha = \pi$ on the passband and $\alpha = 0$ on the stopband, joining them continuously through the separating gap. Any point-to-point pulse for that profile yields, by the exact symmetrization above, a phase-clean π rotation on the passband and the identity on the stopband. Thus spectator refocusing is not a second independent design problem in the ideal two-level model.

Proof. Reverse the time ordering in (127). Conjugation by $e^{-i\pi I_y}$ changes the signs of I_x and I_z and fixes I_y. The inverse propagator has the reversed Hamiltonian with every sign changed, so the two operations together give precisely the controls $(x(T-t), -y(T-t))$. This proves (128).

Using $e^{-i\pi I_y} = \exp(-i\pi I_y)$ and (129),

$$\begin{aligned} V_\nu V_\nu^{\mathrm{tr}} &= V_\nu e^{-i\pi I_y} V_\nu^{-1} e^{i\pi I_y} \\ &= \exp\left[-i\pi\left(I_y \cos\frac{\alpha(\nu)}{2} + I_z \sin\frac{\alpha(\nu)}{2}\right)\right] e^{i\pi I_y} = e^{-i\alpha(\nu)I_x}. \end{aligned}$$

The last equality is the elementary half-angle decomposition of an x-rotation; it is also the point-to-point to universal-rotation identity of [13]. Finally, Duhamel's formula for two skew-Hermitian exponentials gives

$$\|e^{-i\pi H} - e^{-i\pi K}\| \le \pi\|H - K\|,$$

which proves the quantitative assertion. □

Proposition 10.18 (Every continuous ideal point-to-point profile is reachable)**.** *Let $\Omega \subset \mathbb{R}$ be compact and let $\alpha \in C(\Omega;\mathbb{R})$. For every $\varepsilon > 0$ there are $T > 0$ and piecewise-constant real controls x, y in (127) such that*

$$\sup_{\nu\in\Omega}\left\|V_\nu I_y V_\nu^{-1} - \left(I_y \cos\frac{\alpha(\nu)}{2} + I_z \sin\frac{\alpha(\nu)}{2}\right)\right\| < \varepsilon. \tag{131}$$

The controls may be replaced by smooth compactly supported controls after an arbitrarily small increase of the error. In particular, on separated passband and stopband sets, the choices $\alpha = \pi$ and $\alpha = 0$ give, after Proposition 10.17, an arbitrarily accurate selective universal rotation.

This is an ideal ensemble statement only. The construction below gives a finite L^1 and L^∞ norm for each fixed ε, but no bound uniform as $\varepsilon \downarrow 0$. It therefore does not supply the accuracy-independent amplitude budget required by the bounded physical potential.

Proof. Put

$$X = -iI_x, \qquad Y = -iI_y, \qquad Z = -iI_z, \qquad D(\nu) = \nu Z,$$

and work in the uniform topology of $C(\Omega; \mathfrak{su}(2))$. First note that the uniform reachable closure contains the reversible constant flows generated by X and Y. Indeed, for $r \in \mathbb{R}$, a pulse of height $M \operatorname{sgn}(r)$ and duration $|r|/M$ gives

$$\exp\left[\frac{|r|}{M}\{D(\nu) + M \operatorname{sgn}(r) X\}\right] \longrightarrow e^{rX}$$

uniformly on Ω as $M \to \infty$, and similarly for Y. The positive drift flow is obtained with zero control. Conjugating it by $e^{\pi X}$ changes Z to $-Z$, so the drift D is reversible in the same closure.

The standard commutator and Lie–Trotter product formulas now give every flow generated by the real Lie algebra of X, Y, D. Since

$$[D, X] = \nu Y, \qquad [D, Y] = -\nu X, \qquad [\nu^j X, \nu^k Y] = \nu^{j+k} Z,$$

induction gives

$$p(\nu)X, \qquad p(\nu)Y, \qquad p(\nu)Z$$

for every real polynomial p. By the Weierstrass theorem, these polynomial fields are uniformly dense in $C(\Omega; \mathfrak{su}(2))$; this is the polynomial Lie-extension mechanism used in Bloch ensemble control [10].

Apply the preceding conclusion only to the continuous field

$$F(\nu) = -\frac{\alpha(\nu)}{2}\, iI_x.$$

Polynomial approximation of its scalar coefficient, followed by the same finite product formulas, produces a physical pulse whose endpoint is uniformly as close as desired to $G_\nu = \exp[-i\alpha(\nu) I_x/2]$. Since

$$G_\nu I_y G_\nu^{-1} = I_y \cos\frac{\alpha(\nu)}{2} + I_z \sin\frac{\alpha(\nu)}{2},$$

and conjugation of I_y changes by at most the endpoint error, this proves (131). Every finite approximating word has finite amplitude, duration, and L^1 cost. Finally, smoothing finitely many jumps in L^1 changes the unitary endpoint by an arbitrarily small amount, by Duhamel's formula. $\square$

Proposition 10.19 (Broadband two-level inversion has a positive energy floor)**.** *Let $x, y \in C_c^\infty(\mathbb{R}; \mathbb{R})$ and let V_ν be the full-line endpoint of*

$$\dot{V}_\nu(t) = -i\{\nu I_z + x(t) I_x + y(t) I_y\} V_\nu(t), \qquad V_\nu(-\infty) = \mathrm{I}. \tag{132}$$

Write the first column of $V_\nu(+\infty)$ as

$$\begin{pmatrix} a(\nu) \\ b(\nu) \end{pmatrix}, \qquad |a(\nu)|^2 + |b(\nu)|^2 = 1.$$

If $J \subset \mathbb{R}$ is an interval of positive length and

$$\sup_{\nu \in J} |a(\nu)| \le \delta < 1, \tag{133}$$

then
$$\int_{\mathbb{R}} \{x(t)^2 + y(t)^2\}\ \mathrm{d}t \geq \frac{4|J|}{\pi} \log \frac{1}{\delta}. \tag{134}$$
In particular, the same conclusion holds if V_ν is δ-close in operator norm, on J, to any $\mathrm{SU}(2)$ *population swap.*

More generally, let finitely many pulses have inversion intervals J_j whose union contains an interval J. If every pulse obeys the same error $\delta < 1$ on its own interval, then their total quadratic energy satisfies

$$\sum_j \int_{\mathbb{R}} \{x_j(t)^2 + y_j(t)^2\}\ \mathrm{d}t \geq \frac{4|J|}{\pi} \log \frac{1}{\delta}. \tag{135}$$

Thus an isolated two-level passband cover of a fixed detuning band cannot have arbitrarily small total quadratic cost.

Proof. Put $\omega = x + iy$. In the normalization of the continuum selective-excitation transform, its reflection coefficient r is evaluated at $\xi = \nu/2$ and

$$M_z(\nu) = \frac{1 - |r(\nu/2)|^2}{1 + |r(\nu/2)|^2} = 2|a(\nu)|^2 - 1. \tag{136}$$

Consequently
$$1 + |r(\nu/2)|^2 = \frac{1}{|a(\nu)|^2}.$$
The Zakharov–Shabat energy formula, including arbitrary bound-state data, is

$$\int_{\mathbb{R}} |\omega(t)|^2\ \mathrm{d}t = \frac{4}{\pi} \int_{\mathbb{R}} \log(1 + |r(\xi)|^2)\ \mathrm{d}\xi 16 \sum_k \operatorname{Im} \xi_k. \tag{137}$$

The discrete-spectrum term is nonnegative; see [11, Section 1.7, "Energy Formulas"]. Restrict the integral to $\xi \in J/2$. Equation (133) gives

$$\log(1 + |r(\xi)|^2) \geq 2 \log(1/\delta),$$

and $|J/2| = |J|/2$. Substitution in (137) proves (134).

If a unitary is δ-close to a population swap, its corresponding diagonal entry has modulus at most δ, so the preceding argument applies. Finally sum (134) over j and use $\sum_j |J_j| \geq |J|$ to obtain (135). □

Proposition 10.20 (The matrix Zakharov–Shabat energy floor)**.** *Let $Q \in C_c^\infty(\mathbb{R}; \mathrm{Hom}(\mathbb{C}^p, \mathbb{C}^q))$ and consider*
$$\frac{\mathrm{d}}{\mathrm{d}t}\Psi(t,k) = \begin{pmatrix} -ik\mathrm{I}_p & Q(t)^* \\ -Q(t) & ik\mathrm{I}_q \end{pmatrix} \Psi(t,k). \tag{138}$$
Let $A(k)$ be the $p \times p$ transmission block for incidence from the $\mathbb{C}^p$ side. If, for every k in an interval J, some unit input in $\mathbb{C}^p$ has residual norm at most $\delta < 1$ in the same block, then

$$\int_{\mathbb{R}} \mathrm{tr}\{Q(t)^* Q(t)\}\ \mathrm{d}t \geq \frac{2|J|}{\pi} \log \frac{1}{\delta}. \tag{139}$$

The input vector may depend on k. In particular the conclusion applies when one fixed input vector is transferred to the opposite block with error at most δ throughout J.

Proof. The scattering matrix of (138) is unitary on the real axis. If $C(k)$ is the corresponding reflection/transfer block, then

$$A(k)^*A(k) + C(k)^*C(k) = \mathrm{I}_p. \tag{140}$$

All singular values of $A(k)$ are therefore at most one. The transfer hypothesis gives $s_{\min}(A(k)) \leq \delta$, and hence

$$|\det A(k)| \leq \delta, \qquad k \in J. \tag{141}$$

For completeness, we record the first matrix trace identity. The scalar function

$$a(z) = \det A(z)$$

is analytic and of bounded type in the upper half-plane, has contractive boundary values, and tends to one at infinity. Compact support excludes a singular inner factor. The Volterra expansion of the Jost solution gives

$$\log a(z) = -\frac{i}{2z}\int_{\mathbb{R}} \mathrm{tr}\{Q(t)^*Q(t)\}\ \mathrm{d}t + O(z^{-2}), \qquad z \longrightarrow \infty, \quad \operatorname{Im} z > 0. \tag{142}$$

Factor a into its upper-half-plane Blaschke product and outer factor. If z_n are its zeros, counted with multiplicity, comparison of the z^{-1} coefficients gives

$$\int_{\mathbb{R}} \mathrm{tr}\{Q(t)^*Q(t)\}\ \mathrm{d}t = 4\sum_n \operatorname{Im} z_n - \frac{2}{\pi}\int_{\mathbb{R}} \log|\det A(k)|\ \mathrm{d}k. \tag{143}$$

This is the determinant version of (137). One may first prove it when the zero set is finite and then use the canonical-factorization monotone limit; the Blaschke contribution is nonnegative.

Discard that contribution, restrict the real-axis integral to J, and use (141). This gives

$$\int_{\mathbb{R}} \mathrm{tr}(Q^*Q)\ \mathrm{d}t \geq -\frac{2}{\pi}\int_J \log|\det A(k)|\ \mathrm{d}k \geq \frac{2|J|}{\pi}\log\frac{1}{\delta},$$

which is (139). □

Corollary 10.21 (The selected adjacent-block multiband route has a cost floor)**.** *Fix a finite family of ideal adjacent-shell dipole lines and a nondegenerate energy interval J. Suppose finitely many carrier words are assigned passbands $J_j \subset J$ whose union covers J. Assume that the resonant reduction of word j is a rectangular matrix Zakharov–Shabat system with detuning $k = \kappa_j(s)$, where κ_j is affine with nonzero slope drawn from the fixed line family, and that this word achieves a δ-accurate population transfer throughout J_j. Then the total quadratic control cost has a strictly positive lower bound depending only on J, the fixed line family, the Clebsch normalization, and $\delta < 1$. In particular it is independent of the number and widths of the carrier passbands.*

Consequently the adjacent-block tiling requested in Problem 10.56 cannot at the same time have arbitrarily small X_3 cost. Chebyshev tail cancellation and Stark-residue equalization repair operator-norm patching, but they do not repair this nonlinear bandwidth–energy obstruction.

Proof. On line a the ideal resonant potential is

$$Q_a(t) = \alpha_a \sum_{q=-1}^{1} u_q(t) C_{a,q}.$$

The map $u \mapsto Q_a$ is linear on a fixed finite-dimensional control space. Because the line family is fixed and finite, there is one constant C such that every constituent word satisfies

$$\mathrm{tr}(Q_a(t)^*Q_a(t)) \leq C\sum_{q=-1}^{1}|u_q(t)|^2.$$

On the fixed physical profile space, $\sum_q |u_q|^2$ is bounded by a constant times the squared X_3 norm of the scalar mask. Apply Proposition 10.20 to word j after the affine change of variable $k = \kappa_j(s)$. If

$$d_* = \min_j |\kappa_j'| > 0,$$

where the minimum may be taken over the fixed line family, then

$$\int_{\mathbb{R}} \operatorname{tr}(Q_j^* Q_j)\ \mathrm{d}t \geq \frac{2d_*|J_j|}{\pi} \log \frac{1}{\delta}.$$

Sum over j and use $\sum_j |J_j| \geq |J|$. The preceding norm comparison transfers this positive lower bound to the total physical X_3 cost. □

Remark 10.22 (Exact consequence for the weak-layer idea). The estimate is nonlinear and applies to the endpoint of the whole pulse, not to one term of its Dyson expansion. Splitting a π rotation into an arbitrarily large number of weak subpulses therefore cannot lower its energy below (134) while retaining a fixed detuning passband. Absolute carrier start phases remain harmless for a completed population inversion by Proposition 10.16; they do not evade the energy identity. Any surviving arbitrarily small-cost broadband construction must use genuinely multilevel excursions and cannot be a concatenation of isolated two-level selective inversions.

Corollary 10.23 (The swap obstruction is global, not local)**.** *Let $A_3 = -\Delta_{\mathbb{S}^2}$, let $\mathcal{F}$ be a finite span of spherical harmonics, and let P be any unitary on $\mathcal{F}$. For every fixed $s_\star > 0$ and $\delta > 0$, there are a positive-flight scalar-mask word $\mathcal{S}_s$ and a number $0 < h < s_\star$ such that*

$$\sup_{|s-s_\star|<h} \|(\mathcal{S}_s - P)|_{\mathcal{F}}\| < \delta. \tag{144}$$

Thus a phase-clean physical swap always exists on a nonempty parameter window. The unproved content of (117) is a word on the whole prescribed continuum $s \in S$, or a quantitative lower bound on successive local window widths strong enough to make their accumulated hazard diverge.

Proof. Extend P to a unitary on $L^2(\mathbb{S}^2)$. The simultaneous approximate controllability theorem for the rotating molecule [3] applies to the drift A_3 and the three coordinate multiplication controls. It gives, at one fixed scaling, a piecewise-constant scalar-potential evolution approximating P on an orthonormal basis of $\mathcal{F}$ within $\delta/2$. For the scaling $s_\star$, rescale the control time by $s_\star^{-1}$. Finally, on each constant-control interval, the Lie–Trotter product formula replaces $e^{-is_\star t(A_3+M_q)}$ by alternating factors $e^{-is_\star(t/n)A_3}e^{-is_\star M_{tq/n}}$. All flight times are positive, and finiteness of $\mathcal{F}$ makes the accumulated approximation uniform on its unit sphere.

For the resulting fixed finite word, $s \mapsto \mathcal{S}_s\psi$ is continuous for every $\psi \in L^2(\mathbb{S}^2)$: mask factors are norm-continuous because their generators are bounded, and free factors are strongly continuous. Strong continuity is uniform on the compact unit sphere of the finite-dimensional space $\mathcal{F}$. Hence $s \mapsto \mathcal{S}_s|_{\mathcal{F}}$ is operator-norm continuous. Shrinking to some $|s - s_\star| < h$ keeps the additional error below $\delta/2$ and proves (144). □

Proposition 10.24 (Adiabatic scalar monodromy has the wrong phase orientation)**.** *Let $S = [s_0, s_1] \Subset (0, \infty)$ with $s_0 < s_1$, let $e_0 = |\mathbb{S}^2|^{-1/2}$, and let e be an A_3-eigenvector with $A_3 e = \lambda e$, $\lambda > 0$. Consider any finite concatenation of positive-time adiabatic passages for*

$$i\dot\psi = s(A_3 + M_{q(t)})\psi, \qquad q(t) \in C^\infty(\mathbb{S}^2; \mathbb{R}), \tag{145}$$

along loops based at $q = 0$, interspersed with positive free flights. Assume the usual separated-cluster hypotheses, allowing conical crossings only among excited levels, and suppose that the limiting adiabatic monodromy returns both e_0 and e to their rays. If $U_{s,\epsilon}$ denotes the resulting slow propagator, then for every $\tau > 0$,

$$\liminf_{\epsilon\downarrow 0} \inf_{\chi\in C(S;\mathbb{T})} \sup_{s\in S} \left\| U_{s,\epsilon} - \chi(s)e^{is\tau A_3} \right\|_{\operatorname{span}\{e_0,e\}\to L^2} \geq \sin\left(\frac{1}{4}\min\{\tau\lambda(s_1 - s_0), 2\pi\}\right) > 0. \tag{146}$$

Consequently uniform adiabatic population-transfer or tight-binding constructions cannot supply the broadband inverse, even if their leakage is arbitrarily small. A successful scalar word must be genuinely nonadiabatic.

Proof. For every real scalar potential on the connected sphere, the lowest eigenvalue of $A_3 + M_q$ is simple. Along any compact control path it therefore stays separated from the excited cluster. The common factor s changes the eigenvalues but not the eigenvectors. Hence the uniform adiabatic theorem, including its conical-crossing version [5], gives, after division by the phase of the ground branch,

$$\frac{\langle e, U_{s,\epsilon} e\rangle}{\langle e_0, U_{s,\epsilon} e_0\rangle} = \eta_\epsilon e^{-isa_\epsilon} + o(1) \quad \text{uniformly for } s \in S, \tag{147}$$

where $\eta_\epsilon \in \mathbb{T}$ is geometric and independent of s. The relative dynamical action a_ϵ is a sum of positive flight contributions and integrals

$$T_r \int_0^1 (\lambda_{\nu_r(t)}(q_r(t)) - \lambda_0(q_r(t)))\, dt.$$

Every branch $\nu_r(t)$ occurring here is excited, so $a_\epsilon \geq 0$. Excited–excited conical crossings do not change this order inequality.

Modulo a scalar phase, the target in (146) has relative phase $e^{is\tau\lambda}$. Thus the relevant scalar discrepancy is

$$\eta_\epsilon e^{-is(a_\epsilon + \tau\lambda)} - 1.$$

For $b > 0$, an interval of phases $\{\eta e^{-ibs} : s \in [s_0, s_1]\}$ has angular length $b(s_1 - s_0)$. Optimizing its initial phase gives

$$\inf_{\eta\in\mathbb{T}} \sup_{s\in S} |\eta e^{-ibs} - 1| = 2\sin\left(\frac{b(s_1 - s_0)}{4}\right) \quad \text{if } b(s_1 - s_0) \leq 2\pi,$$

and the value is 2 once the arc covers the circle. Since $b = a_\epsilon + \tau\lambda \geq \tau\lambda$, the right side is bounded below by twice the sine in (146). If the operator error were δ, comparison on e_0 and e would make the relative-phase error at most 2δ. Letting $\epsilon \downarrow 0$ proves the claim. □

Proposition 10.25 (A single scalar profile cannot isolate the high-mode line)**.** *Let $S = [s_0, s_1] \Subset (0, \infty)$, put $r = s_1/s_0 > 1$, and write*

$$A_3 Y_{\ell m} = \lambda_\ell Y_{\ell m}, \qquad \lambda_\ell = \ell(\ell + 1).$$

Fix $\ell \geq 1$, $L > \ell$, and magnetic indices m, M, and set

$$\Delta = \lambda_L - \lambda_\ell = (L - \ell)(L + \ell + 1).$$

Suppose that

$$L \geq \ell\,\frac{r + 1}{r - 1}. \tag{148}$$

If a spatial profile $q \in L^2(\mathbb{S}^2)$ cancels every ground-state transition whose detuning band overlaps the target band, in the precise sense that

$$\langle Y_{JN}, M_q Y_{00}\rangle = 0 \quad \textit{whenever} \quad [s_0\lambda_J, s_1\lambda_J] \cap [s_0\Delta, s_1\Delta] \neq \varnothing, \tag{149}$$

for every admissible J, N, then

$$\langle Y_{LM}, M_q Y_{\ell m}\rangle = 0. \tag{150}$$

Thus, in the asymptotic high-mode regime, one fixed spatial profile cannot obtain a spectrally isolated $\ell \to L$ transition by zeroing all resonant ground leakage. In particular this line-isolation strategy cannot turn a single-profile chirped population inversion into the phase-clean high-mode swap required by Proposition 10.15.

Proof. Two positive detuning intervals overlap exactly when

$$[s_0\lambda_J, s_1\lambda_J] \cap [s_0\Delta, s_1\Delta] \neq \varnothing \quad \Longleftrightarrow \quad \frac{\lambda_J}{\Delta} \in [r^{-1}, r].$$

The extreme ratios in the Clebsch–Gordan range are

$$\frac{\lambda_{L-\ell}}{\Delta} = 1 - \frac{2\ell}{L+\ell+1}, \qquad \frac{\lambda_{L+\ell}}{\Delta} = 1 + \frac{2\ell}{L-\ell}. \tag{151}$$

Condition (148) makes the second ratio at most r and, a fortiori, makes the first at least r^{-1}. Monotonicity of $J \mapsto \lambda_J$ then shows that every degree

$$J \in \{L-\ell, \dots, L+\ell\}$$

has a ground-transition band overlapping the target band.

Because Y_{00} is constant,

$$\langle Y_{JN}, M_q Y_{00}\rangle = |\mathbb{S}^2|^{-1/2} \langle Y_{JN}, q\rangle.$$

Hence (149) annihilates every spherical harmonic component of q in degrees $L-\ell, \dots, L+\ell$. On the other hand the Clebsch–Gordan product rule says that $Y_{LM}\overline{Y_{\ell m}}$ has harmonic support only in those degrees. Therefore it is orthogonal to q, which is precisely (150). The argument already holds for complex q; imposing the physical reality condition only restricts the available profiles further. ☐

Proposition 10.26 (Adjacent-shell resonance has full finite control algebra)**.** *For $\ell \geq 0$, let*

$$\mathcal{H}_n = \ker(A_3 - \lambda_n), \qquad \lambda_n = n(n+1),$$

put $\mathcal{G}_\ell = \mathcal{H}_\ell \oplus \mathcal{H}_{\ell+1}$, and let P_ℓ be the orthogonal projection onto $\mathcal{G}_\ell$. Set

$$J = 2\ell + 1, \qquad \Delta_\ell = \lambda_{\ell+1} - \lambda_\ell = 2(\ell+1),$$

and let $\mathcal{Q}_J$ be the real vector space of real spherical harmonics of pure degree J. Then the following two statements hold.

1. *If $b > a \geq 0$, $\lambda_b - \lambda_a = \Delta_\ell$, and*

   $$P_{\mathcal{H}_b} M_q P_{\mathcal{H}_a} \neq 0 \quad \textit{for some } q \in \mathcal{Q}_J,$$

   then $(a, b) = (\ell, \ell+1)$.

2. *With*

   $$Z_\ell = P_\ell A_3 P_\ell - \frac{\operatorname{tr}(P_\ell A_3 P_\ell)}{\dim \mathcal{G}_\ell} P_\ell, \qquad B_q = P_\ell M_q P_\ell,$$

   the real dynamical Lie algebra is

   $$\operatorname{Lie}_{\mathbb{R}}\{\, iZ_\ell, iB_q : q \in \mathcal{Q}_J \,\} = \mathfrak{su}(\mathcal{G}_\ell) \cong \mathfrak{su}(4\ell+4). \tag{152}$$

Thus the phase-clean gate is not obstructed by the exact resonant finite-dimensional algebra. What remains is to lift an ensemble word for this block to the full sphere uniformly in s, with leakage control when distinct transition bands overlap.

Proof. Write $d = b - a \geq 1$. Equality of the gaps gives

$$d(a+b+1) = 2(\ell+1).$$

If $d = 1$, then $a = \ell$ and $b = \ell+1$. If $d \geq 2$, then $a+b \leq \ell < J$. The spherical-harmonic triangle rule requires $J \leq a+b$ for a degree-J multiplier to couple $\mathcal{H}_a$ to $\mathcal{H}_b$, a contradiction. This proves the first claim.

For the second claim, complexify. Under rotations, $\mathcal{H}_n$ is the irreducible spin-n module V_n, and the degree-J profiles form V_J. Since $J = 2\ell+1$, the triangle and parity rules make B_q purely off diagonal on $V_\ell \oplus V_{\ell+1}$. Its lower-left block is the nonzero equivariant irreducible tensor map

$$T : V_J \longrightarrow \mathrm{Hom}(V_\ell, V_{\ell+1}).$$

The two eigenvalues of Z_ℓ on the summands are distinct. Commuting B_q with Z_ℓ therefore separates, in the complexified Lie algebra, each tensor component $T_N : V_\ell \to V_{\ell+1}$ from its adjoint T_N^*.

Coupled commutators $[T_N, T_M^*]$ are block diagonal. The standard tensor recoupling formula shows that their rank-K components on the lower and upper shells have reduced coefficients proportional, respectively, to

$$\begin{Bmatrix} J & J & K \\ \ell & \ell & \ell+1 \end{Bmatrix}, \qquad \begin{Bmatrix} J & J & K \\ \ell+1 & \ell+1 & \ell \end{Bmatrix}. \tag{153}$$

The first is nonzero for $0 \leq K \leq 2\ell$, and the second is nonzero for $0 \leq K \leq 2\ell+2$. Indeed, with $J = 2\ell+1$, the Racah sum for either $6j$ symbol has the single index $z = 4\ell+2+K$; every factorial in that term is nonnegative in the displayed range.

For $K = 2\ell+1$ and $K = 2\ell+2$ there is no lower-shell tensor component. Consequently the complexified Lie algebra contains the two highest irreducible tensor spaces, supported on $V_{\ell+1}$ alone. We use the following elementary matrix-unit fact. On a spin-L module, a Lie algebra containing the complete tensor spaces of ranks $2L$ and $2L-1$ is $\mathfrak{sl}(V_L)$. Indeed, the weight-$2L$ tensor is a nonzero multiple of $E_{L,-L}$. At weight $2L-1$, the rank-$2L$ and rank-$(2L-1)$ tensors are two independent combinations of $E_{L,-L+1}$ and $E_{L-1,-L}$, so both matrix units belong to the algebra. Their adjoints and brackets give the corresponding diagonal differences. Descending the tensor weight, these already obtained diagonal differences separate the nonzero Clebsch–Gordan coefficients in the rank-$2L$ and rank-$(2L-1)$ tensors; induction gives every $E_{m',m}$ with $m' \neq m$. Their brackets give the full traceless diagonal algebra. Taking $L = \ell+1$ therefore yields $\mathfrak{sl}(V_{\ell+1})$ supported on the upper block.

Finally T_0 is injective. In the magnetic basis it maps the vector with index m to a nonzero multiple of the upper-shell vector with the same index; nonvanishing follows from the saturated triangle $J = \ell + (\ell+1)$. Acting on T_0 from the left by $\mathfrak{sl}(V_{\ell+1})$, and adjoining T_0 itself, produces every map $V_\ell \to V_{\ell+1}$. Taking adjoints produces every reverse map, and their commutators produce all block-diagonal matrices of total trace zero. Thus the complexified algebra is $\mathfrak{sl}(4\ell+4, \mathbb{C})$. The original generators are skew-Hermitian and traceless, so their compact real form is exactly (152). $\square$

Proposition 10.27 (Forward drift suffices on the finite adjacent-shell ensemble)**.** *Retain the notation of Proposition 10.26, let $S \Subset (0,\infty)$ be a compact interval, and choose a real basis $q_1, \dots, q_{4\ell+3}$ of $\mathcal{Q}_{2\ell+1}$. Consider*

$$\dot{U}_s(t) = -is\left(Z_\ell + \sum_{\nu=1}^{4\ell+3} u_\nu(t) B_{q_\nu}\right) U_s(t), \qquad U_s(0) = \mathrm{I}, \tag{154}$$

where the controls are real and piecewise constant. For every continuous $G : S \to \mathrm{SU}(\mathcal{G}_\ell)$ and every $\varepsilon > 0$, there are a time $T > 0$ and controls u_ν such that

$$\sup_{s \in S} \|U_s(T) - G(s)\| < \varepsilon. \tag{155}$$

Moreover the same endpoint lies in the uniform closure of finite products of signed compressed scalar masks and positive compressed free flights. Thus positivity of the Talbot drift is not an obstruction on the finite resonant block.

Proof. First identify the control-only algebra. In a real spherical-harmonic basis every B_q has the block form

$$B_q = \begin{pmatrix} 0 & C_q^\top \\ C_q & 0 \end{pmatrix}.$$

Conjugation by $D = \mathrm{diag}(\mathrm{I}_{\mathcal{H}_\ell}, i\mathrm{I}_{\mathcal{H}_{\ell+1}})$ sends iB_q to

$$\begin{pmatrix} 0 & C_q^\top \\ -C_q & 0 \end{pmatrix},$$

so the control-only algebra $\mathfrak{h}_\ell$ is compact and lies in $D^{-1}\mathfrak{so}(4\ell+4)D$. Its action on $\mathcal{G}_\ell$ is irreducible. To see this without assuming a Galerkin invariance, consider the finite matrices only. Products B_qB_p contain the coupled tensors $T_q^*T_p$ on V_ℓ and $T_qT_p^*$ on $V_{\ell+1}$. The profile indices may be coupled into irreducible ranks. For ranks $2\ell+1$ and $2\ell+2$ the lower block has no such tensor, while the upper Racah coefficient in (153) is nonzero. Hence the unital associative algebra generated by the B_q contains, supported on $V_{\ell+1}$ alone, the complete two highest tensor spaces. The matrix-unit induction in the preceding proof shows that these generate $\mathrm{End}(V_{\ell+1})$; in particular, the upper block projection belongs to the associative algebra. Multiplying B_q by the two block projections separates T_q from T_q^*. Since T_0 is injective, left multiplication of T_0 by $\mathrm{End}(V_{\ell+1})$ gives every map $V_\ell \to V_{\ell+1}$. The algebra is closed under adjoints, so it also contains every reverse map, and their products give $\mathrm{End}(V_\ell)$. Thus the associative algebra is $\mathrm{End}(\mathcal{G}_\ell)$ and its commutant consists only of scalars.

The connected control-only group H_ℓ is therefore compact and irreducible. Its Lie algebra is semisimple: the center of a compact irreducible matrix Lie algebra is scalar by Schur's lemma, whereas every element here is traceless. The driftless ensemble generated by the B_q is uniformly controllable on H_ℓ. To apply the semisimple ensemble theorem of [1] with its stated hypotheses, choose a finite cover of $\mathfrak{h}_\ell$ by spin-representation $\mathfrak{su}(2)$ triples. Every member of those triples is a Lie polynomial in the iB_q. Uniform commutator products realize it as a reversible Lie extension with coefficient s^d for some $d \geq 1$. The resulting finite parameter curve $(s^{d_1}, \dots, s^{d_m})$ lies in the positive orthant, and each fixed-s system generates H_ℓ. The cited theorem therefore gives uniform controllability on $C(S, H_\ell)$. These control-only gates remain in the uniform closure of (154): for fixed $t \in \mathbb{R}$,

$$\exp\left[-is\frac{|t|}{M}(Z_\ell + M\,\mathrm{sgn}(t)B_q)\right] \longrightarrow e^{-istB_q}$$

uniformly for $s \in S$ as $M \to \infty$.

The Haar average of $RZ_\ell R^{-1}$ over H_ℓ commutes with H_ℓ and is therefore scalar by irreducibility. Its trace is zero, so the average vanishes. The orbit spans an H_ℓ-invariant real space containing Z_ℓ. If zero were on the relative boundary of its convex hull, a nonzero supporting functional would be nonnegative on the orbit; its zero Haar average would force it to vanish on the entire orbit and hence on its span, a contradiction. Thus zero is a relative interior point. By Carathéodory's theorem there are $R_1, \dots, R_m \in H_\ell$, weights $w_j > 0$, and $c > 0$ such that

$$-cZ_\ell = \sum_{j=1}^{m} w_j R_j Z_\ell R_j^{-1}. \tag{156}$$

Approximate each constant R_j and its inverse by the control-only ensemble. Positive free flights conjugated by these gates, followed by the Lie product formula and (156), uniformly synthesize

e^{+istZ_ℓ} for every $t>0$. Thus both signs of the drift, as well as both signs of every control, belong to the uniform reachable closure.

The system is now effectively driftless with reversible generators whose finite Lie algebra is $\mathfrak{su}(4\ell+4)$ by Proposition 10.26. Repeat the preceding spin-triple Lie-extension construction for $\mathfrak{su}(4\ell+4)$. Its parameter curve again consists of positive powers of s, every fixed-s system is controllable, and the semisimple ensemble theorem supplies every target in $C(S, \mathrm{SU}(\mathcal{G}_\ell))$. This proves (155). Finally, a uniform Lie–Trotter splitting of every constant-control interval in (154) gives signed scalar masks separated by positive free flights. A signed constant mask removes the scalar trace phase that distinguishes $P_\ell A_3 P_\ell$ from Z_ℓ. $\square$

Proposition 10.28 (The physical resonant cluster grows with angular momentum)**.** *Let $S = [s_0, s_1] \Subset (0,\infty)$, $r = s_1/s_0 > 1$, fix $\ell \geq 0$, and put $J = 2\ell+1$. For every integer*

$$\ell \leq a \leq \lfloor r(\ell+1)\rfloor - 1 \tag{157}$$

and every nonzero $q \in \mathcal{Q}_J$,

$$P_{\mathcal{H}_{a+1}} M_q P_{\mathcal{H}_a} \neq 0, \qquad [s_0\Delta_a, s_1\Delta_a] \cap [s_0\Delta_\ell, s_1\Delta_\ell] \neq \varnothing, \quad \Delta_a = 2(a+1). \tag{158}$$

In particular, if

$$\ell + 1 \geq \frac{1}{r-1}, \tag{159}$$

then every nonzero target profile also couples the next pair $\mathcal{H}_{\ell+1} \leftrightarrow \mathcal{H}_{\ell+2}$, whose detuning band overlaps the target band. The number of obligatorily coupled adjacent pairs is

$$\lfloor r(\ell+1)\rfloor - \ell,$$

which grows linearly in ℓ. Consequently an isolated two-shell rotating-wave estimate cannot lift Proposition 10.27 on any fixed nondegenerate energy interval. A valid lift must control the entire growing resonant cluster, whose Hilbert-space dimension is of order ℓ^2.

Proof. For $a \geq \ell$, the degree $J = 2\ell+1$ satisfies

$$|(a+1)-a| \leq J \leq a + (a+1), \qquad a + (a+1) + J \in 2\mathbb{Z}.$$

Thus V_J occurs once in $V_{a+1} \otimes V_a^*$. The map

$$q \longmapsto P_{\mathcal{H}_{a+1}} M_q P_{\mathcal{H}_a}$$

is a nonzero equivariant map from the irreducible module V_J, hence is injective. This proves the first assertion in (158). The two positive detuning bands overlap exactly when

$$\frac{\Delta_a}{\Delta_\ell} = \frac{a+1}{\ell+1} \in [r^{-1}, r].$$

The lower inequality is automatic for $a \geq \ell$, and the upper inequality is exactly (157). Taking $a = \ell+1$ gives (159). The count and the quadratic dimension estimate follow by summing $\dim \mathcal{H}_a = 2a+1$ over the displayed range. $\square$

Proposition 10.29 (Exact adjacent-shell band-isolation threshold)**.** *Retain the notation of Proposition 10.28, let $q \in \mathcal{Q}_J$ be nonzero, and write $\Delta_\ell = 2(\ell+1)$. Among all positive spectral gaps*

$$\delta_{a,b} = \lambda_b - \lambda_a, \qquad b > a \geq 0,$$

for which $P_{\mathcal{H}_b} M_q P_{\mathcal{H}_a} \neq 0$, the smallest is Δ_ℓ, attained only by $(a,b) = (\ell, \ell+1)$, and the next smallest is

$$\Delta_{\ell+1} = 2(\ell+2) = \left(1 + \frac{1}{\ell+1}\right)\Delta_\ell.$$

Every nonadjacent admissible gap is at least $3\Delta_\ell$. Consequently

$$[s_0\Delta_\ell, s_1\Delta_\ell] \cap [s_0\delta_{a,b}, s_1\delta_{a,b}] = \varnothing \quad \textit{for every other coupled pair } (a,b) \tag{160}$$

if and only if

$$\frac{s_1}{s_0} < 1 + \frac{1}{\ell+1}. \tag{161}$$

For a centred window $S = [c-h, c+h] \Subset (0,\infty)$ this is exactly

$$h < \frac{c}{2\ell+3}. \tag{162}$$

In particular, the growing cluster in Proposition 10.28 is unavoidable on a fixed nondegenerate interval as $\ell \to \infty$, but it disappears on windows of the sharp relative scale $h \asymp (\ell+1)^{-1}$. If c_j remains in one compact subset of $(0,\infty)$ and $h_j = \vartheta c_j/(2\ell_j+3)$ with fixed $0 < \vartheta < 1$, then

$$\sum_j h_j = \infty \quad\iff\quad \sum_j \frac{1}{\ell_j+1} = \infty. \tag{163}$$

Thus an isolated-block implementation would reduce the chromatic stopping test to a quantitative bound on the growth of the auxiliary angular momenta; mere existence of a local gate gives no such bound.

Proof. Put $d = b-a$. The Clebsch–Gordan rules for a degree-J multiplier require

$$d \le J \le a+b, \qquad a+b+J \in 2\mathbb{Z}. \tag{164}$$

Conversely, for every pair satisfying these rules, the map

$$\mathcal{Q}_J \longrightarrow \mathrm{Hom}(\mathcal{H}_a, \mathcal{H}_b), \qquad q \longmapsto P_{\mathcal{H}_b} M_q P_{\mathcal{H}_a},$$

is a nonzero equivariant map out of the irreducible module V_J and hence is injective. Thus every nonzero q couples every admissible pair.

Because $J = 2\ell+1$ is odd, the parity condition in (164) makes d odd. If $d = 1$, then $J \le 2a+1$ gives $a \ge \ell$, and

$$\delta_{a,a+1} = 2(a+1).$$

The first two such gaps are therefore Δ_ℓ and $\Delta_{\ell+1}$. If $d \ge 3$, then $a+b \ge J$ and

$$\delta_{a,b} = d(a+b+1) \ge d(J+1) \ge 3\Delta_\ell.$$

This proves the ordering and also recovers the uniqueness assertion of Proposition 10.26.

For every other coupled pair the gap satisfies $\delta_{a,b} > \Delta_\ell$. Its positive detuning band meets the target band exactly when

$$s_0\delta_{a,b} \le s_1\Delta_\ell \quad\iff\quad \frac{\delta_{a,b}}{\Delta_\ell} \le \frac{s_1}{s_0}.$$

The smallest ratio on the left is $(\ell+2)/(\ell+1)$, proving (160)–(161). Substituting $s_0 = c-h$ and $s_1 = c+h$ and cross-multiplying gives $(2\ell+3)h < c$, which is (162). On a fixed compact set of centres, c_j is bounded above and below by positive constants, so comparison with the harmonic series gives (163). □

Remark 10.30 (Finite growing-cluster audit)*.* For

$$\mathcal{C}_{\ell,A} = \bigoplus_{a=\ell}^{A} \mathcal{H}_a, \qquad N_{\ell,A} = \dim \mathcal{C}_{\ell,A},$$

direct finite Lie closure gives the following checks for all real degree-$(2\ell+1)$ profiles:

(ℓ, A)	$N_{\ell,A}$	$\dim \mathfrak{h}_{\mathrm{control}}$	$\dim \mathfrak{h}_{\mathrm{drift+control}}$
$(0,2)$	9	36	80
$(1,3)$	15	105	224
$(2,4)$	21	210	440
$(3,5)$	27	351	728

Thus in every checked growing cluster the control-only algebra has dimension $N_{\ell,A}(N_{\ell,A}-1)/2$, and adjoining the compressed drift gives $N_{\ell,A}^2-1$. This is evidence for conjugates of $\mathfrak{so}(N_{\ell,A})$ and $\mathfrak{su}(N_{\ell,A})$, respectively; it is not used as a theorem for arbitrary ℓ and A.

There is also a decisive warning. The noncommutativity of the compressed controls comes from inserting $P_{\mathcal{C}_{\ell,A}}$ between scalar multipliers. On the full sphere the multiplication operators themselves commute. Therefore even a proof of the displayed finite pattern would not be the physical lift: one must retain drift-mediated noncommutativity and prove that excursions outside the growing cluster refocus uniformly.

Remark 10.31 (Exact scope of the available infinite-dimensional control theorems). Two quantitative ingredients needed for a physical lift are already known for one fixed system. The multi-input theorem of [4] proves phase-sensitive approximate simultaneous controllability of any prescribed finite frame for the three-dimensional rotating molecule, with an L^1 control bound independent of the final accuracy. The bounded-variation Galerkin theorem of [16] says that, after a total-variation bound and a finite input frame are fixed, one Galerkin cutoff approximates the full evolution uniformly in time for every control inside that bound. Thus, at one fixed scaling, leakage tolerance does not force a circular choice between the control budget and the cutoff.

Neither result contains the required continuum quantifier. The partial ensemble theorem of [2] keeps the drift fixed, places the uncertain parameter only in the control coupling, and prescribes moduli of matrix entries on a selected two-level block rather than a phased unitary frame. Here the family is

$$\dot{U}_s = -is\left(A_3 + \sum_{\nu} u_\nu(t) M_{q_\nu}\right) U_s, \qquad s \in S,$$

so the drift frequencies and all control strengths scale together. The small-time theorem of [17] also does not remove this mismatch: its reachable maps for a general state are multiplication phases $e^{i\phi}$, and its sphere corollary transfers only the two extremal harmonics Y^j_{-j} and Y^j_j inside one degenerate eigenspace. It does not provide a simultaneous finite-frame gate.

Accordingly, the missing theorem is now quantitative and unambiguous. One must either prove a phased, bounded-budget ensemble extension for the fully scaled family above, strong enough to invoke the uniform Galerkin estimate, or construct isolated adjacent-shell windows with indices ℓ_j satisfying the nonsummability condition (163). Pointwise control, local continuity, and small-time phase synthesis do not imply either alternative.

Proposition 10.32 (Uniform Galerkin approximation for the fully scaled family)**.** *Let $A = A_3$, let $B_1, \ldots, B_m$ be multiplication by fixed smooth real functions on $\mathbb{S}^2$, and let P_N be the projection onto the first N eigenspaces of A. Let $S = [s_0, s_1] \Subset (0,\infty)$. Given $L < \infty$, $\varepsilon > 0$, and finitely many smooth input vectors $\psi_1, \ldots, \psi_r$, there is N such that, for every $s \in S$, every $T > 0$, and every piecewise integrable real control satisfying*

$$\sum_{\nu=1}^{m} \int_0^T |u_\nu(t)| \,\mathrm{d}t \le L,$$

the full propagator U_s^u of

$$i\dot{U}_s^u = s\left(A + \sum_{\nu=1}^{m} u_\nu(t) B_\nu\right) U_s^u$$

and its N-dimensional Galerkin propagator $U^u_{s,N}$ satisfy

$$\sup_{\substack{s\in S\\ 0\le t\le T\\ 1\le p\le r}} \left\| U^u_s(t)\psi_p - U^u_{s,N}(t)P_N\psi_p \right\| < \varepsilon. \tag{165}$$

Thus the common-scale parameter creates no additional leakage/cutoff circularity once one common control with an a priori L^1 bound has been constructed.

Proof. For fixed s, set $\tau = st$ and $v_{\nu,s}(\tau) = u_\nu(\tau/s)$. In the τ variable both the full and Galerkin equations have the unscaled drift A and controls $v_{\nu,s}$. Moreover

$$\sum_{\nu=1}^{m} \int_0^{sT} |v_{\nu,s}(\tau)|\, \mathrm{d}\tau = s \sum_{\nu=1}^{m} \int_0^{T} |u_\nu(t)|\, \mathrm{d}t \le s_1 L.$$

Apply the bounded-variation Galerkin theorem of [16] with the single budget s_1L. Its cutoff is uniform in time and over every control inside that budget, so the same N works for every $s \in S$. Replacing τ by st gives (165). The stated finite-control version follows by the same estimate with the variation norm $\sum_\nu |v_\nu|$; the proof of the cited theorem is unchanged for finitely many bounded control operators. ☐

Proposition 10.33 (Full-operator bandpass filter on an isolated shell pair)**.** *Let $J = 2\ell+1$, let $0 \ne q \in \mathcal{Q}_J$, and let*

$$C_q = P_{\mathcal{H}_{\ell+1}} M_q P_{\mathcal{H}_\ell}, \qquad \Delta_\ell = 2(\ell+1).$$

Suppose $S = [s_0, s_1]$ satisfies (161). For every $f \in C(S;\mathbb{C})$ and every $\varepsilon > 0$, there are $T > 0$ and a real $u \in C_c^\infty((0,T))$ such that

$$\sup_{s\in S} \left\| \int_0^T u(t) e^{istA_3} M_q e^{-istA_3}\, \mathrm{d}t - \{f(s)C_q + \overline{f(s)}C_q^*\} \right\| < \varepsilon. \tag{166}$$

The norm is the full operator norm on $L^2(\mathbb{S}^2)$; no Galerkin projection is inserted.

Consequently, if $U_{s,\eta}$ solves

$$i\dot{U}_{s,\eta} = s(A_3 + \eta u(t) M_q) U_{s,\eta}, \qquad U_{s,\eta}(0) = \mathrm{I},$$

and $V_{s,\eta}(T) = e^{isTA_3} U_{s,\eta}(T)$, then for every $g \in C(S;\mathbb{C})$ the filter may be chosen with $f(s) = g(s)/s$, and

$$\sup_{s\in S} \left\| V_{s,\eta}(T) - \mathrm{I} + i\eta\{g(s)C_q + \overline{g(s)}C_q^*\} \right\| \le s_1|\eta|\varepsilon + \tfrac{1}{2} X_\eta^2 e^{X_\eta}, \tag{167}$$

where

$$X_\eta = s_1|\eta|\, \|q\|_\infty \|u\|_{L^1}.$$

Hence every continuous complex quadrature of the target transition is a genuine full-sphere interaction-picture Lie direction. By first sending the filter error to zero and then taking η sufficiently small relative to $\|u\|_{L^1}$, one obtains a sequence with full operator-norm leakage $o(|\eta|)$.

Proof. Use the Fourier convention

$$\widehat{u}(\omega) = \int_{\mathbb{R}} u(t) e^{i\omega t}\, \mathrm{d}t.$$

By Proposition 10.29, the target band

$$K_0 = [s_0\Delta_\ell, s_1\Delta_\ell]$$

is separated by a positive gap from every other positive frequency $s(\lambda_b - \lambda_a)$ coupled by a degree-J multiplier. In fact all such frequencies belong to $[\omega_1, \infty)$ for

$$\omega_1 = s_0 \Delta_{\ell+1} > s_1 \Delta_\ell .$$

We first construct a real time filter. On K_0 prescribe $F(\omega) = f(\omega/\Delta_\ell)$. Approximate F uniformly by a smooth function, extend it smoothly to a compactly supported function on $(0, \omega_1)$, and impose Hermitian symmetry $F(-\omega) = \overline{F(\omega)}$. Its inverse Fourier transform is real and Schwartz. Truncation in time approximates F uniformly on the whole real axis, because the discarded inverse transform has arbitrarily small L^1 norm. To put the truncation in positive time, first multiply the prescribed frequency function by $e^{-it_0\omega}$, perform the symmetric truncation, and then translate the time function by t_0. For t_0 larger than the truncation radius this gives a real $u \in C_c^\infty((0,T))$ satisfying, for an arbitrarily small $\delta > 0$,

$$\sup_{\omega \in K_0} |\widehat{u}(\omega) - F(\omega)| < \delta, \tag{168}$$

$$\sup_{|\omega| \geq \omega_1} |\widehat{u}(\omega)| < \delta. \tag{169}$$

Decompose the degree-J multiplier into its finitely many shell shifts,

$$M_q = \sum_{d=-J}^{J} B_q^{(d)}, \qquad B_q^{(d)} = \sum_{a \geq \max\{0,-d\}} P_{\mathcal{H}_{a+d}} M_q P_{\mathcal{H}_a}.$$

Only shifts of the correct parity occur. For fixed d, the domain blocks and range blocks are mutually orthogonal, so $\|B_q^{(d)}\| \leq \|M_q\| = \|q\|_\infty$. On the block $\mathcal{H}_a \to \mathcal{H}_{a+d}$ the integral in (166) multiplies $B_q^{(d)}$ by

$$\widehat{u}(s(\lambda_{a+d} - \lambda_a)) .$$

Equations (168)–(169), the reality relation at negative frequencies, and the exact band isolation therefore leave $f(s)C_q + \overline{f(s)}C_q^*$ on the target pair and give a residual norm at most $(2J+1)\delta\|q\|_\infty$. Choosing δ sufficiently small proves (166).

Finally the interaction propagator satisfies

$$i\dot{V}_{s,\eta} = s\eta u(t) e^{istA_3} M_q e^{-istA_3} V_{s,\eta}.$$

The first Dyson term, with $f = g/s$, is the displayed target in (167); its filter error is at most $s_1|\eta|\varepsilon$. The sum of all Dyson terms of order at least two is bounded by $e^{X_\eta} - 1 - X_\eta \leq \frac{1}{2}X_\eta^2 e^{X_\eta}$. □

Corollary 10.34 (The physical isolated-band tangent algebra is full)**.** *Under the hypotheses of Proposition 10.33, put*

$$H_{q,\theta} = e^{i\theta} C_q + e^{-i\theta} C_q^*, \qquad q \in \mathcal{Q}_J, \quad \theta \in \mathbb{R}.$$

Then

$$\mathrm{Lie}_{\mathbb{R}}\{\, iH_{q,\theta} : q \in \mathcal{Q}_J,\ \theta \in \mathbb{R}\,\} = \mathfrak{su}(\mathcal{G}_\ell). \tag{170}$$

Every generator on the left, with an arbitrary continuous scalar coefficient in s, is obtained as a full-sphere interaction-picture tangent direction with $o(|\eta|)$ operator-norm leakage. Thus the isolated-band physical lift has no Galerkin, first-variation, or finite-dimensional Lie-algebra obstruction. The remaining step is nonlinear: integrate these causal time-dependent tangent directions into one drift-decorated swap with a finite quantitative control budget.

Proof. The complexification of the displayed real Lie algebra contains C_q and C_q^* separately, because the phases $\theta = 0$ and $\theta = \pi/2$ give their two real quadratures. The proof of Proposition 10.26, starting after the separation of T_q from T_q^*, then applies verbatim: the coupled commutators give the two highest tensor ranks on the upper shell, the matrix-unit induction gives $\mathfrak{sl}(\mathcal{H}_{\ell+1})$, and the injective tensor C_q connects both shells. The complexified algebra is $\mathfrak{sl}(4\ell+4, \mathbb{C})$, and its skew-Hermitian real form is (170). The physical tangent assertion is Proposition 10.33, with the filter accuracy chosen along a diagonal sequence. □

Proposition 10.35 (Causal phase correction has superpolynomial filter cost)**.** *Let $K = [a, b] \Subset (0, \infty)$ be nondegenerate and fix $0 < \varepsilon < 1$. For every integer $N \geq 1$ there is $c_{K,\varepsilon,N} > 0$ with the following property. If $\tau \geq 1$ and $u \in L^1(\mathbb{R})$ is supported in $[\tau, \infty)$ and satisfies*

$$\sup_{\omega \in K} |\widehat{u}(\omega) - 1| \leq \varepsilon, \qquad \widehat{u}(\omega) = \int_{\mathbb{R}} u(t) e^{i\omega t} \, \mathrm{d}t, \tag{171}$$

then

$$\|u\|_{L^1} \geq c_{K,\varepsilon,N} \tau^N. \tag{172}$$

Thus the fixed-phase delayed filters required by a sequential interaction-picture Euler construction cannot have a uniform L^1 budget. More precisely, if such a filter is used in (167), the perturbative requirement $|\eta| \|u\|_{L^1} \to 0$ forces

$$|\eta| \tau^N \longrightarrow 0 \qquad \textit{for every fixed } N.$$

Consequently Corollary 10.34 cannot be integrated by repeating fixed-size first-order gates at successively later times. This is a no-go for that proof method, not for a one-shot nonperturbative broadband gate.

Proof. Choose a nonnegative $\phi \in C_c^\infty((a, b))$ with $\int_K \phi(\omega) \, \mathrm{d}\omega = 1$, and put

$$\Phi(t) = \int_K \phi(\omega) e^{i\omega t} \, \mathrm{d}\omega.$$

By Fubini and (171),

$$1 - \varepsilon \leq \left| \int_K \phi(\omega) \widehat{u}(\omega) \, \mathrm{d}\omega \right| = \left| \int_\tau^\infty u(t) \Phi(t) \, \mathrm{d}t \right| \leq \|u\|_{L^1} \sup_{t \geq \tau} |\Phi(t)|.$$

Integrating by parts N times and using the compact support of ϕ gives

$$\sup_{t \geq \tau} |\Phi(t)| \leq \|\phi^{(N)}\|_{L^1} \tau^{-N}.$$

This proves (172) with $c_{K,\varepsilon,N} = (1 - \varepsilon)/\|\phi^{(N)}\|_{L^1}$. The final assertions follow from the definition of X_η in (167). □

Proposition 10.36 (Power-of-two saturated gaps are uniquely resonant)**.** *Fix $\ell \geq 0$ and $k \geq 1$, and put*

$$d = 2^k, \qquad L = \ell + d, \qquad J = \ell + L, \qquad N = \lambda_L - \lambda_\ell = d(2\ell + d + 1), \qquad \lambda_a = a(a+1). \tag{173}$$

For every nonzero $q \in \mathcal{Q}_J$,

$$P_{\mathcal{H}_L} M_q P_{\mathcal{H}_\ell} \neq 0.$$

Moreover, among all positive coupled gaps

$$\lambda_b - \lambda_a, \qquad b > a \geq 0, \qquad P_{\mathcal{H}_b} M_q P_{\mathcal{H}_a} \neq 0,$$

the value N occurs only for $(a,b)=(\ell,L)$.

All spherical Laplacian gaps are even integers. Consequently, if $S=[c-h,c+h]\Subset(0,\infty)$ and

$$h\le\frac{c}{2(N+1)},\tag{174}$$

then every carrier mismatch

$$s(\lambda_b-\lambda_a)\pm cN$$

which is not one of the two resonant orientations of the target pair is bounded away from zero uniformly in $s\in S$ and in the coupled pair (a,b). The lower bound may be chosen to depend only on c.

Proof. The triangle and parity rules are satisfied by the target pair because $J=\ell+L$ is saturated and $2J$ is even. For uniqueness, suppose that a coupled pair (a,b), $b>a$, has gap N. Set

$$d'=b-a,\qquad n'=a+b+1.$$

Then $N=d'n'$. Since J is even, the coupling parity rule makes $a+b$ even, and hence n' is odd. Write

$$n=2\ell+d+1=J+1,$$

which is also odd. Since $N=2^kn$ and n' is odd, n' divides n. On the other hand the triangle rule $J\le a+b$ gives $n'\ge J+1=n$. Thus $n'=n$, then $d'=d$, and the identities defining d' and n' give $(a,b)=(\ell,L)$.

Because every λ_a is even, distinct positive gaps differ by at least two. Under (174), the nearest possible gaps $N-2$ and $N+2$ already have a carrier separation bounded below by c; gaps farther away have a larger separation. The counter-rotating target orientation has size at least cN. The same argument after changing signs handles negative gaps. ☐

Proposition 10.37 (Saturated carrier quadratures generate the full block)**.** *Retain* (173)*, put*

$$\mathcal{G}_{\ell,L}=\mathcal{H}_\ell\oplus\mathcal{H}_L,\qquad C_q=P_{\mathcal{H}_L}M_qP_{\mathcal{H}_\ell},$$

and define the carrier quadratures

$$H_{q,\theta}=e^{i\theta}C_q+e^{-i\theta}C_q^*,\qquad q\in\mathcal{Q}_J,\quad\theta\in\mathbb{R}.\tag{175}$$

Then

$$\mathrm{Lie}_{\mathbb{R}}\{\,iH_{q,\theta}:q\in\mathcal{Q}_J,\ \theta\in\mathbb{R}\,\}=\mathfrak{su}(\mathcal{G}_{\ell,L}).\tag{176}$$

Proof. Complexification separates C_q and C_q^* by taking $\theta=0$ and $\theta=\pi/2$. Under rotations, $\mathcal{H}_a$ is the spin-a module V_a, while the profiles form V_J. The coupled commutators $[C_q,C_p^*]$ are block diagonal. Their rank-K component on the upper block has reduced coefficient proportional to

$$\begin{Bmatrix}J&J&K\\L&L&\ell\end{Bmatrix}.\tag{177}$$

For $0\le K\le 2L$ the Racah sum for (177) has the single index $2J+K$. Every factorial in that term is nonnegative, so the coefficient is nonzero. The lower block has no tensor component of rank $K>2\ell$. In particular, the complete tensor spaces of ranks $2L$ and $2L-1$ occur on the upper block alone. The matrix-unit argument in the proof of Proposition 10.26 therefore gives $\mathfrak{sl}(V_L)$ supported on that block.

For the zonal profile $q_0\in V_J$, the saturated Clebsch–Gordan coefficients show that $C_{q_0}:V_\ell\to V_L$ maps every magnetic basis vector to a nonzero multiple of the vector with the same magnetic index. Thus C_{q_0} is injective. Left multiplication by $\mathrm{End}(V_L)$ now spans all of $\mathrm{Hom}(V_\ell,V_L)$; the identity multiple is supplied by C_{q_0} itself and the traceless multiples by $\mathfrak{sl}(V_L)$. Adjoints give every reverse map, and commutators give all block-diagonal matrices of total trace zero. Hence the complexified algebra is

$$\mathfrak{sl}(V_\ell \oplus V_L).$$

The original generators are skew-Hermitian and traceless, so the real compact form is exactly (176). ☐

Remark 10.38 (Finite algebra audit). The two-quadrature Lie closures give

(ℓ, L)	computed dimension	target algebra
$(0,2)$	35	$\mathfrak{su}(6)$
$(1,3)$	99	$\mathfrak{su}(10)$
$(2,4)$	195	$\mathfrak{su}(14)$

This is a finite audit only; Proposition 10.37 is the proof for all ℓ and k.

Theorem 10.39 (Shrinking-window carrier lift in full operator norm)**.** *Retain* (173)*, let* $q_1, \dots, q_m$ *be a real basis of* $\mathcal{Q}_J$*, and fix* $c > 0$*,* $R < \infty$*, and*

$$P \in \mathrm{SU}(\mathcal{G}_{\ell,L}).$$

For every $\varepsilon > 0$*,* $B > 0$*, and* $\zeta > 0$*, there are* $T < \infty$*,* $\lambda > 0$*, and smooth real controls* $u_{1,\lambda}, \dots, u_{m,\lambda}$ *supported in* $[0, T/\lambda]$ *such that the full propagator*

$$i\dot{U}_s(t) = s\left(A_3 + \sum_{r=1}^{m} u_{r,\lambda}(t) M_{q_r}\right) U_s(t), \qquad U_s(0) = \mathrm{I}, \tag{178}$$

satisfies, on

$$S_\lambda = \left[c - \frac{\lambda R}{N}, c + \frac{\lambda R}{N}\right], \tag{179}$$

the estimate

$$\sup_{s \in S_\lambda} \left\| e^{is(T/\lambda)A_3} U_s(T/\lambda) - \left(P \oplus \mathrm{I}_{\mathcal{G}_{\ell,L}^\perp}\right) \right\| < \varepsilon. \tag{180}$$

If

$$Q_\lambda(t, \omega) = \sum_{r=1}^{m} u_{r,\lambda}(t) q_r(\omega),$$

the controls may simultaneously be chosen so that

$$\sup_t \|Q_\lambda(t, \cdot)\|_\infty \le B, \tag{181}$$

$$\int_0^{T/\lambda} \|Q_\lambda(t, \cdot)\|_\infty^2 \, \mathrm{d}t \le \zeta, \tag{182}$$

$$\int_0^{T/\lambda} \|Q_\lambda(t, \cdot)\|_\infty \, \mathrm{d}t \le C_P < \infty, \tag{183}$$

where the finite L^1 *budget is fixed before any outgoing Galerkin cutoff.*

Proof. For smooth real slow-time envelopes x_r, y_r supported in $[0, T]$, set

$$u_{r,\lambda}(t) = \frac{2\lambda}{c} \left\{ x_r(\lambda t) \cos(cNt) - y_r(\lambda t) \sin(cNt) \right\}. \tag{184}$$

Write $s = c + \lambda\nu/N$, so $|\nu| \le R$, and pass to slow time $\tau = \lambda t$ and the interaction picture

$$V_{\lambda,\nu}(\tau) = e^{is(\tau/\lambda)A_3} U_s(\tau/\lambda).$$

The resonant part of its Hamiltonian is, up to the factor $s/c = 1 + O(\lambda)$,

$$K_\nu(\tau) = \sum_{r=1}^{m} \Big\{ (x_r(\tau) - iy_r(\tau))e^{i\nu\tau} C_{q_r} + (x_r(\tau) + iy_r(\tau))e^{-i\nu\tau} C_{q_r}^* \Big\}. \tag{185}$$

We first choose the envelopes. By Proposition 10.37, a finite product of exponentials of the constant Hamiltonians (175) equals P. Multiply each Hamiltonian in this word by M and divide the length of its time interval by M. The control-only endpoint is unchanged, whereas the total slow time tends to zero. Uniformly for $|\nu| \le R$, the factors $e^{\pm i\nu\tau}$ in (185) then tend to one throughout the word. Thus, for M sufficiently large, the endpoint of (185) is within $\varepsilon/3$ of P, uniformly in ν. Smoothing the finitely many junctions changes the endpoint by an arbitrarily small amount. Notice that this strong-pulse compression increases the slow-time peak but leaves

$$\int_0^T \sum_r (|x_r| + |y_r|)\ \mathrm{d}\tau$$

bounded by the finite word length.

It remains to justify the full-sphere lift. Decompose each pure-degree multiplier into its finitely many shell shifts as in the proof of Proposition 10.33. On the block $\mathcal{H}_a \to \mathcal{H}_b$, every nonresonant term in slow time has a phase

$$\exp\Big(i\frac{s(\lambda_b - \lambda_a) \pm cN}{\lambda}\tau \Big).$$

After decreasing λ, the window (179) satisfies (174). Proposition 10.36 therefore gives a common nonzero lower bound for every off-target carrier mismatch. Integration by parts for the smooth envelopes, together with the orthogonality of the domain and range blocks for each fixed shell shift, gives

$$\sup_{\substack{|\nu|\le R \\ 0\le\tau\le T}} \Big\| \int_0^\tau \mathcal{R}_{\lambda,\nu}(r)\ \mathrm{d}r \Big\| \le C\lambda, \tag{186}$$

where $\mathcal{R}_{\lambda,\nu}$ is the sum of all nonresonant terms. The constant is finite because the number of spatial profiles and shell shifts is finite; it may depend on their smooth variation, but not on λ or ν.

For completeness, if $\mathcal{Q}(\tau) = \int_0^\tau \mathcal{R}_{\lambda,\nu}(r)\ \mathrm{d}r$, the Duhamel integral between the resonant and full propagators may be integrated by parts. Unitarity and the differential equations bound the endpoint error by

$$\sup_\tau \|\mathcal{Q}(\tau)\| \left[2 + \int_0^T (2\|K_\nu(\tau)\| + \|\mathcal{R}_{\lambda,\nu}(\tau)\|)\ \mathrm{d}\tau \right].$$

The bracket is finite and independent of λ. Equation (186), and the $O(\lambda)$ error from $s/c - 1$, therefore make the full interaction endpoint uniformly $O(\lambda)$-close to the ideal endpoint. Choosing λ after the strong-pulse word proves (180).

Finally put

$$F(\tau) = \sum_r (|x_r(\tau)| + |y_r(\tau)|)\|q_r\|_\infty.$$

Equation (184) gives

$$\sup_t \|Q_\lambda(t)\|_\infty \le \frac{2\lambda}{c}\|F\|_\infty,$$
$$\int_0^{T/\lambda} \|Q_\lambda(t)\|_\infty\ \mathrm{d}t \le \frac{2}{c}\int_0^T F(\tau)\ \mathrm{d}\tau,$$
$$\int_0^{T/\lambda} \|Q_\lambda(t)\|_\infty^2\ \mathrm{d}t \le \frac{4\lambda}{c^2}\int_0^T F(\tau)^2\ \mathrm{d}\tau.$$

After the envelopes are fixed, a further decrease of λ proves (181)–(183). □

Corollary 10.40 (Local high-gap population swaps are physical)**.** *Let $\mathcal{E}$ be a finite A_3-invariant space with an orthonormal eigenbasis $e_1, \dots, e_d$, and fix $c > 0$. For every $\delta, B, \zeta > 0$ there are mutually orthogonal auxiliary eigenvectors $f_1, \dots, f_d \perp \mathcal{E}$, a nonempty interval $S \ni c$, and physical scalar-control words $\mathcal{S}_{j,s}$ such that:*

1. *their gaps satisfy*
$$\Delta_j = \mu_j - \lambda_j > d\lambda_*;$$
2. *for an* $\mathrm{SU}(2)$ *population inversion R_j on* $\mathrm{span}\{e_j, f_j\}$ *which fixes its orthogonal complement,*
$$\sup_{s \in S} \|(\mathcal{S}_{j,s} - e^{-isT_j A_3} R_j)|_{\mathcal{F}}\| < \delta, \qquad \mathcal{F} = \mathrm{span}\{e_1, \dots, e_d, f_1, \dots, f_d\}; \tag{187}$$
3. *the pointwise control height is at most B, the total quadratic control cost is at most ζ, and every word has a finite L^1 budget fixed before the safety and outgoing cutoffs.*

Consequently the hypothesis of Proposition 10.16 holds on some nonempty window about every prescribed centre c. On that window, positive flights and scalar masks synthesize the inverse Talbot drift on $\mathcal{E}$, up to arbitrary error and scalar phase.

Proof. Write $e_j \in \mathcal{H}_{\ell_j}$. Choose distinct powers 2^{k_j} so large that, with $L_j = \ell_j + 2^{k_j}$,
$$N_j = \lambda_{L_j} - \lambda_{\ell_j} > d\lambda_*,$$
and choose $f_j \in \mathcal{H}_{L_j}$. Distinct upper shells make the f_j orthogonal. On $\mathcal{H}_{\ell_j} \oplus \mathcal{H}_{L_j}$ choose a determinant-one population swap sending e_j to f_j and f_j to $-e_j$, while fixing the remaining vectors. Apply Theorem 10.39 to this target. It gives (187) on a nonempty window centred at c, with full-operator rather than projected leakage control. Restrict all finitely many constructions to the intersection of their windows and allocate the height and cost budgets among them. Each smooth controlled interval may be approximated, uniformly for $s \in S$ on the finite smooth space $\mathcal{F}$, by Lie–Trotter products of positive free flights and signed scalar masks. Thus the controls have the word form required by Proposition 10.15. Proposition 10.16, followed by Proposition 10.15, gives the inverse. $\square$

Remark 10.41 (What the carrier theorem does and does not close)*.* Theorem 10.39 is a one-shot nonlinear gate, not a delayed concatenation of first-order filters. It therefore does not contradict Proposition 10.35: the whole Lie word is compressed into one fixed slow-time interval before λ is sent to zero. Its long physical free-flight time costs no potential norm, while (182) makes the weak-mask cost arbitrarily small. Lemmas 9.3 and 9.4 may therefore be applied after the finite word has been fixed.

The theorem closes the previously missing *local* selective-pulse lift. It does not give a uniform lower bound on the window $|S_\lambda| = 2\lambda R/N$. Hence it does not by itself prove the adaptive hazard divergence (98) or the fixed-prescribed-band Problem 10.58. The remaining obstruction is global chromatic coverage, not physical realization of one local drift-decorated swap.

Proposition 10.42 (Dipole carriers separate every finite adjacent-line family)**.** *For $a \geq 0$, put*
$$\mathcal{G}_a = \mathcal{H}_a \oplus \mathcal{H}_{a+1}, \qquad C_{a,q} = P_{\mathcal{H}_{a+1}} M_q P_{\mathcal{H}_a}, \qquad q \in \mathcal{Q}_1,$$
and
$$H_{a,q,\theta} = e^{i\theta} C_{a,q} + e^{-i\theta} C_{a,q}^*.$$
Then
$$\mathrm{Lie}_{\mathbb{R}}\{\, iH_{a,q,\theta} : q \in \mathcal{Q}_1,\ \theta \in \mathbb{R} \,\} = \mathfrak{su}(\mathcal{G}_a) \cong \mathfrak{su}(4a + 4). \tag{188}$$

More generally, let $\mathcal{A} \subset \mathbb{N}_0$ be finite and let $\alpha_a \neq 0$. One common dipole control has the direct-sum algebra

$$\mathrm{Lie}_{\mathbb{R}} \left\{ i \bigoplus_{a \in \mathcal{A}} \alpha_a H_{a,q,\theta} : q \in \mathcal{Q}_1,\ \theta \in \mathbb{R} \right\} = \bigoplus_{a \in \mathcal{A}} \mathfrak{su}(\mathcal{G}_a). \tag{189}$$

Proof. Complexify and use the standard spherical components $q = -1, 0, 1$ of the degree-one tensor. In magnetic bases,

$$C_{a,q}|a, m\rangle = c_{m,q}|a + 1, m + q\rangle, \tag{190}$$

where, up to one positive factor depending only on a,

$$\begin{aligned} |c_{m,1}|^2 &= (a + m + 1)(a + m + 2), \\ |c_{m,0}|^2 &= 2((a + 1)^2 - m^2), \\ |c_{m,-1}|^2 &= (a - m + 1)(a - m + 2). \end{aligned} \tag{191}$$

Every displayed coefficient is nonzero on its allowed edge. The three diagonal commutators

$$D_q = [C_{a,q}, C^*_{a,q}]$$

belong to the complexified algebra. For an edge $e = (m, q)$ in (190), form its three-component signature

$$d_e^{(r)} = D_r(a + 1, m + q) - D_r(a, m), \qquad r = -1, 0, 1, \qquad \mathbf{d}_e = (d_e^{(-1)}, d_e^{(0)}, d_e^{(1)}).$$

Substitution of (191) shows that the signatures are pairwise distinct. Choose a real linear combination $D = \gamma_{-1}D_{-1} + \gamma_0 D_0 + \gamma_1 D_1$ outside the finitely many hyperplanes $\gamma \cdot (\mathbf{d}_e - \mathbf{d}_{e'}) = 0$. The eigenvalue differences of ad_D on all edges are then distinct. For each fixed q, the vectors

$$C_{a,q},\ \mathrm{ad}_D\, C_{a,q},\ \ldots,\ \mathrm{ad}_D^{2a}\, C_{a,q}$$

give a Vandermonde system and isolate every individual matrix unit on the q-edge set. Their adjoints are also present. The bipartite graph in (190), using $q = -1, 0, 1$, is connected. Brackets along paths therefore give every off-diagonal matrix unit, and brackets with adjoints give every traceless diagonal matrix. The complexified algebra is $\mathfrak{sl}(4a + 4, \mathbb{C})$, proving (188).

For (189), the projection of the generated algebra onto each factor is surjective by the first part. The factors $\mathfrak{su}(4a + 4)$ are simple and pairwise nonisomorphic. The Lie-algebra Goursat lemma therefore rules out a diagonal identification between two factors; induction over the finite set $\mathcal{A}$ gives the full direct sum. □

Corollary 10.43 (Ideal dipole profiles on finitely many lines are reachable)**.** *Let $\mathcal{A} \subset \mathbb{N}_0$ be finite, let $R < \infty$, and let $\alpha_a \neq 0$. On*

$$\prod_{a \in \mathcal{A}} \mathrm{SU}(\mathcal{G}_a)$$

consider the common-control ensemble whose ath component has bounded detuning drift νZ_a, $|\nu| \leq R$, and dipole carrier quadratures $\alpha_a H_{a,q,\theta}$. Every continuous target family

$$\nu \longmapsto \big(G_a(\nu)\big)_{a \in \mathcal{A}} \in \prod_{a \in \mathcal{A}} \mathrm{SU}(\mathcal{G}_a)$$

belongs to the uniform reachable closure under one finite piecewise-constant common control. The control has finite amplitude and finite L^1 budget, although no bound uniform in (R, G, ε) is asserted.

Proof. Proposition 10.42 supplies the control-only direct sum of the simple factors. Haar convexification in each factor puts the negative traceless detuning drift in the uniform reachable closure using positive time. Bracketing independently supported factor generators with νZ_a gives every monomial $\nu^k X_a,\ X_a \in \mathfrak{su}(\mathcal{G}_a)$. Polynomial approximation on $[-R, R]$, followed by the standard Lie-product construction, gives every continuous Lie-algebra profile. Finite products of local exponential charts cover the compact target image and give the asserted group-valued profile. $\square$

Proposition 10.44 (Dipole parity postpones off-line leakage to third order)**.** *Put $\lambda_j = j(j+1)$ and $\Delta_a = 2(a+1)$. Fix a dipole carrier $\omega > 0$ and one of its resonance centres*

$$s_a = \frac{\omega}{\Delta_a}.$$

In the interaction picture, every elementary degree-one carrier term maps $\mathcal{H}_j$ to $\mathcal{H}_{j+\delta}$, $\delta \in \{-1, 1\}$, and has a phase with frequency

$$\Gamma_{j,\delta,\sigma}(s) = s(\lambda_{j+\delta} - \lambda_j) + \sigma\omega, \qquad \sigma \in \{-1, 1\}. \tag{192}$$

At $s = s_a$, every resonant product of two such terms preserves the spherical shell. Equivalently, if $\mathcal{N}_{2,a}$ denotes the resonant quadratic mean produced by the first near-identity averaging step, then the order-one effective Hamiltonian in the slow-scale normal form commutes with A_3:

$$[A_3, \mathcal{N}_{2,a}(\tau)] = 0. \tag{193}$$

The assertion is uniform on a sufficiently small scaled-detuning tube about each member of any fixed finite family of resonance lines.

This statement is sharp in order. If $a \geq 1$, the three-step path

$$\mathcal{H}_{a-1} \longrightarrow \mathcal{H}_a \longrightarrow \mathcal{H}_{a+1} \longrightarrow \mathcal{H}_{a+2}$$

has the exact three-photon resonance

$$\lambda_{a+2} - \lambda_{a-1} = 6(a+1) = 3\Delta_a, \tag{194}$$

so a nonadjacent resonant term may first occur at order two in the slow-scale effective Hamiltonian.

Proof. For a two-step path $j_0 \to j_1 \to j_2$, multiplication of the two interaction-picture phases telescopes the spectral part. Its total frequency is

$$s(\lambda_{j_2} - \lambda_{j_0}) + (\sigma_1 + \sigma_2)\omega.$$

At $s = s_a$, vanishing of this frequency is equivalent to

$$\lambda_{j_2} - \lambda_{j_0} + (\sigma_1 + \sigma_2)\Delta_a = 0. \tag{195}$$

Here $j_2 - j_0 \in \{-2, 0, 2\}$ and $\sigma_1 + \sigma_2 \in \{-2, 0, 2\}$. If $j_2 = j_0$, the equation may hold, and the resulting operator preserves that shell. If $j_2 = j_0 + 2$, then

$$\lambda_{j_0+2} - \lambda_{j_0} = 4j_0 + 6 \equiv 2 \pmod 4,$$

whereas every nonzero value of $(\sigma_1 + \sigma_2)\Delta_a$ is divisible by four. Thus (195) is impossible. The case $j_2 = j_0 - 2$ is identical after reversing the path. This proves (193).

The nonzero frequency gaps just used are separated from zero. Continuity in s, followed by a finite minimum over any prescribed finite resonance family, gives the asserted uniform tube. For smooth envelopes supported away from the slow-time endpoints, the standard near-identity integration by parts has no endpoint kick. Its quadratic mean is precisely the sum of the resonant two-step products classified above, so it enters with one power of the slow scale and is shell preserving.

Finally,

$$\lambda_{a+2} - \lambda_{a-1} = (a+2)(a+3) - (a-1)a = 6(a+1).$$

Taking all three carrier signs opposite to the positive shell shifts makes the total frequency zero at $s = s_a$. Three interactions contribute at order two after slow-time rescaling, proving (194). □

Proposition 10.45 (Uniform first dipole normal form on a finite safety space)**.** *Fix one resonance line a, a compact scaled-detuning interval $|\nu| \le R$, and smooth dipole-carrier envelopes supported in the interior of a slow-time interval $[0,T]$. Let U_ν be the corresponding ideal resonant propagator on $\mathcal{G}_a$, extended by the identity, and let $V_{\lambda,\nu}$ be the exact full-sphere interaction propagator with*

$$s = s_a + \frac{\lambda\nu}{\Delta_a}.$$

If $\mathcal{F}$ is any fixed finite sum of spherical shells containing $\mathcal{G}_a$, then there are continuous bounded self-adjoint families $D_{\mathrm{res},\nu}(\tau)$ and $D_{\mathrm{St},\nu}(\tau)$ such that

$$D_{\mathrm{res},\nu}(\tau) = P_{\mathcal{G}_a} D_{\mathrm{res},\nu}(\tau) P_{\mathcal{G}_a}, \tag{196}$$

$$[A_3, D_{\mathrm{St},\nu}(\tau)] = 0, \tag{197}$$

and, with

$$E_\nu = \int_0^T U_\nu(\tau)^* \{D_{\mathrm{res},\nu}(\tau) + D_{\mathrm{St},\nu}(\tau)\} U_\nu(\tau)\, \mathrm{d}\tau, \tag{198}$$

one has

$$\sup_{|\nu|\le R} \big\| \{V_{\lambda,\nu}(T) - U_\nu(T)(\mathrm{I} - i\lambda E_\nu)\}|_{\mathcal{F}} \big\| \le C_{\mathcal{F},R}\lambda^2 \tag{199}$$

for all sufficiently small $\lambda > 0$. In particular $E_\nu \mathcal{F} \subseteq \mathcal{F}$: the entire $O(\lambda)$ endpoint defect is a finite-dimensional phase/frame error on the prescribed safety space, not leakage out of it.

Proof. Write every degree-one multiplier as the sum of its raising and lowering shell shifts. In slow time, the exact interaction Hamiltonian is the resonant Hamiltonian K_ν plus a sum of off-resonant terms

$$R_{\lambda,\nu}(\tau) = \sum_\gamma e^{i\gamma(s)\tau/\lambda} B_{\gamma,\nu}(\tau), \tag{200}$$

where the frequencies and blocks are those in (192). On the fixed detuning compactum, every frequency retained in (200) is separated from zero. For each shell shift the domain blocks and range blocks are mutually orthogonal. Hence division by $\gamma(s)$ and differentiation of the smooth envelopes define bounded banded operators uniformly in λ and ν; the infinite shell sum costs only the finite number of shifts.

Set

$$S_{\lambda,\nu}(\tau) = \lambda \sum_\gamma \frac{e^{i\gamma(s)\tau/\lambda}}{i\gamma(s)} B_{\gamma,\nu}(\tau)$$

and combine adjoint frequencies so that $S_{\lambda,\nu}$ is self-adjoint. The envelopes are supported in $(0,T)$, so $S_{\lambda,\nu}(0) = S_{\lambda,\nu}(T) = 0$. Conjugating by $e^{-iS_{\lambda,\nu}}$ removes $R_{\lambda,\nu}$ to first order. The resulting Hamiltonian is

$$K_\nu + \lambda D_{\mathrm{res},\nu} + \lambda D_{\mathrm{St},\nu} + \lambda R^{(1)}_{\lambda,\nu} + O(\lambda^2).$$

Here $D_{\mathrm{res},\nu}$ contains the explicit $s/s_a - 1$ correction to the resonant control and is supported on $\mathcal{G}_a$. The mean quadratic term is $D_{\mathrm{St},\nu}$. Proposition 10.44 shows that every one of its resonant two-step products preserves a shell, proving (197). The remaining $R^{(1)}_{\lambda,\nu}$ has zero fast mean. A second integration by parts, again with a near-identity transformation equal to the identity at both endpoints, removes it with an $O(\lambda^2)$ remainder.

All operators in these two transformations have only finitely many shell shifts and are uniformly bounded on the stated detuning compactum. Duhamel's formula therefore gives the expansion (199), with the first variation around U_ν equal to (198). Equations (196) and (197) show that conjugation by U_ν cannot take a vector in $\mathcal{F}$ out of $\mathcal{F}$. □

Proposition 10.46 (The first dipole defect is locally correctable)**.** *Retain the hypotheses of Proposition 10.45, and enlarge the finite safety space if necessary so that*

$$\mathcal{F}_L = \bigoplus_{j=0}^{L} \mathcal{H}_j$$

for some $L \geq a+1$. On every fixed compact detuning interval $|\nu| \leq R$, there is a sequence $\lambda_n \downarrow 0$ and finite additional dipole-carrier words, using the adjacent carrier frequencies $s_a\Delta_j$, $0 \leq j < L$, such that the corrected exact propagators satisfy

$$\sup_{|\nu|\leq R} \left\| \left\{ V^{\mathrm{corr}}_{\lambda_n,\nu} - \chi_n(\nu) U_\nu(T) \right\} \Big|_{\mathcal{F}_L} \right\| = o(\lambda_n) \tag{201}$$

for continuous scalar phases $\chi_n(\nu) \in \mathbb{T}$. The correction word has slow-time L^1 size $o(1)$; consequently its physical height is bounded and its additional physical L^1 budget tends to zero after the carrier pullback.

Proof. The overlapping adjacent-pair algebras generate the full algebra on the finite shell chain:

$$\mathrm{Lie}\{\mathfrak{su}(\mathcal{G}_0), \ldots, \mathfrak{su}(\mathcal{G}_{L-1})\} = \mathfrak{su}(\mathcal{F}_L). \tag{202}$$

Indeed, every factor supplies all matrix units within $\mathcal{H}_j \oplus \mathcal{H}_{j+1}$, and brackets along the connected shell chain give every off-diagonal matrix unit on $\mathcal{F}_L$; brackets with adjoints give the traceless diagonal algebra.

For the carrier centred at $s_a\Delta_j$, the scaled detuning on the same energy tube is a fixed nonzero multiple of ν. Applying the polynomial Lie-extension argument of Corollary 10.43 to (202) therefore gives every continuous profile in $C([-R,R];\mathfrak{su}(\mathcal{F}_L))$ in the uniform Lie closure. In particular it gives the traceless part

$$E^\circ_\nu = E_\nu - \frac{\mathrm{tr}_{\mathcal{F}_L} E_\nu}{\dim \mathcal{F}_L} \mathrm{I}_{\mathcal{F}_L}$$

of the continuous endpoint defect in (198).

To retain the small scale, first approximate E°_ν uniformly by one finite Lie polynomial in the available ensemble generators. Let r be its largest bracket depth. The standard commutator product formulas synthesize

$$\exp(i\lambda E^\circ_\nu)$$

with elementary slow-control size $O(\lambda^{1/r})$ and total slow L^1 size $O(\lambda^{1/r})$; lower-depth monomials use the corresponding powers $\lambda^{1/d}$. Subdivision makes the finite Lie-product error $o(\lambda)$. The carrier integration-by-parts estimate used in Theorem 10.39 now has a remainder bounded by λ times this vanishing slow-control size. Thus the exact physical lift of the correction differs from its ideal endpoint by $o(\lambda)$ on $\mathcal{F}_L$.

Choose successively finer Lie-polynomial approximations to E°_ν, and after each finite word has been fixed choose λ_n small enough for the Lie-product and physical-lift errors to be $o(\lambda_n)$.

Concatenating the correction with (199) cancels $-i\lambda_n E^\circ_\nu$. The trace part is the continuous scalar phase χ_n, proving (201). The stated budget assertions follow from the $O(\lambda_n^{1/r})$ slow-control size and the carrier pullback. ☐

Remark 10.47 (What the parity calculation settles). The $O(\lambda)$ bound in Theorem 10.39 is not, by itself, a leakage lower bound for a dipole carrier. Proposition 10.44 shows that its resonant first correction is an AC–Stark-type operator which preserves every spherical shell. On a fixed finite safety space it is therefore a finite-dimensional phase/frame defect; leakage to a nonadjacent shell begins at $O(\lambda^2)$.

Proposition 10.46 cancels this defect on every fixed safety space and fixed detuning compactum. What it does not control is the compactified tail as the scaled detuning grows like λ^{-1} across a prescribed energy band. The next proposition decides whether that tail can be controlled by absolute-error patching.

Proposition 10.48 (Every nontrivial dipole pulse has a nonsummable Stark tail)**.** *Let $\mathcal{H}_-$ and $\mathcal{H}_+$ be finite-dimensional, let $C \in C_c^\infty((0,T); \mathrm{Hom}(\mathcal{H}_-, \mathcal{H}_+))$, and put*

$$B(\tau) = \begin{pmatrix} 0 & 0 \\ C(\tau) & 0 \end{pmatrix} \quad \text{on } \mathcal{H}_- \oplus \mathcal{H}_+.$$

Let W_ν solve

$$i\dot{W}_\nu(\tau) = \{e^{i\nu\tau}B(\tau) + e^{-i\nu\tau}B(\tau)^*\}W_\nu(\tau), \qquad W_\nu(0) = \mathrm{I}. \tag{203}$$

Then, as $|\nu| \to \infty$,

$$W_\nu(T) = \mathrm{I} - \frac{i}{\nu}\mathcal{S}_C + O(\nu^{-2}), \qquad \mathcal{S}_C = \int_0^T \begin{pmatrix} C(\tau)^*C(\tau) & 0 \\ 0 & -C(\tau)C(\tau)^* \end{pmatrix} \mathrm{d}\tau. \tag{204}$$

The remainder is uniform on bounded subsets of $C_c^\infty((0,T))$. If C is not identically zero, then $\mathcal{S}_C \neq 0$ and

$$\|W_\nu(T) - \mathrm{I}\| = \frac{\|\mathcal{S}_C\|}{|\nu|} + O(\nu^{-2}). \tag{205}$$

Consequently, for every $h > 0$ and every nonzero such pulse,

$$\sum_{k \in \mathbb{Z}\setminus\{0\}} \|W_{kh}(T) - \mathrm{I}\| = \infty. \tag{206}$$

Proof. Set

$$S_\nu(\tau) = \frac{1}{i\nu}\{e^{i\nu\tau}B(\tau) - e^{-i\nu\tau}B(\tau)^*\}.$$

It is self-adjoint, $S_\nu(0) = S_\nu(T) = 0$, and

$$\dot{S}_\nu = e^{i\nu\tau}B + e^{-i\nu\tau}B^* + O(\nu^{-1})$$

with the last term oscillatory and containing B'. Conjugating (203) by e^{-iS_ν} cancels its order-one oscillation. The quadratic mean in the transformed Hamiltonian is

$$\frac{1}{\nu}[B(\tau)^*, B(\tau)] = \frac{1}{\nu}\begin{pmatrix} C(\tau)^*C(\tau) & 0 \\ 0 & -C(\tau)C(\tau)^* \end{pmatrix}.$$

All remaining order-ν^{-1} terms have zero fast mean. A second integration by parts removes them with an $O(\nu^{-2})$ endpoint error; there are no endpoint kicks because every derivative of B vanishes near 0 and T. Duhamel's formula gives (204).

If $C \not\equiv 0$, then

$$\int_0^T C(\tau)^*C(\tau)\,\mathrm{d}\tau$$

is a nonzero positive operator, so $\mathcal{S}_C \neq 0$. Equation (205) follows from the norm expansion, and comparison with the harmonic series proves (206). ☐

Corollary 10.49 (The absolute-error multiband patch is impossible)**.** *No nontrivial family of isolated dipole population-transfer pulses can close the fixed-band problem by assigning each pulse an absolutely summable operator-norm tail and then telescoping the errors over an $O(\lambda^{-1})$ window cover. Any successful fixed-band construction must instead cancel the signed $1/\nu$ Stark terms collectively between different carriers, or use a genuinely one-shot broadband mechanism.*

Proposition 10.50 (Unconstrained matrix inverse scattering is not the physical dipole inverse)**.** *For one adjacent line put*

$$\mathcal{D}_a = \operatorname{span}_{\mathbb{C}}\{C_{a,-1}, C_{a,0}, C_{a,1}\} \subset \operatorname{Hom}(\mathcal{H}_a, \mathcal{H}_{a+1}).$$

Then

$$\dim_{\mathbb{C}} \mathcal{D}_a = 3, \qquad \dim_{\mathbb{C}} \operatorname{Hom}(\mathcal{H}_a, \mathcal{H}_{a+1}) = (2a+1)(2a+3). \tag{207}$$

In particular the inclusion is proper for every $a \geq 1$.

More precisely, let $u_q \in C_c^\infty((0,T);\mathbb{C})$ and let $W_{\nu,\epsilon}$ solve

$$i\dot{W}_{\nu,\epsilon} = \epsilon \left\{ e^{i\nu t} \sum_{q=-1}^{1} u_q(t) C_{a,q} + e^{-i\nu t} \sum_{q=-1}^{1} \overline{u_q(t)} C_{a,q}^* \right\} W_{\nu,\epsilon}, \qquad W_{\nu,\epsilon}(0) = \mathrm{I}. \tag{208}$$

Uniformly on every compact ν-set,

$$P_{\mathcal{H}_{a+1}} \frac{W_{\nu,\epsilon}(T) - \mathrm{I}}{\epsilon} P_{\mathcal{H}_a} = -i \sum_{q=-1}^{1} \widehat{u}_q(\nu) C_{a,q} + O(\epsilon). \tag{209}$$

Consequently the differential at the zero pulse of the physical scattering map has its off-diagonal range pointwise inside $\mathcal{D}_a$. If $a \geq 1$, $D \notin \mathcal{D}_a$, and $0 \neq \phi \in C_c^\infty(\mathbb{R})$, the admissible infinitesimal matrix reflection profile

$$R_\epsilon(\nu) = \epsilon\phi(\nu) D + O(\epsilon^2) \tag{210}$$

cannot be the image, to first order, of any physical dipole potential. Thus no differentiable local right inverse for arbitrary matrix scattering data can take values in the physical dipole-control class.

For a finite family $\mathcal{A}$, the obstruction is stronger. A common physical control has

$$Q_a(t) = \alpha_a \sum_{q=-1}^{1} u_q(t) C_{a,q}, \qquad a \in \mathcal{A}, \tag{211}$$

so the same three Fourier coefficients occur in every factor. Independent matrix reflection data on the different lines are therefore not independent physical inputs.

Proof. The map

$$\mathcal{Q}_1 \longrightarrow \operatorname{Hom}(\mathcal{H}_a, \mathcal{H}_{a+1}), \qquad q \longmapsto C_{a,q},$$

is a nonzero equivariant map from the irreducible three-dimensional SO(3) module $\mathcal{Q}_1$. Its kernel is invariant and hence zero. This proves the first equality in (207); the second follows from $\dim \mathcal{H}_j = 2j+1$.

Duhamel's formula applied to (208) gives

$$W_{\nu,\epsilon}(T) = \mathrm{I} - i\epsilon \int_0^T \left\{ e^{i\nu t} \sum_q u_q(t) C_{a,q} + e^{-i\nu t} \sum_q \overline{u_q(t)} C_{a,q}^* \right\} \mathrm{d}t + O(\epsilon^2).$$

The Dyson remainder is uniform when ν stays in a compact set. Taking the upper–lower block proves (209). Its right-hand side lies in $\mathcal{D}_a$ for every ν, whereas (210) does not at any point where $\phi \neq 0$. The last assertion follows by taking the direct sum of the first-Born formula and observing the common coefficients in (211). □

Remark 10.51 (Exact consequence for the one-shot route). The classical selective-excitation inverse theorem is a 2×2 Zakharov–Shabat theorem with one unrestricted complex potential [11]. Matrix AKNS inverse scattering similarly starts from an arbitrary matrix potential in its prescribed off-diagonal block class; see, for example, [12]. Proposition 10.50 shows that neither theorem may be applied and then merely "reinterpreted" as a physical scalar dipole pulse. The exception $a = 0$ is only one vector line: $\mathcal{D}_0 = \mathrm{Hom}(\mathcal{H}_0, \mathcal{H}_1)$. It does not solve the higher-line or simultaneous common-control problem.

Nor can the missing matrix directions be manufactured by a drift-free control factorization on the full sphere: scalar multiplication operators commute,

$$[M_q, M_r] = 0.$$

The full special-unitary algebras in Proposition 10.42 arise only after resonant compression, and their physical lift uses drift-mediated carrier averaging. Therefore a surviving one-shot theorem must solve the genuinely constrained simultaneous inverse problem (211), including the full-sphere leakage and cost ledgers. The proposition rules out an off-the-shelf matrix inverse-scattering closure; it does not rule out such a new constrained theorem.

Proposition 10.52 (Chebyshev carriers cancel a common Stark tail)**.** *Let $S = S^*$ act on a finite-dimensional Hilbert space and suppose $W : \mathbb{R} \to \mathrm{U}(d)$ satisfies, for $|\nu| \geq \nu_0$,*

$$\left\| W(\nu) - \exp\left(-\frac{iS}{\nu}\right) \right\| \leq \frac{C}{\nu^2}. \tag{212}$$

For $N \geq 1$ put

$$h_N = \frac{\pi}{N}, \qquad \theta_{j,N} = \left(j - \frac{1}{2}\right) h_N, \qquad x_{j,N} = \cos\theta_{j,N}, \qquad 1 \leq j \leq N.$$

Fix $0 < \delta < \pi/2$. For $\theta \in [\delta, \pi - \delta]$, $x = \cos\theta$, and an integer M satisfying

$$2 \leq M, \qquad M h_N \leq \delta/2, \tag{213}$$

call a node local when $|\theta_{j,N} - \theta| < (M+1)h_N$ and remote otherwise. Then

$$\left| \frac{1}{N} \sum_{\text{remote}} \frac{1}{x - x_{j,N}} \right| \leq C_\delta \left(\frac{1}{M} + \frac{M}{N} \right), \tag{214}$$

$$\frac{1}{N^2} \sum_{\text{remote}} \frac{1}{|x - x_{j,N}|^2} \leq \frac{C_\delta}{M}. \tag{215}$$

Consequently, in any fixed order,

$$\sup_{\theta \in [\delta, \pi - \delta]} \left\| \prod_{\text{remote}} W(N(x - x_{j,N})) - \mathrm{I} \right\| \leq C_{\delta,S,C} \left(\frac{1}{M} + \frac{M}{N} \right) \tag{216}$$

whenever M is large enough in terms of (δ, ν_0). In particular the remote product converges uniformly to the identity for every diagonal choice

$$M = M_N \longrightarrow \infty, \qquad M_N/N \longrightarrow 0.$$

Proof. The arcsine measure becomes normalized Lebesgue measure under $t = \cos\varphi$, and its interior Cauchy transform vanishes:

$$\mathrm{p.\,v.}\ \frac{1}{\pi} \int_0^\pi \frac{\mathrm{d}\varphi}{\cos\theta - \cos\varphi} = 0, \qquad 0 < \theta < \pi. \tag{217}$$

For completeness, this also follows from

$$\frac{1}{\cos\theta-\cos\varphi}=-\frac{1}{2\sin\theta}\left\{\cot\frac{\theta-\varphi}{2}+\cot\frac{\theta+\varphi}{2}\right\}.$$

The first cotangent has zero principal value and the second supplies the opposite regular contribution.

Remove the symmetric interval $|\varphi-\theta|<(M+1)h_N$. On its complement the total variation of $(\cos\theta-\cos\varphi)^{-1}$ is at most $C_\delta/(Mh_N)$. The midpoint quadrature error is therefore at most C_δ/M. The principal-value integral over the removed symmetric interval is $O_\delta(Mh_N)$: its odd $(\sin\theta(\varphi-\theta))^{-1}$ singularity cancels, and the remaining part is bounded. Adjusting the two boundary midpoint cells costs another $O_\delta(M^{-1})$. This proves (214).

For a remote node,

$$|\cos\theta-\cos\theta_{j,N}|\geq c_\delta\min\{|\theta-\theta_{j,N}|,1\}.$$

Summing by angular distance from θ gives

$$\frac{1}{N^2}\sum_{\text{remote}}\frac{1}{|x-x_{j,N}|^2}\leq C_\delta\sum_{r\geq M}\frac{1}{r^2}+\frac{C_\delta}{N}\leq\frac{C_\delta}{M},$$

which is (215).

Put

$$E_{j,N}=\exp\left(-\frac{iS}{N(x-x_{j,N})}\right).$$

All $E_{j,N}$ commute and are unitary. Equations (212) and (215), followed by a unitary telescoping estimate, give

$$\left\|\prod_{\text{remote}}W(N(x-x_{j,N}))-\prod_{\text{remote}}E_{j,N}\right\|\leq\frac{C_\delta}{M}.$$

On the other hand,

$$\prod_{\text{remote}}E_{j,N}=\exp\left\{-iS\,\frac{1}{N}\sum_{\text{remote}}\frac{1}{x-x_{j,N}}\right\},$$

whose distance from the identity is bounded by $\|S\|$ times (214). This proves (216). ☐

Corollary 10.53 (A quantitative triangular-array stopping criterion)**.** *For every N, let $S_N=S_N^*$ and let $W_{j,N}:\mathbb{R}\to\mathrm{U}(d)$, $1\leq j\leq N$, satisfy*

$$\left\|W_{j,N}(\nu)-\exp\left(-\frac{iS_N}{\nu}\right)\right\|\leq\frac{C_N}{\nu^2},\qquad|\nu|\geq\nu_{0,N},\tag{218}$$

with constants independent of j. Use the Chebyshev centres and the same local/remote convention as in Proposition 10.52. Then, whenever $M\geq c_\delta\nu_{0,N}$ and $Mh_N\leq\delta/2$,

$$\sup_{\theta\in[\delta,\pi-\delta]}\left\|\prod_{\text{remote}}W_{j,N}(N(x-x_{j,N}))-\mathrm{I}\right\|\leq C_\delta\left[\|S_N\|\left(\frac{1}{M}+\frac{M}{N}\right)+\frac{C_N}{M}\right].\tag{219}$$

In particular, if

$$K_N:=1+\|S_N\|+C_N+\nu_{0,N}=o(\sqrt{N}),\tag{220}$$

then $M_N=\lfloor\sqrt{N}\rfloor$ makes the remote product converge uniformly to the identity.

Proof. The proof of Proposition 10.52 is unchanged. The telescoping remainder is $C_\delta C_N/M$, while the commuting exponentials contribute $C_\delta\|S_N\|(M^{-1}+M/N)$. A remote node has $|N(x-x_{j,N})|\geq c_\delta M$, which explains the threshold on M. Substitution of $M_N=\lfloor\sqrt{N}\rfloor$ and (220) proves the last assertion. ☐

Proposition 10.54 (Symmetric sidebands equalize every finite Stark family)**.** *Let $C_1, \dots, C_m \in \mathrm{Hom}(\mathcal{H}_-, \mathcal{H}_+)$ and, for a smooth compactly supported vector control $u = (u_1, \dots, u_m)$, put*

$$C_u(t) = \sum_{q=1}^{m} u_q(t) C_q, \qquad \Gamma(u)_{qr} = \int_{\mathbb{R}} \overline{u_q(t)} u_r(t)\, \mathrm{d}t.$$

The Stark residue in Proposition 10.48 is the linear function of this positive Gramian

$$\mathcal{S}(\Gamma) = \begin{pmatrix} \sum_{q,r} \Gamma_{qr} C_q^* C_r & 0 \\ 0 & -\sum_{q,r} \Gamma_{rq} C_q C_r^* \end{pmatrix}. \tag{221}$$

Let $u^{(1)}, \dots, u^{(J)}$ be any finite family of such controls. For every $R < \infty$ and $\eta > 0$ there are one positive matrix Γ_* and padded controls $\widetilde{u}^{(1)}, \dots, \widetilde{u}^{(J)}$ with the following properties:

$$\Gamma(\widetilde{u}^{(j)}) = \Gamma_*, \qquad 1 \le j \le J, \tag{222}$$

$$\sup_{|\nu| \le R} \left\| W_\nu[\widetilde{u}^{(j)}] - W_\nu[u^{(j)}] \right\| < \eta. \tag{223}$$

Here the original pulse is kept first and the padding is concatenated after it, so the two end-point propagators are compared in the same absolute interaction frame. For each fixed padding construction,

$$W_\nu[\widetilde{u}^{(j)}] = \mathrm{I} - \frac{i}{\nu} \mathcal{S}(\Gamma_*) + O_j(\nu^{-2}), \qquad |\nu| \longrightarrow \infty. \tag{224}$$

Thus common-residue equalization has no algebraic obstruction on a fixed finite carrier family. The statement does not assert a remainder or cost bound uniform as J grows without additional hypotheses. Quantitatively, if a triangular input family has uniformly bounded Gramians and its unpadded quadratic-tail constants and thresholds are $o(\sqrt{N})$ uniformly in the carrier index, then the padding can be chosen so that the equalized family still satisfies (220).

Proof. Choose

$$\gamma > \max_{1 \le j \le J} \|\Gamma(u^{(j)})\|, \qquad \Gamma_* = \gamma \mathrm{I}_m, \qquad H_j = \Gamma_* - \Gamma(u^{(j)}).$$

Every H_j is positive definite. By a Cholesky decomposition and disjoint smooth bump intervals, construct finitely many vector controls $p_{j,\ell}$ such that

$$2 \sum_{\ell} \Gamma(p_{j,\ell}) = H_j. \tag{225}$$

For a parameter $\Omega > 0$, append for every ℓ two disjoint copies of $p_{j,\ell}$, modulated respectively by $e^{i\Omega t}$ and $e^{-i\Omega t}$. Modulation by a scalar phase does not change a control Gramian, and disjoint concatenation adds Gramians. Hence (225) proves (222) exactly.

It remains to check that the padding is locally invisible. In (203), modulation of a fixed bump by $e^{\pm i\Omega t}$ replaces its detuning by $\nu \pm \Omega$. Translating the bump to its appended interval only conjugates the off-diagonal block by a scalar block phase; this leaves its Stark residue invariant. Proposition 10.48, applied to the two unmodulated copies, therefore gives uniformly for $|\nu| \le R$,

$$W_{\nu,+\Omega} = \mathrm{I} - \frac{i\mathcal{S}_\ell}{\nu + \Omega} + O_R(\Omega^{-2}), \qquad W_{\nu,-\Omega} = \mathrm{I} - \frac{i\mathcal{S}_\ell}{\nu - \Omega} + O_R(\Omega^{-2}).$$

The leading terms cancel because

$$\frac{1}{\nu + \Omega} + \frac{1}{\nu - \Omega} = \frac{2\nu}{\nu^2 - \Omega^2} = O_R(\Omega^{-2}).$$

Thus every appended pair has endpoint $\mathrm{I} + O_R(\Omega^{-2})$. There are only finitely many pairs, so their product has the same bound with a finite constant. Taking Ω large proves (223).

Finally, Proposition 10.48 and (221) give (224). In the physical dipole controller, the complex components $u_q = x_q - iy_q$ are precisely the two available real carrier quadratures. The two scalar sideband modulations therefore remain admissible real dipole controls.

For the quantitative last assertion, choose γ once above the uniform Gramian bound and use one fixed bump template in the Cholesky construction. The coefficients of all padding bumps are then uniformly bounded. For $|\nu| \geq 2\Omega$, the shifted expansions above and

$$\left|\frac{1}{\nu+\Omega} + \frac{1}{\nu-\Omega} - \frac{2}{\nu}\right| = \frac{2\Omega^2}{|\nu|\,|\nu^2-\Omega^2|} \leq \frac{C\Omega}{\nu^2}$$

show that the padding adds a tail threshold $O(\Omega)$ and a quadratic-tail constant $O(1+\Omega)$, uniformly in the carrier index. Choose $\Omega = \Omega_N \to \infty$ so slowly that the original growth bound plus Ω_N remains $o(\sqrt{N})$. The local padding error is then $O_R(\Omega_N^{-2}) \to 0$, while the equalized family satisfies (220). $\square$

Corollary 10.55 (Under a common residue, the harmonic tail is not the global obstruction)**.** *Consider a family of rescaled dipole words whose far-detuned propagators have the common-residue expansion*

$$W_{j,N}(x) = \exp\left\{-\frac{iS}{N(x-x_{j,N})}\right\} + O\left(\frac{1}{N^2|x-x_{j,N}|^2}\right)$$

on one fixed finite safety space, with constants independent of j and N. Place their centres at the affine images of the Chebyshev nodes in an interval strictly larger than the target energy band. If, for every fixed M, the $O(M)$ local words can be jointly synthesized with an error tending to zero after $N \to \infty$, then a diagonal choice $M_N \to \infty$, $M_N/N \to 0$ gives a uniformly compactified multiband word.

Thus Proposition 10.48 rules out absolute telescoping but does not rule out the multiband theorem. Under the displayed common-residue hypothesis, its leading tail is cancelled by (216). The remaining physical obligation is now quantitative and explicit. Proposition 10.54 supplies a gate-preserving common residue for every fixed finite tiling. What is not supplied is a bound uniform as the tiling and its local cluster grow: the padded residue, the $O(\nu^{-2})$ constant, and the joint local synthesis error must still admit one compatible diagonal limit.

Remaining proof obligation 10.56 (The single remaining multiband tail estimate)**.** *Fix $K = [s_0, s_1] \Subset (0,\infty)$ and a finite harmonic safety space. For a dipole carrier ω, the only positive coupled gaps are*

$$\Delta_a = 2(a+1), \qquad a \geq 0,$$

and the resonance centres in K form the finite set

$$\mathcal{A}(\omega, K) = \{\, a \geq 0 : \omega/\Delta_a \in K \,\}.$$

Prove a quantitative physical lift of Corollary 10.43 with the following order of quantifiers:

1. *first prescribe continuous target profiles on the finitely many resonance tubes, taking the identity on every unwanted line;*
2. *then choose one slow carrier scale λ so that the exact full-sphere interaction propagator realizes those profiles uniformly on* all *of K, including the unbounded scaled-detuning tails between the tubes;*
3. *retain one finite L^1 budget before the safety and outgoing cutoffs, bounded physical height, and arbitrarily small X_3 cost.*

Such a theorem gives compactly energy-supported adjacent-shell gates. Overlapping adjacent blocks generate the full special-unitary group on every finite harmonic safety space. A finite partition of unity on K then produces the broadband drift-decorated population swaps in Proposition 10.16, and hence solves Problem 10.58. The obstruction left in this problem is quantitative rather than algebraic: Corollary 10.43 controls every fixed detuning compactum, while the physical rescaling makes the detuning range across K grow like λ^{-1}. Moreover a cover by windows of width $O(\lambda)$ uses $O(\lambda^{-1})$ pulses. Proposition 10.46 gives the required $o(\lambda)$ endpoint error on every fixed detuning compactum, but its diagonal construction is not uniform as that compactum grows. Proposition 10.48 shows that an absolute summable tail modulus for each nontrivial pulse is impossible. Proposition 10.52 supplies the required correlated cancellation whenever the elementary words have one common Stark residue, and Proposition 10.54 imposes that residue without changing any fixed finite family of ideal gates in the limit. Hence the unresolved statement is no longer an unspecified tail bound or a finite-dimensional equalization problem. It is the growing-cluster uniformity in Corollary 10.55: after the tiling is equalized, prove the sub-square-root controller growth (220) and make the joint $O(\sqrt{N})$ local synthesis error tend to zero. Corollary 10.53 then fixes $M_N = \lfloor\sqrt{N}\rfloor$ and closes the remote product. Independent operator-norm telescoping remains ruled out.

Proposition 10.19 imposes an additional stopping test on this last step. On every fixed finite harmonic safety space, the effective two-level Rabi envelope is a bounded linear functional of the scalar profile, so its quadratic energy is bounded above by a constant times the physical X_3 cost. Hence a partition of K into isolated two-level inversion passbands has a strictly positive total X_3 cost and cannot prove item 3 above. A positive solution of the present problem must therefore use genuinely multilevel excursions in an essential way and prove that their matrix scattering cost escapes the scalar trace bound. If the corresponding matrix Zakharov–Shabat trace identity yields the same lower bound, the selected multiband route fails structurally. Proposition 10.20 and Corollary 10.21 prove precisely that lower bound. Thus the adjacent-block resonant version of this problem is answered negatively.

Remark 10.57 (Stopping decision for the selected carrier route). The failure is now structural, not a missing estimate. The common-residue Chebyshev construction solves the global operator-norm tail, but any fixed-band population-transfer tiling inside the matrix resonant block spends a positive amount of quadratic control energy. It therefore cannot be inserted into generations whose X_3 costs are required to be summable and arbitrarily prescribed. In accordance with the stated stopping rule, this adjacent-block multiband route must be abandoned. A future positive Lemma B would need a genuinely full-sphere mechanism whose essential excursions do not reduce to a matrix Zakharov–Shabat scattering word.

Remaining proof obligation 10.58 (The lens-assisted inverse stopping test). *Let $A_3 = -\Delta_{\mathbb{S}^2}$, let $S \Subset (0,\infty)$ be a nondegenerate interval, and let $\mathcal{E} \subset C^\infty(\mathbb{S}^2)$ be finite dimensional and invariant under conjugation. Decide whether, for every $\tau > 0$ and $\varepsilon > 0$, there are smooth real scalar masks $q_1, \dots, q_N$ and positive flight times $t_1, \dots, t_N$ such that the physically ordered word*

$$\mathcal{W}_s = \overleftarrow{\prod}_{j=1}^{N} \left(e^{-ist_jA_3}e^{-isM_{q_j}}\right) \tag{226}$$

satisfies, for some continuous $\chi : S \to \mathbb{T}$,

$$\sup_{s\in S} \left\| \mathcal{W}_s - \chi(s)e^{is\tau A_3} \right\|_{\mathcal{E}\to L^2(\mathbb{S}^2)} < \varepsilon, \tag{227}$$

with the same estimate on the finite compact-output safety enlargement needed by Lemma 8.1.

A positive answer converts the Talbot drift into an effectively signed broadband control; the semisimple ensemble-control theorem may then be applied only after the compressed scalar controls are shown to generate the required finite-dimensional Lie algebra and the word is realized with the radial residual and X_3 ledgers. A negative answer is a structural failure of the only *remaining broadband scalar mechanism. Proposition 10.11 proves that masks are essential: free-flight recurrence alone cannot solve this problem. Proposition 10.12 proves, conversely, that one exact signed noncommuting* $\mathrm{SU}(2)$ *mixer would be sufficient at the ideal two-level level. The unresolved physical issue is whether smooth scalar multiplication supplies such mixers on the required finite safety bundle with uniform detuning control and without treating a Galerkin compression as an invariant physical subspace. Proposition 10.13 rules out that shortcut: any successful mixer must make controlled excursions into the complementary angular modes and refocus them uniformly on* S*. Proposition 10.14 further rules out the direct positive-average lift of the Fejér proof. Any successful word must therefore use genuinely higher-order noncommutative composite dynamics, not only first-order convexification of scalar-mask conjugates. Proposition 10.15 gives an exact constructive reduction of that remaining task: it is enough to realize drift-decorated transpositions*

$$\mathcal{S}_{j,s} \simeq e^{-isT_jA_3}P_j$$

between the target harmonics and sufficiently high auxiliary harmonics, uniformly for $s \in S$*. The terminal free factor is not an error: Proposition 10.15 absorbs every* T_j *exactly into modified positive echo times. Pointwise controllability extends by Corollary 10.23 to some nonempty neighbourhood, but with no quantitative lower bound. Proposition 10.24 excludes a simple gapped adiabatic loop as a phase-reversal shortcut. It does not exclude a quasi-adiabatic population permutation through a deliberately collapsing gap: after such a permutation, the echo rather than the adiabatic phase reverses the relative phase orientation. The finite-dimensional scalar-control theorem of [9] proves population inversion of one initial eigenstate up to an uncontrolled phase, under separated gap intervals. Proposition 10.16 shows that this internal phase is harmless: an isolated* $\mathrm{SU}(2)$ *inversion suffices. Proposition 10.17 goes further at the ideal two-level level: time-and-phase reversal converts a point-to-point passband profile into a full universal rotation, while a zero-angle stopband is refocused to the identity. Thus uncontrolled transfer phase and spectator refocusing are not separate ideal pulse-design obstructions. Proposition 10.18 proves that this joint passband–stopband profile is reachable in the ideal Bloch ensemble. Its strong-pulse closure has no amplitude bound uniform in the target accuracy, and the cited finite-dimensional scalar theorem gives no dimension-free infinite-sphere leakage estimate. Neither result therefore implies* (124)*. Proposition 10.25 shows more in the asymptotic high-mode regime used to enlarge the echo gaps: every spatial harmonic that can drive the target line also drives an overlapping transition from the ground state. Zeroing those unwanted couplings therefore zeros the target coupling itself. A surviving construction must use multiple spatial profiles and genuinely noncommutative interference, not a single isolated chirped line. Propositions 10.26 and 10.27 identify a concrete algebraically viable replacement. Pure degree-*$2\ell+1$ *profiles isolate the exact adjacent-shell resonance, generate the full special-unitary algebra on* $\mathcal{H}_\ell \oplus \mathcal{H}_{\ell+1}$*, and synthesize any continuous compressed ensemble gate with forward time. Proposition 10.28 then locates the exact fixed-window obstruction: on* $S = [s_0, s_1]$*, the same profiles obligatorily couple every adjacent pair from* ℓ *through* $\lfloor (s_1/s_0)(\ell+1) \rfloor - 1$*. The selected route avoids that cluster by Proposition 10.29: on a centred window the sharp condition is* $h < c/(2\ell+3)$*. Proposition 10.33 then performs the missing full-sphere spectral selection in operator norm, without any Galerkin projection, and Corollary 10.34 proves that the resulting physical tangent algebra is* $\mathfrak{su}(4\ell+4)$*. Proposition 10.32 separately shows that a fixed* L^1 *budget would make the leakage cutoff uniform in* s*.*

The local selective-pulse lift is now settled by Propositions 10.36 and 10.37, Theorem 10.39, and Corollary 10.40. They produce, about every prescribed centre, a bounded-height finite-budget physical scalar word whose full-sphere interaction propagator is an $\mathrm{SU}(2)$ *population swap on the selected pair and the identity on its complement, up to arbitrary operator-norm error. Proposition 10.35 is bypassed because the complete nonlinear Lie word is compressed into one slow-time pulse before the window scale is sent to zero.*

What these results do not prove is the quantifier in the present problem: one word on an arbitrarily prescribed fixed *nondegenerate interval S, or a quantitative lower bound on the successive local windows sufficient for* (98)*. Thus the remaining stopping test is global chromatic coverage. It may be closed either by a genuinely fixed-band composite word, or by an adaptive factorization whose certified local widths have divergent total hazard.*

10.1 The last lens mechanism: quasi-adiabatic population permutations

The preceding carrier analysis does not exhaust smooth scalar potentials. There is one qualitatively different mechanism. A slowly moved, high, narrow potential wall can make two spectral gaps exponentially small and can alternate adiabatic and effectively diabatic passages. On an interval this mechanism realizes finite permutations of Schrödinger eigenmodes [8]. The parameter s in (226) multiplies the *entire* Hamiltonian. Therefore the worst adiabatic error on a gapped portion occurs at $s = s_0$, whereas the worst error for an intentionally diabatic passage through a very small avoided crossing occurs at $s = s_1$. Uniformity on $S = [s_0, s_1] \Subset (0, \infty)$ is possible only if the wall construction opens a two-scale window: the outer gaps must be adiabatic already at s_0 while the selected minimum gap remains diabatic even at s_1. The following propositions separate what is already rigorous from the one sphere-specific lemma that remains.

Proposition 10.59 (Exact positive-speed identity; no projective positivity obstruction)**.** *Let*

$$W_s = e^{-isH_N} \cdots e^{-isH_1},$$

where the H_r are self-adjoint and nonnegative on a common invariant core. Then

$$K_W(s) := iW_s^* \partial_s W_s = \sum_{r=1}^{N} (e^{-isH_{r-1}} \cdots e^{-isH_1})^* H_r (e^{-isH_{r-1}} \cdots e^{-isH_1}) \;\geq\; 0 \tag{228}$$

in quadratic-form sense, with the $r = 1$ conjugating product interpreted as the identity. Every physical lens word (226) *has this property after multiplication by a scalar phase.*

Nevertheless (228) *does not obstruct* (227)*. If $\mathcal{E}$ is a finite A_3-invariant harmonic space and $c \geq \tau \lambda_{\max}(A_3|_{\mathcal{E}})$, then*

$$e^{-ics} e^{is\tau A_3}|_{\mathcal{E}} = e^{-is(cI - \tau A_3|_{\mathcal{E}})}, \qquad cI - \tau A_3|_{\mathcal{E}} \geq 0.$$

Thus the desired inverse drift is itself a positive-speed path after the allowed projective phase is chosen.

Proof. Differentiate the product. In the term in which the rth factor is differentiated, all factors to its left cancel against their adjoints in W_s^*, leaving the rth Hamiltonian conjugated by the factors to its right. This gives (228). For a mask q, choose $a \geq -\min_{\mathbb{S}^2} q$. Replacing q by $q + a$ makes $M_{q+a} \geq 0$ and changes e^{-isM_q} only by the scalar e^{isa}. The free Hamiltonians $t_r A_3$ are nonnegative. The last assertion follows from the displayed finite-dimensional spectral inequality. □

Proposition 10.60 (Why the conical-intersection shortcut stops at the ground mode)**.** *For every smooth real q on the connected sphere, the lowest eigenvalue of*

$$H(q) = A_3 + M_q$$

is simple and its eigenfunction may be chosen strictly positive. Hence no finite-parameter family of smooth real scalar potentials has an intersection between its first and second ordered eigenvalues. In particular, the conically connected-spectrum hypothesis of the infinite-dimensional adiabatic controllability theorem of [6] can never hold for the full scalar Schrödinger spectrum on $\mathbb{S}^2$.

Proof. The heat semigroup of $A_3 + M_q$ is positivity improving on the connected compact manifold. Equivalently, the strong maximum principle and the variational characterization of the lowest eigenvalue show that a ground state has one sign and that two linearly independent ground states are impossible. Thus $\lambda_1(q) < \lambda_2(q)$ for every q. Conical connectedness requires, in particular, an intersection of λ_1 and λ_2, so its hypothesis fails. □

Remark 10.61 (What Proposition 10.60 does and does not say). The proposition falsifies a tempting direct application of generic conical intersection theory; it does not prove dynamical noncontrollability. The pointwise theorem of [7] supplies generic nonresonant spectra and connected scalar couplings, but gives a control for one system, not one control uniform under the common dilation $H \mapsto sH$. The driftless semisimple ensemble theorem, Chambrion's control-amplitude dispersion theorem, and the finite-dimensional isolated-gap theorem likewise have different dispersion or truncation hypotheses. None therefore settles Problem 10.58. A high-wall protocol evades Proposition 10.60 by using a gap that becomes arbitrarily small in a singular wall limit, rather than an exact ground-state intersection at a finite smooth potential.

Proposition 10.62 (Uniform common-dilation adiabatic estimate and physical lens compilation)**.** *Let $S = [s_0, s_1] \Subset (0, \infty)$ and let*

$$H(r) = A_3 + M_{V(r)}, \qquad 0 \le r \le 1,$$

where V is a C^2 path of smooth real potentials. Suppose that a finite-rank spectral projection $P(r)$ is separated from the rest of the spectrum by a gap $g > 0$ along the path. If $U_{s,T}$ is the propagator of

$$i\partial_t\psi = sH(t/T)\psi, \qquad 0 \le t \le T,$$

then, after the usual dynamical and geometric unitary inside $\operatorname{Ran} P(r)$ *is removed,*

$$\sup_{s\in S} \|(I - P(1))U_{s,T}P(0)\| \le \frac{C(H,P)}{s_0 T}. \tag{229}$$

For fixed T and V, and for every finite compact set of smooth initial data, $U_{s,T}$ is the uniform limit, for $s \in S$, of physical lens words

$$\overleftarrow{\prod}_{k=0}^{n-1} \left(e^{-is\Delta A_3} e^{-isM_{\Delta V(k/n)}}\right), \qquad \Delta = T/n > 0. \tag{230}$$

Consequently, any uniform population-permutation protocol produced by a smooth time-dependent scalar potential can be inserted into Problem 10.58; no extra negative free flight is needed.

Proof. Apply the quantitative adiabatic theorem to the rescaled equation $i(sT)^{-1}\partial_r\psi = H(r)\psi$. Its constant is uniform on the compact path, and $sT \ge s_0T$, which gives (229). For (230), use the time-sliced Lie–Trotter formula. On a finite compact set of smooth data the local splitting error is uniform for $s \in [s_0, s_1]$; telescoping and compactness then give uniform strong convergence. Each time slice has exactly the ordering and signs allowed in (226). □

Proposition 10.63 (A finite smooth multiwell has an ideal Dirichlet permutation limit)**.** *Fix $N \ge 2$. There are pairwise disjoint congruent geodesic discs $D_1, \dots, D_N \Subset \mathbb{S}^2$, functions*

$$b, \eta_1, \dots, \eta_N \in C^\infty(\mathbb{S}^2; \mathbb{R}), \qquad b \ge 0,$$

and numbers $\nu_1 < \nu_2$ with the following properties:

$$\{b = 0\} = \bigcup_{j=1}^{N} \overline{D}_j, \qquad \eta_j = 1 \text{ near } \overline{D}_j, \qquad \eta_j = 0 \text{ near } \overline{D}_k \quad (k \ne j),$$

and ν_1, ν_2 are the first two Dirichlet eigenvalues of every D_j. For

$$H_{B,c} = A_3 + BM_b + \sum_{j=1}^{N} c_j M_{\eta_j}, \qquad c = (c_1, \dots, c_N), \tag{231}$$

let ϕ_j be the positive Dirichlet ground state of D_j, extended by zero. If

$$\operatorname{osc}(c) := \max_j c_j - \min_j c_j < \nu_2 - \nu_1, \tag{232}$$

then, as $B \to \infty$, the first N eigenvalues and their spectral projections converge, uniformly for c in compact subsets of (232), to the ordered collection

$$\{\nu_1 + c_j : 1 \le j \le N\}, \qquad \{\mathbb{C}\phi_j : 1 \le j \le N\}.$$

The $(N+1)$st limiting eigenvalue is at least $\nu_2 + \min_j c_j$. In particular, when the coordinates of c are distinct, the first N finite-B eigenvalues are simple for all sufficiently large B, and their normalized eigenfunctions are uniformly close, up to signs, to the corresponding ϕ_j.

Proof. Choose the discs small and disjoint, choose b flat on their closures and strictly positive off their union, and choose the η_j in disjoint collars. The closed forms

$$\mathfrak{h}_{B,c}[u] = \|\nabla u\|_2^2 + B\|\sqrt{b}\,u\|_2^2 + \sum_j c_j \|\sqrt{\eta_j}\,u\|_2^2$$

increase with B. Their monotone limit has form domain $\bigoplus_j H_0^1(D_j)$ and is the direct sum

$$\bigoplus_{j=1}^{N} \left(-\Delta_{D_j}^{\mathrm{D}} + c_j\right).$$

Kato's monotone form theorem gives strong resolvent convergence. Compact resolvent, the min–max principle, and compactness of the parameter set upgrade this to convergence of each fixed finite spectral cluster and its Riesz projection, uniformly in c. The first two levels in the jth summand are $\nu_1 + c_j$ and $\nu_2 + c_j$. Condition (232) puts all N ground levels strictly below every second level. The final assertions follow from the spectral projection convergence and the gap at the endpoints. $\square$

Proposition 10.64 (Barrier-independent fast floor sweep)**.** *Retain Proposition 10.63, let $S = [s_0, s_1] \Subset (0, \infty)$, and let $c : [0, \tau] \to \mathbb{R}^N$ be smooth. Denote by $U_{s,B,c}(t)$ the propagator of*

$$i\partial_t u = sH_{B,c(t)}u.$$

For every $\varepsilon > 0$ there is $\tau_\varepsilon > 0$, independent of B and of the height and speed of the floor sweep $c(t)$, such that whenever $0 < \tau < \tau_\varepsilon$,

$$\sup_{\substack{s\in S\\ 1\le j\le N}} \left\| U_{s,B,c}(\tau)\phi_j - \exp\left[-is \int_0^{\tau} (\nu_1 + c_j(t))\,\mathrm{d}t\right] \phi_j \right\|_2 < \varepsilon. \tag{233}$$

The same assertion holds if ϕ_j at the two endpoints is replaced by the corresponding finite-B eigenfunction, after B is chosen sufficiently large.

Proof. Given $\delta > 0$, choose $\phi_{j,\delta} \in C_c^\infty(D_j)$ with

$$\|\phi_{j,\delta} - \phi_j\|_2 < \delta, \qquad R_\delta := \max_j \|(A_3 - \nu_1)\phi_{j,\delta}\|_2 < \infty.$$

On the support of $\phi_{j,\delta}$ one has $b = 0$, $\eta_j = 1$, and $\eta_k = 0$ for $k \neq j$. Hence

$$\psi_{j,s}(t) = \exp\left[-is \int_0^t (\nu_1 + c_j(r))\, \mathrm{d}r\right] \phi_{j,\delta}$$

satisfies

$$\|(i\partial_t - sH_{B,c(t)})\psi_{j,s}(t)\|_2 \leq s_1 R_\delta.$$

Duhamel's formula and unitarity give, uniformly in B, c, and s,

$$\|U_{s,B,c}(\tau)\phi_j - \psi_{j,s}(\tau)\|_2 \leq 2\delta + s_1 \tau R_\delta.$$

First choose δ, then τ. The endpoint replacement follows from Proposition 10.63. $\square$

Theorem 10.65 (Uniform signed permutations of finitely many spherical eigenmodes)**.** *Let $S = [s_0, s_1] \Subset (0, \infty)$ and let*

$$\mathcal{F} = \bigoplus_{\ell=0}^{L} \mathcal{H}_\ell$$

be a finite harmonic safety space. There is a real harmonic eigenbasis $e_1, \ldots, e_N$ of $\mathcal{F}$ with the following property. Given a permutation π of $\{1, \ldots, N\}$, $\varepsilon > 0$, and any prescribed finite harmonic enlargement of $\mathcal{F}$, there are a smooth real time-dependent potential $V(t, \omega)$, a duration $T > 0$, a real number C, and signs $\epsilon_j \in \{-1, 1\}$ such that the full-sphere propagator

$$i\partial_t U_s(t) = s(A_3 + M_{V(t)})U_s(t), \qquad U_s(0) = \mathrm{I},$$

satisfies

$$\sup_{\substack{s\in S\\ 1\leq j\leq N}} \left\|U_s(T)e_j - \epsilon_j e^{-is(C+T\lambda_{\pi(j)})} e_{\pi(j)}\right\|_2 < \varepsilon, \qquad A_3 e_j = \lambda_j e_j. \tag{234}$$

The estimate is an $L^2(\mathbb{S}^2)$ estimate, not a Galerkin estimate; in particular it includes leakage into the whole complementary spherical spectrum. The construction permits π to transpose the constant ground harmonic with an arbitrarily high auxiliary mode while fixing every listed spectator.

Proof. Step 1: remove the endpoint degeneracies. For each $\ell \leq L$, the real Clebsch–Gordan map

$$\bigoplus_{\substack{0\leq r\leq 2\ell\\ r \text{ even}}} \mathcal{H}_r^{\mathbb{R}} \longrightarrow \mathrm{Sym}(\mathcal{H}_\ell^{\mathbb{R}}), \qquad q \longmapsto P_\ell M_q P_\ell,$$

is an isomorphism: both sides have dimension $(\ell+1)(2\ell+1)$, and the multiplicity-one Clebsch–Gordan decomposition makes the map nonzero on every displayed irreducible summand. Consequently a generic real q_0 makes every $P_\ell M_{q_0} P_\ell$, $\ell \leq L$, simple. Choose $e_1, \ldots, e_N$ to be the resulting real eigenbases inside the harmonic shells. Analytic degenerate perturbation theory gives simple eigenvectors $e_j(a)$ of $A_3 + aM_{q_0}$ with

$$\max_{j\leq N} \|e_j(a) - e_j\|_2 \longrightarrow 0 \quad (a \downarrow 0).$$

A smooth ramp from zero to aq_0 performed in a time tending to zero changes the finitely many e_j by $o(1)$, uniformly in $s \in S$. Thus the exactly degenerate endpoint costs only an arbitrarily small error.

Step 2: encode the ordered modes in separated wells. Apply Proposition 10.63 with N discs. Choose two vectors c^-, c^+ with distinct coordinates, both satisfying (232), such that the rank of the jth well ground level for c^+ is π of its rank for c^-. Choose B so large that the endpoint eigenvectors are as close to the ϕ_j as the error budget requires.

The set of smooth scalar potentials for which the first N eigenvalues are simple is residual. More precisely, compression of a multiplication perturbation to a multiple eigenspace supplies the real symmetric splitting directions; a two-parameter transverse detour therefore avoids the codimension-two discriminant. It follows that the potentials aq_0 and $Bb + \sum c_j^- \eta_j$ can be joined by a smooth path on which the first N eigenvalues are simple. The same is true from $Bb + \sum c_j^+ \eta_j$ back to aq_0. Compactness of the two paths gives a positive minimum gap for the finite cluster. Traverse these paths slowly. Proposition 10.62, with worst parameter s_0, transports all N modes to and from the well ground states with arbitrarily small full-space leakage.

Step 3: cross all desired ranks without tunnelling. Join c^- to c^+ in a time τ. Proposition 10.64, with worst parameter s_1, shows that every population stays in its spatial well. Since the endpoint ordering has changed, lowering the barrier maps the jth incoming mode to the $\pi(j)$th outgoing mode. This proves a uniform population permutation. Notice the genuine two-scale choice:

$$T_{\mathrm{ad}} \gg (s_0 g^2)^{-1}, \qquad \tau \ll (s_1 R_\delta)^{-1};$$

the slow and fast intervals are independent and therefore compatible.

Step 4: calibrate the dynamical phases. Only a finite phase ledger remains. At large B the Hellmann–Feynman matrix for the well floors is

$$G_{jk}(B,c) = \frac{\partial \lambda_j(H_{B,c})}{\partial c_k} = \langle \varphi_j(B,c), M_{\eta_k} \varphi_j(B,c) \rangle \longrightarrow \delta_{jk}.$$

It is therefore invertible. Insert, while the populations are localized, two long gapped holds with nearby distinct floor vectors. By the inverse function theorem their two eigenvalue vectors can be chosen to have any prescribed average modulo the common vector $(1, \dots, 1)$. Taking the holds long makes the required floor perturbations arbitrarily small, so (232) and the cluster gap are preserved. Their common component is changed freely by adding a scalar constant to the potential.

The Hamiltonians are real. A continuous normalized real eigenvector has zero Berry connection; its only possible closed-path holonomy is a sign. Choose the holds so that the total dynamical action of the branch starting at e_j is

$$C + T\lambda_{\pi(j)}$$

modulo an arbitrarily small error. All nonadiabatic ramps used to insert the holds may instead be included in the finite action ledger; the invertibility of (10.1) is unchanged. Thus the remaining geometric factors are precisely signs ϵ_j, and (234) follows. Enlarging the initial finite cluster incorporates any prescribed finite safety list; all estimates above are full-space estimates. □

Proposition 10.66 (Signed spectator phases disappear after a doubled echo)**.** *In Proposition 10.15, replace P_j by a signed transposition*

$$Q_j = D_j P_j,$$

where D_j is diagonal with entries in $\{-1, 1\}$ in the displayed eigenbasis. Then the same inverse conclusion holds, without requiring Q_j to fix the spectators pointwise.

Proof. Every signed transposition satisfies

$$Q_j A Q_j^{-1} = P_j A P_j, \qquad Q_j^4 = \mathrm{I},$$

and Q_j^2 is diagonal, equals the identity on every spectator, and is a common sign on $\mathrm{span}\{e_j, f_j\}$. Use two echo blocks for the jth pair:

$$(e^{-isT_jA} Q_j) e^{-isd_{j,1}A} (e^{-isT_jA} Q_j)(e^{-isT_jA} Q_j) e^{-isd_{j,2}A} (e^{-isT_jA} Q_j).$$

The signs square away and the resulting generator is

$$2T_jA + x_jP_jAP_j, \qquad x_j = d_{j,1} + d_{j,2} + 2T_j.$$

The linear calculation in (119)–(121) is unchanged after replacing T_Σ by $2T_\Sigma$ and imposing $x_j > 2T_j$. The free parameter D can again be taken arbitrarily large, so this inequality holds and $x_j - 2T_j$ splits into two positive numbers. The same unitary telescoping estimate gives the approximate assertion. □

Theorem 10.67 (The lens-assisted inverse on a prescribed band)**.** *Problem 10.58 has a positive answer for every finite harmonic safety space and every prescribed nondegenerate $S \Subset (0,\infty)$. Hence the signed-drift input required by the full-block Talbot–Riesz route is available.*

Proof. Choose the real harmonic basis supplied by Theorem 10.65. Enlarge the safety space to contain mutually orthogonal, sufficiently high auxiliary basis vectors and all intermediate harmonic shells. Apply that theorem to each isolated target–auxiliary transposition, fixing the other displayed modes. Equation (234) gives, uniformly on S,

$$\mathcal{S}_{j,s} \simeq e^{-isC_j} e^{-isT_jA_3} Q_j.$$

Proposition 10.62 turns these propagators into physical positive-flight scalar-mask words. Proposition 10.66 absorbs all signed spectator holonomies and all terminal drifts into positive echo times and produces $\chi(s)e^{is\tau A_3}$ on the requested space. Allocate the finitely many permutation, adiabatic, floor-sweep, and Trotter errors below the telescoping budget. This is (227). □

Proposition 10.68 (The double-well scale window gives a physical broadband SU(2) module)**.** *Let $S = [s_0, s_1] \Subset (0,\infty)$. For every $\varepsilon > 0$ there are a smooth real double-well Hamiltonian $H_0 = A_3 + M_{V_0}$, a rank-two spectral projection P separated from the remaining spectrum by a gap $G_{\mathrm{ext}} > 0$, and a smooth real floor-imbalance profile q such that the following two operations hold on $PL^2(\mathbb{S}^2)$, uniformly for $s \in S$ and with full-space leakage below ε:*

1. *equal-floor holds realize every forward tunnel flight e^{-istX}, $t \geq 0$, up to a scalar phase;*
2. *signed imbalance holds realize e^{-isuZ} for every $u \in \mathbb{R}$, up to a scalar phase.*

Here X, Z are two anticommuting Pauli generators in a localized well basis. Consequently finite positive-time scalar-control words realize every continuous target $G : S \to \mathrm{SU}(2)$ uniformly on the doublet.

Proof. Take two congruent discs separated by a high smooth barrier and, if necessary, add a common floor so that their ground doublet is the chosen spectral cluster. Let its eigenvalues be $c - \gamma$ and $c + \gamma$, $\gamma > 0$. In the localized basis obtained by adding and subtracting the two real eigenfunctions,

$$PH_0P = c\mathrm{I} + \gamma X.$$

The external gap can be increased by shrinking the discs, while γ can be decreased independently by raising or widening only the separating barrier.

Choose q equal to 1 on one well and -1 on the other, with its transition inside the barrier. Norm-resolvent convergence gives

$$PM_qP = Z + o(1)$$

in the localized basis. Use a small imbalance δM_q and choose the three scales in the compatible order

$$\gamma \ll \delta \ll G_{\mathrm{ext}}. \tag{235}$$

For an equal-floor hold, P is exactly invariant and a duration t/γ gives e^{-istX} after the scalar $e^{-isct/\gamma}$ is removed.

For an imbalance hold, let P_δ be the rank-two spectral projection of $H_0 + \delta M_q$. The Riesz projection formula and the Schur complement give

$$\|P_\delta - P\| \le C\delta/G_{\mathrm{ext}}, \tag{236}$$

$$\big\|P_\delta(H_0 + \delta M_q)P_\delta - (c\mathrm{I} + \gamma X + \delta Z)\big\| \le C\delta^2/G_{\mathrm{ext}} + o(\delta). \tag{237}$$

The subspace $P_\delta L^2$ is invariant for the entire static hold. Therefore these estimates do not grow linearly through leakage when the hold is long. A duration $|u|/\delta$, using $\operatorname{sgn}(u)q$, gives

$$\sup_{s\in S}\Big\|U_s(|u|/\delta)|_{PL^2} - \xi_u(s)e^{-isuZ}\Big\| \le C_{S,u}\left(\frac{\gamma}{\delta} + \frac{\delta}{G_{\mathrm{ext}}}\right) + o(1). \tag{238}$$

This proves the two claimed operations by (235).

Apply Proposition 10.12 with Z relabelled as the forward tunnel generator and X as the signed floor generator. It supplies the missing negative tunnel flights uniformly on S. We now have signed flows of two noncommuting generators. Their iterated commutators give odd powers of s in the real-symmetric part of $\mathfrak{su}(2)$ and even powers in the real-skew part. On $S \Subset (0,\infty)$, polynomials $sp(s^2)$ and $s^2q(s^2)$ are respectively dense in both required coefficient spaces. Standard commutator products and a finite exponential factorization of G finish the uniform $\mathrm{SU}(2)$ synthesis. □

Theorem 10.69 (Full-sphere finite-frame ensemble compiler)**.** *Let $S \Subset (0,\infty)$, let $\mathcal{E} \subset C^\infty(\mathbb{S}^2)$ be finite dimensional and invariant under conjugation, and let $G : S \to \mathcal{U}(\mathcal{E})$ be continuous. For every $\varepsilon > 0$ and every prescribed finite compact-output safety list, a smooth real time-dependent scalar potential has a common-dilation propagator satisfying*

$$\sup_{s\in S}\|(U_s - \chi(s)G(s))|_{\mathcal{E}}\|_{\mathcal{E}\to L^2(\mathbb{S}^2)} < \varepsilon \tag{239}$$

for a continuous scalar phase χ. The estimate includes leakage into the entire complementary spherical spectrum. The propagator is uniformly approximable on the safety list by a finite word of positive A_3-flights and signed smooth scalar masks.

Proof. Approximate the given smooth frame, its conjugates, the compact family of target images, and the safety list in one finite harmonic space and polar-correct the finite Gram matrices. Use the generic small endpoint splitting from Theorem 10.65 to choose a real basis.

Construct $N = \dim\mathcal{E}$ separated wells. At any one gate, lower only the barrier between a selected pair, leaving every other inter-well tunnelling splitting smaller than the selected γ by an arbitrarily prescribed factor. The switching time can be chosen in the nonempty window

$$(s_0 G_{\mathrm{ext}})^{-1} \ll T_{\mathrm{switch}} \ll (s_1\gamma)^{-1}.$$

Thus cluster adiabaticity controls the full complement while the finite populations do not move appreciably during switching. Independent well floors detune and phase-calibrate the spectators. Proposition 10.68 then realizes a broadband $\mathrm{SU}(2)$ gate on the selected pair.

The pairwise Pauli generators and diagonal floor generators generate $\mathfrak{su}(N)$. The same parity-aware polynomial argument as in the last paragraph of Proposition 10.68 gives the uniform closure $C(S, \mathrm{SU}(N))$. Equivalently, decompose a finite piecewise-exponential approximation of $G(s)$ into pairwise Givens gates and diagonal phases. There are only finitely many modules, so choose the inter-well splittings, external gaps, switching errors, and spectator phase errors successively below one telescoping budget.

Slow gapped entrance and exit paths map the free harmonic basis to the well frame and back. Their continuous dynamical phases and real holonomy signs are included in the prescribed finite-dimensional target; independent floor holds provide the required diagonal calibration. This proves (239). The time-sliced Lie–Trotter argument of Proposition 10.62 produces the claimed physical word. □

Proposition 10.70 (The outward angular-time budget)**.** *An outward radial controller supported beyond radius R has total positive angular kinetic time at most*

$$\int_R^\infty \frac{\mathrm{d}r}{r^2} = \frac{1}{R} \tag{240}$$

before the harmless common energy factor is inserted. In particular, an abstract positive-flight/mask word with total A_3-flight bounded below independently of R cannot be placed on successive exterior shells by the weak-slab scale ledger alone.

Proof. In the outgoing radial reduction the angular generator is A_3/r^2 (and the common factor $(2k)^{-1}$ is the dilation parameter s). Positive radial ordering adds its coefficient, so any subdivision into free collars and weak slabs has total coefficient bounded by the displayed integral. Long weak slabs do reduce the charged cost of a mask, but they also consume their share of the same positive angular time. Thus the bound cannot be evaded by subdividing the shell. $\square$

Proposition 10.71 (Direct scaled slabs fit the outward time and cost ledgers)**.** *Let $A_3 = -\Delta_{\mathbb{S}^2}$, let $S = [s_0, s_1] \Subset (0, \infty)$, and fix $T < \infty$. For each $N \to \infty$ choose*

$$\sqrt{N} \ll R_N \ll N, \qquad L_N = \frac{R_N^2}{N}. \tag{241}$$

Let $Q_N \in C^\infty([0,T] \times \mathbb{S}^2; \mathbb{R})$, and compare the scaled angular evolution

$$i\partial_x \mathcal{U}_{s,N}(x) = s\left(\frac{A_3}{N} + M_{Q_N(x)}\right)\mathcal{U}_{s,N}(x) \tag{242}$$

with the outward forward evolution on $r = R_N + L_N x$,

$$i\partial_r U_{s,N}(r) = s\left(\frac{A_3}{r^2} + M_{V_N(r)}\right)U_{s,N}(r), \qquad V_N(R_N + L_N x, \omega) = L_N^{-1} Q_N(x, \omega). \tag{243}$$

Suppose a prescribed compact safety family $\mathfrak{C}_N$ and both propagators obey the trajectory bound

$$\sup_{\substack{s \in S,\ 0 \le x \le T \\ f \in \mathfrak{C}_N}} (\|A_3 \mathcal{U}_{s,N}(x) f\|_2 + \|A_3 U_{s,N}(R_N + L_N x) f\|_2) \le C_{\mathfrak{C}} N. \tag{244}$$

Then

$$\sup_{\substack{s \in S,\ 0 \le x \le T \\ f \in \mathfrak{C}_N}} \|U_{s,N}(R_N + L_N x) f - \mathcal{U}_{s,N}(x) f\|_2 \le C_{S,T,\mathfrak{C}} \frac{R_N}{N}. \tag{245}$$

Moreover,

$$\int_{R_N}^{R_N + L_N T} \|V_N(r)\|_2^2 \,\mathrm{d}r = \frac{N}{R_N^2} \int_0^T \|Q_N(x)\|_2^2 \,\mathrm{d}x, \tag{246}$$

$$\|V_N\|_\infty = \frac{N}{R_N^2} \|Q_N\|_\infty, \tag{247}$$

$$\int_{R_N}^{R_N + L_N T} \frac{\mathrm{d}r}{r^2} = \frac{T}{N} + O_T\left(\frac{R_N}{N^2}\right) = o(R_N^{-1}). \tag{248}$$

Thus any family (242) with uniformly bounded quadratic control action and the trajectory bound (244) has a forward radial realization with vanishing charged cost, vanishing height, and sufficient outward angular time. Angular motion inside the slab is used deliberately; it is not treated as the weak-mask error in Proposition 10.81.

Proof. After the change of variables $r = R_N + L_N x$, (243) becomes

$$i\partial_x U_{s,N} = s\left(\frac{L_N}{(R_N + L_N x)^2} A_3 + M_{Q_N(x)}\right) U_{s,N}.$$

Uniformly for $0 \le x \le T$,

$$\left|\frac{L_N}{(R_N + L_N x)^2} - \frac{1}{N}\right| \le \frac{C_T R_N}{N^2},$$

because $L_N/R_N = R_N/N \to 0$. Duhamel's formula, unitarity, and (244) give (245). The change of variables gives (246) and (247) exactly. Expanding $(R_N + L_N x)^{-2}$ and using (241) gives (248). ☐

Proposition 10.72 (Direct scaled slabs have vanishing counter-propagation)**.** *Retain Proposition 10.71, let $K = [k_0, k_1] \Subset (0, \infty)$, and let P_N be an A_3-invariant finite-rank projection such that*

$$\|A_3 P_N\| \le C_A N. \tag{249}$$

Assume, uniformly in N,

$$\sup_{0 \le x \le T}\Big(\|P_N M_{Q_N(x)} P_N\| + \|P_N M_{\partial_x Q_N(x)} P_N\|\Big) \le C_Q. \tag{250}$$

Let $\mathcal{U}^{(P)}_{s,N}$ denote the propagator of the compression of (242) to $P_N L^2(\mathbb{S}^2)$. Consider on $P_N L^2(\mathbb{S}^2)$ the exact truncated radial equation

$$-u''(r) + \left(\frac{P_N A_3 P_N}{r^2} + P_N M_{V_N(r)} P_N\right) u(r) = k^2 u(r). \tag{251}$$

After the harmless plane-wave phases at the two endpoints are removed, let $\mathsf{T}_N(k)$ and $\mathsf{R}_N(k)$ denote its forward transmission and backward reflection blocks. Then, uniformly for $k \in K$,

$$\|\mathsf{R}_N(k)\| \le C_{K,T,A,Q}\left(\frac{1}{L_N} + \frac{R_N}{N}\right), \tag{252}$$

$$\left\|\mathsf{T}_N(k) - \mathcal{U}^{(P)}_{(2k)^{-1},N}(T)\right\| \le C_{K,T,A,Q}\left(\frac{1}{L_N} + \frac{R_N}{N}\right). \tag{253}$$

Both errors tend to zero in the scale window (241). Consequently, once the scaled angular gate has a full-space safety enlargement and the outgoing Schur cutoff is taken after that enlargement, counter-propagation does not create an additional radial obstruction.

Proof. Write

$$W_N(r) = \frac{P_N A_3 P_N}{r^2} + P_N M_{V_N(r)} P_N.$$

Equations (249), (250), and $L_N = R_N^2/N$ give, on the slab,

$$\|W_N(r)\| \le \frac{C}{L_N}, \qquad \|W_N'(r)\| \le \frac{C}{L_N^2}. \tag{254}$$

For the centrifugal derivative, use $N/R_N^3 = (L_N R_N)^{-1} \le L_N^{-2}$, which follows from $L_N/R_N = R_N/N < 1$.

Use the exact variation-of-constants representation

$$u(r) = e^{ikr}a(r) + e^{-ikr}b(r), \qquad e^{ikr}a'(r) + e^{-ikr}b'(r) = 0.$$

Equation (251) is equivalent to

$$\begin{aligned} a' &= -\frac{i}{2k} W_N(r)\big(a + e^{-2ikr}b\big), \\ b' &= \frac{i}{2k} W_N(r)\big(e^{2ikr}a + b\big). \end{aligned}$$

Remove the two diagonal propagators generated by $\mp iW_N/(2k)$. Their norms and the norms of their derivatives are bounded uniformly by (254). Every remaining off-diagonal coefficient has the form

$$e^{\pm 2ikr}B_N(r), \qquad \|B_N(r)\| \le C/L_N, \qquad \|B_N'(r)\| \le C/L_N^2.$$

Integration by parts, using $2k \ge 2k_0$, gives

$$\sup_{R_N \le y \le R_N + L_N T} \left\| \int_{R_N}^{y} e^{\pm 2ikr} B_N(r)\, \mathrm{d}r \right\| \le \frac{C}{L_N}.$$

The standard primitive-form Duhamel estimate, obtained by one further integration by parts against the diagonal propagators, now gives (252) with its L_N^{-1} term and shows that the forward block differs by the same amount from the diagonal equation

$$ia'(r) = \frac{1}{2k}W_N(r)a(r).$$

After $r = R_N + L_N x$, that diagonal equation is the forward evolution in Proposition 10.71. Apply (245) to obtain the additional R_N/N term in (253). The same comparison absorbs the harmless R_N/N allowance in (252). □

Proposition 10.73 (Every cost-compatible direct slab freezes spatial densities)**.** *Let $\mathcal{U}_{s,N}$ solve (242), and let $\mathfrak{C}$ contain normalized trajectories from one fixed finite smooth input frame. Put*

$$E_N = \sup_{\substack{s\in S,\ 0\le x\le T\\ f\in\mathfrak{C}}} \|A_3\mathcal{U}_{s,N}(x)f\|_2. \tag{255}$$

Suppose radii R_N are chosen so that the direct radial realization has vanishing control cost and vanishing frozen-radius error:

$$\frac{N}{R_N^2} \longrightarrow 0, \qquad \frac{R_N E_N}{N^2} \longrightarrow 0. \tag{256}$$

Then, for every real $X \in C^\infty(\mathbb{S}^2)$,

$$\sup_{\substack{s\in S,\ 0\le x\le T\\ f\in\mathfrak{C}}} |\langle \mathcal{U}_{s,N}(x)f, M_X\mathcal{U}_{s,N}(x)f\rangle - \langle f, M_X f\rangle| \le \frac{C_{S,T,X}(1+\sqrt{E_N})}{N} \longrightarrow 0. \tag{257}$$

In particular, let $e_0 = (4\pi)^{-1/2}$ and let e_λ be a normalized real spherical harmonic with $A_3 e_\lambda = \lambda e_\lambda$, $\lambda > 0$. On any nondegenerate interval S and for every $\tau > 0$, no such family can converge, up to scalar phase, to $e^{is\tau A_3}$ on $\operatorname{span}\{e_0, e_\lambda\}$. Hence the direct scaled-slab formulation (276) is false for the lens inverse required in Problem 10.78.

Proof. For a normalized solution $\psi(x) = \mathcal{U}_{s,N}(x)f$, multiplication by $Q_N(x)$ commutes with M_X. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\langle \psi, M_X\psi\rangle = \frac{is}{N}\langle \psi, [A_3, M_X]\psi\rangle.$$

Since

$$[A_3, M_X]\psi = -(\Delta X)\psi - 2\nabla X\cdot\nabla\psi,$$

Cauchy–Schwarz and $\|\nabla\psi\|_2^2 = \langle\psi, A_3\psi\rangle \le \|A_3\psi\|_2$ give

$$\left|\frac{\mathrm{d}}{\mathrm{d}x}\langle \psi, M_X\psi\rangle\right| \le \frac{C_{S,X}}{N}\left(1+\sqrt{E_N}\right).$$

Integration proves the displayed bound. The first condition in (256) implies $R_N \to \infty$, and the second gives

$$\frac{E_N}{N^2} = \frac{1}{R_N}\frac{R_N E_N}{N^2} \longrightarrow 0.$$

Thus the right side of (257) tends to zero.

For the final assertion take

$$f = \frac{e_0 + e_\lambda}{\sqrt{2}}, \qquad X = e_\lambda.$$

The scalar phase is irrelevant to expectations, and

$$\left\langle e^{is\tau A_3} f, M_{e_\lambda} e^{is\tau A_3} f \right\rangle - \langle f, M_{e_\lambda} f \rangle = \frac{1}{\sqrt{4\pi}} \{\cos(s\tau\lambda) - 1\}.$$

This continuous function is not identically zero on a nondegenerate interval S. Uniform operator convergence to the inverse drift would therefore contradict (257). □

Proposition 10.74 (Bounded-depth multiscale slabs also freeze spatial densities)**.** *For each q, concatenate m_q scaled angular slabs. On slab j, of scaled duration $T_{q,j}$, let the actual normalized trajectory solve*

$$i\partial_x \psi_{q,j} = s\left(\frac{A_3}{N_{q,j}} + M_{Q_{q,j}(x)}\right)\psi_{q,j}, \qquad 0 \le x \le T_{q,j}, \tag{258}$$

with the final state of one slab used as the initial state of the next. Let $R_{q,j}$ be its physical radius and put

$$E_{q,j} = \sup_{\substack{s\in S,\ 0\le x\le T_{q,j} \\ f\in\mathfrak{C}}} \|A_3 \psi_{q,j}(x; s, f)\|_2,$$

where $\mathfrak{C}$ is one fixed finite smooth input frame. Define the three resource sums

$$\alpha_q = \sum_{j=1}^{m_q} \frac{T_{q,j}}{N_{q,j}}, \qquad \beta_q = \sum_{j=1}^{m_q} \frac{T_{q,j} R_{q,j} E_{q,j}}{N_{q,j}^2}, \qquad \gamma_q = \sum_{j=1}^{m_q} \frac{T_{q,j}}{R_{q,j}}. \tag{259}$$

Then every real $X \in C^\infty(\mathbb{S}^2)$ satisfies

$$\sup_{\substack{s\in S \\ f\in\mathfrak{C}}} |\langle \psi_{q,m_q}(T_{q,m_q}), M_X \psi_{q,m_q}(T_{q,m_q})\rangle - \langle f, M_X f\rangle| \le C_{S,X}\left(\alpha_q + \sqrt{\beta_q\gamma_q}\right). \tag{260}$$

Consequently, if $\mathfrak{C}$ contains $f = (e_0 + e_\lambda)/\sqrt{2}$ and

$$\alpha_q \longrightarrow 0, \qquad \beta_q\gamma_q \longrightarrow 0, \tag{261}$$

the cascade cannot converge, up to scalar phase, to $e^{is\tau A_3}$ on $\operatorname{span}\{e_0, e_\lambda\}$ for any $\tau > 0$ and any nondegenerate interval S.

In particular, every bounded-depth family with

$$\sup_q m_q < \infty, \qquad \sup_{q,j} T_{q,j} < \infty, \qquad \min_j N_{q,j} \longrightarrow \infty, \qquad \min_j R_{q,j} \longrightarrow \infty,$$

and vanishing summed frozen-radius error $\beta_q \to 0$ is excluded. More generally, any surviving multiscale compiler must violate the explicit product criterion

$$\left(\sum_j \frac{T_{q,j} R_{q,j} E_{q,j}}{N_{q,j}^2}\right)\left(\sum_j \frac{T_{q,j}}{R_{q,j}}\right) \longrightarrow 0.$$

Proof. On slab j, the proof of Proposition 10.73 gives

$$|\langle \psi_{q,j}(T_{q,j}), M_X \psi_{q,j}(T_{q,j})\rangle - \langle \psi_{q,j}(0), M_X \psi_{q,j}(0)\rangle| \le C_{S,X} \frac{T_{q,j}(1 + \sqrt{E_{q,j}})}{N_{q,j}}.$$

Telescoping over the concatenation leaves α_q and the sum

$$\sum_j \frac{T_{q,j}\sqrt{E_{q,j}}}{N_{q,j}} = \sum_j \sqrt{\frac{T_{q,j}R_{q,j}E_{q,j}}{N_{q,j}^2}}\sqrt{\frac{T_{q,j}}{R_{q,j}}}.$$

Cauchy–Schwarz bounds the latter by $\sqrt{\beta_q\gamma_q}$, proving (260). The two-harmonic expectation computed in the proof of Proposition 10.73 is not constant in s, so (261) rules out the lens inverse. Under the bounded-depth assumptions, $\alpha_q \le C/\min_j N_{q,j} \to 0$ and $\gamma_q \le C/\min_j R_{q,j} \to 0$, which proves the stated special case. ☐

Proposition 10.75 (A cheap logarithmic shell must develop $R^{-1/2}$ angular microstructure)**.** *Consider the forward angular model on $R \le r \le Re^T$,*

$$i\partial_r U_{s,R}(r) = s\left(\frac{A_3}{r^2} + M_{V_R(r)}\right)U_{s,R}(r), \qquad U_{s,R}(R) = I, \tag{262}$$

where $S \Subset (0,\infty)$ and

$$V_R(Re^t,\omega) = \frac{1}{Re^t}W_R(t,\omega).$$

After the exact change of variables $r = Re^t$,

$$i\partial_t U_{s,R} = s\left(\frac{e^{-t}}{R}A_3 + M_{W_R(t)}\right)U_{s,R}, \qquad 0 \le t \le T, \tag{263}$$

and the physical quadratic ledger is

$$\int_R^{Re^T} \|V_R(r)\|_2^2 \,\mathrm{d}r = \frac{1}{R}\int_0^T e^{-t}\|W_R(t)\|_2^2 \,\mathrm{d}t. \tag{264}$$

Let

$$\Phi_R(t,\omega) = \int_0^t W_R(x,\omega)\,\mathrm{d}x, \qquad Z_{s,R}(t) = e^{-isM_{\Phi_R(t)}}.$$

If

$$\sup_R \left\{\int_0^T \|W_R(t)\|_2^2 \,\mathrm{d}t + \int_0^T \|W_R(t)\|_{W^{2,\infty}(\mathbb{S}^2)}\,\mathrm{d}t\right\} < \infty, \tag{265}$$

then for every fixed finite smooth frame $\mathfrak{C}$,

$$\sup_{\substack{s\in S,\ 0\le t\le T\\ f\in\mathfrak{C}}} \|U_{s,R}(Re^t)f - Z_{s,R}(t)f\|_2 \le \frac{C_{S,T,\mathfrak{C}}}{R}. \tag{266}$$

Consequently, uniformly smooth logarithmic shells cannot converge, up to scalar phase, to $e^{is\tau A_3}$ on $\operatorname{span}\{e_0,e_\lambda\}$ for any $\tau > 0$ and any nondegenerate interval S.

More quantitatively, if such a logarithmic-shell family does approximate that lens on the two-harmonic frame, then along a subsequence

$$\sup_{0\le t\le T}\left(\|\Delta\Phi_R(t)\|_\infty + \|\nabla\Phi_R(t)\|_\infty^2\right) \ge cR. \tag{267}$$

Thus a bounded-amplitude phase must contain angular frequencies of order at least $\sqrt{R}$; the cheap $1/r$ tail survives the cost ledger only by leaving every uniformly smooth control class.

Proof. The rescaled equation and (264) follow from $\partial_t = r\partial_r$ and $\mathrm{d}r = Re^t\,\mathrm{d}t$. Since multiplication operators commute at different times, $Z_{s,R}$ is the propagator obtained from (263) after deleting its kinetic term. Duhamel's formula gives

$$\|U_{s,R}(Re^t)f - Z_{s,R}(t)f\|_2 \le \frac{|s|}{R}\int_0^t e^{-x}\|A_3 Z_{s,R}(x)f\|_2\,\mathrm{d}x.$$

For fixed smooth f,

$$A_3(e^{-is\Phi}f) = e^{-is\Phi}\left[A_3 f + is(\Delta\Phi)f + 2is\,\nabla\Phi\cdot\nabla f + s^2|\nabla\Phi|^2 f\right],$$

up to the immaterial sign convention for Δ. Condition (265) therefore bounds the Duhamel integrand uniformly and proves (266).

Take $f = (e_0 + e_\lambda)/\sqrt{2}$. Every $Z_{s,R}(T)$ preserves $|f|$ pointwise, whereas

$$e^{is\tau A_3}f = \frac{e_0 + e^{is\tau\lambda}e_\lambda}{\sqrt{2}}$$

has a different pointwise modulus whenever $\cos(s\tau\lambda) \ne 1$. The elementary inequality

$$\|g-h\|_2 \ge \big\||g|-|h|\big\|_2$$

then gives a positive lower bound, for one s in every nondegenerate interval on which the target is nontrivial, between the target and every phase multiplier of f. This contradicts (266).

Finally, the displayed formula for $A_3(e^{-is\Phi}f)$ shows that if the left side of (267) were $o(R)$, the Duhamel error would be $o(1)$ on the two-harmonic frame. The same modulus argument would again contradict lens convergence. This proves the quantitative necessity. ☐

Proposition 10.76 (Bounded-action logarithmic shells require ballistic angular degree)**.** *Retain the logarithmic-shell model (263). Let $P_{\le L}$ denote the sum of the spherical-harmonic shells of degrees at most L. Suppose*

$$W_R(t) \in \bigoplus_{j=0}^{D_R}\mathcal{H}_j, \qquad \int_0^T \|W_R(t)\|_\infty\,\mathrm{d}t \le M \tag{268}$$

for all R. Then, for every fixed L_0,

$$\sup_{\substack{s\in S,\ 0\le t\le T\\ f\in P_{\le L_0}L^2,\ \|f\|_2=1}} \|\nabla_\omega U_{s,R}(Re^t)f\|_2 \le C_{S,M,L_0}(1+D_R). \tag{269}$$

Consequently, every real $X \in C^\infty(\mathbb{S}^2)$ satisfies

$$\sup_{\substack{s\in S,\ 0\le t\le T\\ f\in P_{\le L_0}L^2,\ \|f\|_2=1}} \left|\langle U_{s,R}(Re^t)f, M_X U_{s,R}(Re^t)f\rangle - \langle f, M_X f\rangle\right| \le C_{S,T,M,L_0,X}\frac{1+D_R}{R}. \tag{270}$$

In particular, if $D_R = o(R)$, the logarithmic shell cannot converge, up to scalar phase, to $e^{is\tau A_3}$ on $\mathrm{span}\{e_0, e_\lambda\}$ *for any $\tau > 0$ and any nondegenerate interval S.*

Thus, under the bounded rescaled action (268), every successful logarithmic-shell lens must have

$$\liminf_{R\to\infty}\frac{D_R}{R} > 0. \tag{271}$$

At radius R this is angular wavelength $O(R^{-1})$, hence physical tangential wavelength $O(1)$. It is precisely the regime in which A_3/r^2 is order one on the active modes and the paraxial small-centrifugal approximation is unavailable.

Proof. Put

$$\vartheta_R(t) = \frac{1 - e^{-t}}{R}$$

and pass to the interaction picture

$$Y_{s,R}(t) = e^{is\vartheta_R(t)A_3} U_{s,R}(Re^t).$$

Its generator is

$$B_{s,R}(t) = s\, e^{is\vartheta_R(t)A_3} M_{W_R(t)} e^{-is\vartheta_R(t)A_3}.$$

Conjugation by the free group preserves every harmonic shell and $\|B_{s,R}(t)\| \le s_1 \|W_R(t)\|_\infty$. The Clebsch–Gordan triangle rule gives

$$M_{W_R(t)} P_{\le L} \subseteq P_{\le L + D_R} L^2.$$

Hence the mth Dyson term applied to $P_{\le L_0} L^2$ lies in $P_{\le L_0 + mD_R} L^2$ and has operator norm at most

$$\frac{a^m}{m!}, \qquad a = s_1 M.$$

Since $\|A_3^{1/2} P_{\le L}\| \le L + 1$, summing the Dyson series yields

$$\begin{aligned} \|A_3^{1/2} Y_{s,R}(t) f\|_2 &\le \sum_{m=0}^{\infty} (L_0 + mD_R + 1) \frac{a^m}{m!} \\ &= e^a (L_0 + 1 + aD_R). \end{aligned}$$

The free interaction-picture factor commutes with A_3, proving (269).

For $\psi(t) = U_{s,R}(Re^t) f$, multiplication by $W_R(t)$ commutes with M_X, and therefore

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi, M_X \psi \rangle = \frac{ise^{-t}}{R} \langle \psi, [A_3, M_X] \psi \rangle.$$

Using

$$[A_3, M_X]\psi = -(\Delta X)\psi - 2\nabla X \cdot \nabla \psi$$

and (269), then integrating in t, proves (270). The two-harmonic expectation computed in Proposition 10.73 is not constant in s. Fix one $s_* \in S$ for which its absolute change is $d_* > 0$. Uniform lens convergence and (270) imply, for all sufficiently large R,

$$\frac{C_{S,T,M,L_0,X}(1 + D_R)}{R} \ge \frac{d_*}{2}.$$

This proves the positive lower limit in (271). ☐

Proposition 10.77 (An isolated ballistic carrier has a positive fixed-band cost floor)**.** *Let $q_R \in \mathcal{H}_{D_R}$ be real with $\|q_R\|_2 = 1$, put $\Lambda_R = D_R(D_R + 1)$, and consider in (263) a single control*

$$W_R(t, \omega) = w_R(t) q_R(\omega).$$

Let $e_0 = (4\pi)^{-1/2}$. In the interaction picture of the kinetic term, the first Born coefficient from e_0 to q_R at logarithmic time T is

$$b_R(s) = -\frac{is}{\sqrt{4\pi}} \int_0^T w_R(t) \exp\Big(\frac{is\Lambda_R}{R} (1 - e^{-t}) \Big)\, \mathrm{d}t. \tag{272}$$

If, for some $\eta > 0$,

$$\inf_{s \in S} |b_R(s)| \ge \eta, \tag{273}$$

then the physical quadratic ledger obeys

$$\int_R^{Re^T} \|V_R(r)\|_2^2 \,\mathrm{d}r \geq c_{S,T}\eta^2 \frac{\Lambda_R}{R^2}. \tag{274}$$

Consequently, if $\liminf D_R/R > 0$*, every isolated Born-level ballistic transfer that is uniform on the fixed nondegenerate interval* S *has a strictly positive physical cost. It cannot be one cell of an arbitrarily small-cost radial lens compiler.*

Proof. The free logarithmic propagator contributes the relative phase $s\Lambda_R(1-e^{-t})/R$, while

$$\langle q_R, M_{q_R} e_0\rangle = \frac{1}{\sqrt{4\pi}},$$

which proves (272). Set

$$x = 1 - e^{-t}, \qquad a = 1 - e^{-T}, \qquad g_R(x) = \frac{w_R(-\log(1-x))}{1-x}.$$

With the Fourier convention

$$\widehat{g}_R(\xi) = \int_0^a g_R(x) e^{i\xi x}\,\mathrm{d}x,$$

condition (273) gives

$$|\widehat{g}_R(\xi)| \geq \frac{\sqrt{4\pi}\eta}{s_1} \quad \text{for} \quad \xi \in \left[\frac{s_0\Lambda_R}{R}, \frac{s_1\Lambda_R}{R}\right].$$

Plancherel therefore yields

$$\int_0^a |g_R(x)|^2\,\mathrm{d}x \geq \frac{2\eta^2(s_1-s_0)}{s_1^2}\frac{\Lambda_R}{R}.$$

On the other hand, the exact cost identity (264) and $\mathrm{d}x = e^{-t}\,\mathrm{d}t$ give

$$\int_R^{Re^T} \|V_R(r)\|_2^2\,\mathrm{d}r = \frac{1}{R}\int_0^a |w_R(-\log(1-x))|^2\,\mathrm{d}x.$$

Since $w_R(-\log(1-x)) = (1-x)g_R(x)$ and $1 - x \geq e^{-T}$, the last two displays prove (274). □

Remaining proof obligation 10.78 (Short-time radialization of the finite-frame compiler)**.** *Strengthen Theorem 10.69 as follows. Given* $\eta > 0$*, require the approximating positive-flight/signed-mask word to have*

$$\sum_j t_j < \eta \tag{275}$$

while retaining uniform common-dilation accuracy, full-sphere leakage control, a finite mask ledger, and a radial realization with arbitrarily small charged cost and one pointwise height bound.

The double-well gate core can be accelerated by shrinking the wells: its absolute tunnel splitting and external gap then grow while the relative window $\gamma \ll \delta \ll G_{\mathrm{ext}}$ *remains available. What is not proved above is an equally short entrance and exit between the fixed low harmonic frame and the localized well frame. The adiabatic construction in Theorems 10.65 and 10.69 may use arbitrarily large total* A_3*-flight, so Lemma 9.3 cannot be cited to obtain* (275)*.*

Propositions 10.80, 10.81, and 10.82 below give a sharp audit of one natural shortcut. Rotationally averaged twist conjugation really does amplify A_3 *by an arbitrarily large factor, but the known small-time gradient-flow implementation uses* $1/\tau$ *phase kicks. Those kicks cannot simultaneously satisfy the present weak-slab error and arbitrarily small* X_3 *ledgers when* $\tau \lesssim R^{-1}$*. Taking the phase modulo* 2π *does make its representative bounded, but it preserves angular oscillation of order* τ^{-1} *and therefore leaves a positive physical cost floor. Thus abstract small-time* *controllability alone is not the missing theorem: the compiler must also be quantitatively cost-efficient under radialization.*

Proposition 10.71 isolates one direct sufficient formulation which initially survives that audit. It asks for controls Q_N in (242), on one fixed x interval, which approximate the required common-dilation finite-frame gate while satisfying

$$\sup_N \int_0^T \|Q_N(x)\|_2^2 \, \mathrm{d}x < \infty, \qquad \sup_N \|Q_N\|_\infty < \infty, \qquad \sup_{\text{trajectories}} \|A_3 u\|_2 = O(N). \tag{276}$$

Proposition 10.72 proves that the counter-propagating coefficient is $o(1)$ on every active block obeying the same scale and regularity bounds. Proposition 10.73 then decides the formulation negatively. Vanishing radial cost and frozen-radius error force every position density to remain asymptotically fixed, whereas the required inverse drift changes the density of a two-harmonic superposition. Thus the single-scale direct slab is closed as decisively as the standard fast-gradient implementation. Proposition 10.74 further excludes every bounded-depth multiscale cascade. Any positive solution within this concatenated frozen-radius model must therefore use an unbounded number of genuinely interacting scales and must violate the reciprocal-radius product criterion (261); no one bounded collection of energy parameters may control all intermediate trajectories and the radial freezing estimate. Proposition 10.75 isolates the other exact escape, logarithmic radial scaling. Although a $1/r$ control tail on $[R, Re^T]$ costs only $O(R^{-1})$, every uniformly smooth rescaled control tends to a pointwise phase multiplier. Hence a successful logarithmic-shell compiler must also create angular frequencies at least of order $\sqrt{R}$. Proposition 10.76 sharpens this for bounded rescaled action: every degree-$o(R)$ controller still freezes densities. The remaining logarithmic target is therefore a degree-$\asymp R$ scalar grating in the exact second-order radial equation, with simultaneous control of the growing resonant cluster, radial reflection, and the full fixed coupling interval. Proposition 10.77 excludes the perturbative isolated carrier version by a positive Plancherel cost floor. The surviving candidate is a nonperturbative collective gate controlling the entire growing resonant cluster.

A positive solution makes the reference-energy full-block concentration and reverse refocusing words uniform on all of K. It then proves the broadband cell claimed in the preceding draft, solves Problem 10.8 by global phase placement, and activates Proposition 11.1. A lower bound excluding (275) would close the present lens implementation structurally.

Proposition 10.79 (What the available small-time control theorems do and do not supply)**.** *The phase-saturation theorem of Chambrion–Pozzoli [17] supplies strong small-time limits for multiplication by phases on a closed manifold. On $\mathbb{S}^2$ its stated consequences are arbitrary phase multiplication of one initial state and a particular transfer between the extremal harmonics in one degenerate eigenspace. The Beauchard–Pozzoli framework [18] adds small-time phase and weighted-diffeomorphism operations in its developed torus and Euclidean examples. Neither cited result states the following assertion required in Problem 10.78: one control, independent of the input vector, which approximates a prescribed unitary on an entire finite harmonic frame, uniformly for a continuum of common dilations $s \in S$.*

Proof. The basic limit in [17, Theorem 2] is strong operator convergence for each fixed phase construction. Hence, after the construction has been fixed, it is automatically uniform on the unit sphere of a fixed finite-dimensional input space. Its limiting operators are nevertheless phase multipliers. The saturation theorem [17, Theorem 6] is formulated as reachability from one initial state, and the sphere corollary [17, Corollary 4] only transfers Y_j^j and Y_{-j}^j up to a sign.

Definition 2.1 of [18] calls a unitary L small-time approximately reachable when, for every initial state ψ_0, there is a control which approximates $L\psi_0$; the control is allowed to depend on ψ_0. Its controlled phase and diffeomorphism operators have the form

$$e^{i\varphi} f, \qquad \mathcal{L}_P f = |\det DP|^{1/2} (f \circ P).$$

Their finite compositions are again weighted-composition operators. This does not, merely from the cited theorem, produce a common control for a prescribed collection of orthogonal inputs or the non-spatial finite-frame unitary used in Theorem 10.69. Finally, neither statement contains the compact-parameter uniformity in s required by (239). Thus the literature supplies important fast generators but does not close the entrance/exit assertion. $\square$

Proposition 10.80 (An exact rotational twist amplifier). *Let $A_3 = -\Delta_{\mathbb{S}^2}$ and let X_z be the Killing field generating rotations about the z axis. Fix $\epsilon \neq 0$, put*

$$F(z) = 1 + \epsilon z, \qquad f = F(z)x, \qquad g = F(z)y,$$

and define the area-preserving twist

$$P_r = \mathrm{Fl}^r_{[\nabla f, \nabla g]}\,.$$

If $\mathcal{L}_P u = u \circ P$ and normalized Haar measure on SO(3) *is denoted by* dR*, then, as a quadratic-form identity on $H^1(\mathbb{S}^2)$,*

$$\int_{\mathrm{SO}(3)} \mathcal{L}^{-1}_{R^{-1}P_rR} A_3 \mathcal{L}_{R^{-1}P_rR}\,\mathrm{d}R = \left(1 + \frac{16\epsilon^2}{15} r^2\right) A_3. \tag{277}$$

Consequently, if the two twist gates with parameters $\pm r$ were available with negligible positive A_3-time and negligible charged cost, finite rotation quadrature and Lie–Trotter products would compress any prescribed positive A_3-flight by a factor of order r^{-2} on every fixed smooth safety compactum.

Proof. Using $\mathrm{Hess}\, x = -x g_{\mathbb{S}^2}$ and the analogous identities for y, z, or computing in spherical coordinates, gives

$$[\nabla(Fx), \nabla(Fy)] = \left(F^2 + (1 - z^2)(F')^2\right) X_z = (1 + \epsilon^2 + 2\epsilon z) X_z. \tag{278}$$

Thus, in coordinates (θ, ϕ),

$$P_r(\theta, \phi) = \left(\theta, \phi + r(1 + \epsilon^2 + 2\epsilon\cos\theta)\right).$$

It preserves spherical area. In the orthonormal frame $e_\theta = \partial_\theta$, $e_\phi = (\sin\theta)^{-1}\partial_\phi$, its differential is a shear, and

$$|DP_r|^2_{\mathrm{HS}} = 2 + 4\epsilon^2 r^2 \sin^4\theta.$$

For any smooth area-preserving diffeomorphism P, the rotational average of $\mathcal{L}_P^{-1} A_3 \mathcal{L}_P$ is a rotation-invariant, symmetric second-order differential operator. It annihilates constants, so it equals $a(P)A_3$. The addition theorem for degree-one harmonics, or equivalently averaging the principal quadratic form, gives

$$a(P) = \frac{1}{8\pi} \int_{\mathbb{S}^2} |DP|^2_{\mathrm{HS}}\,\mathrm{d}\omega.$$

Since

$$\int_{\mathbb{S}^2} \sin^4\theta\,\mathrm{d}\omega = 2\pi \int_{-1}^{1} (1 - z^2)^2\,\mathrm{d}z = \frac{32\pi}{15},$$

we obtain (277). Positive finite quadrature approximates the Haar form uniformly on any prescribed finite smooth safety compactum, and the form version of the Lie–Trotter theorem gives the final conditional assertion. $\square$

Proposition 10.81 (The standard fast-gradient implementation violates the radial cost ledger)**.** *Consider the known conjugation construction of a nontrivial gradient-flow gate in angular kinetic time τ. It contains phase masks* $\exp(\pm i\varphi/\tau)$ *for some fixed nonconstant smooth real φ. Suppose such a mask is realized near radius R by one or more weak slabs of total frozen-radius length T.*

If the radial-freezing error requires $T/R \le \varepsilon$ and the outward budget requires $\tau R \le c < 1$, then its weak-mask contribution satisfies

$$\mathrm{Cost}_{X_3} \ge \frac{\|\varphi\|^2_{L^2(\mathbb{S}^2)}}{\tau^2 T} \ge \frac{\|\varphi\|^2_{L^2(\mathbb{S}^2)}}{\varepsilon c^2} R. \tag{279}$$

Even if the freezing condition is discarded and only the angular-motion condition $D^2T/R^2 \le \varepsilon$ is retained, where $D > 0$ is any fixed nonzero angular frequency in φ, one still has the positive floor

$$\mathrm{Cost}_{X_3} \ge \frac{D^2\|\varphi\|^2_{L^2(\mathbb{S}^2)}}{\varepsilon c^2}. \tag{280}$$

The same bounds hold after arbitrary finite subdivision of the phase mask inside the same frozen-radius cell, provided the aggregate length obeys the displayed scale constraints. Hence the Chambrion–Pozzoli fast-gradient construction cannot be used to radialize Proposition 10.80 with arbitrarily small charged cost under the present weak-slab scale ledger.

Proof. A weak slab of length T_j carrying integrated mask w_j has quadratic cost $T_j^{-1}\|w_j\|_2^2$. If $\sum_j w_j = \varphi/\tau$, Cauchy–Schwarz gives

$$\sum_j \frac{\|w_j\|_2^2}{T_j} \ge \frac{\|\varphi/\tau\|_2^2}{\sum_j T_j}.$$

Use $\sum_j T_j \le \varepsilon R$ and $\tau \le c/R$ to obtain (279). Alternatively, $D^2 \sum_j T_j/R^2 \le \varepsilon$ gives (280). The lower bounds already apply to one of the two conjugating masks, so cancellations elsewhere in the formal transport word cannot remove them. ☐

Proposition 10.82 (Phase wrapping preserves the radial cost floor)**.** *Let φ be a fixed nonconstant smooth real function on $\mathbb{S}^2$, and put*

$$q_\tau = \exp(i\varphi/\tau), \qquad \tau \downarrow 0.$$

Suppose that smooth real aggregate phase profiles w_τ produce $m_\tau = e^{iw_\tau}$ with

$$\|m_\tau - q_\tau\|_{L^2(\mathbb{S}^2)} \longrightarrow 0. \tag{281}$$

There are constants $a_\varphi, b_\varphi > 0$ such that

$$\liminf_{\tau\downarrow 0} \|w_\tau\|_2 \ge a_\varphi, \qquad \liminf_{\tau\downarrow 0} \tau\|\nabla m_\tau\|_2 \ge b_\varphi. \tag{282}$$

Define the intrinsic angular frequency

$$D_\tau^2 = \frac{\|\nabla m_\tau\|_2^2}{\|m_\tau\|_2^2}.$$

If one or more weak slabs of aggregate frozen-radius length T_τ realize this phase at radius R_τ, and the angular-motion ledger requires

$$\frac{D_\tau^2 T_\tau}{R_\tau^2} \le \varepsilon, \qquad \tau R_\tau \le c, \tag{283}$$

then their quadratic charged cost has a strictly positive asymptotic floor:

$$\liminf_{\tau\downarrow 0} \mathrm{Cost}_{X_3} \geq \frac{a_\varphi^2 b_\varphi^2}{4\pi\varepsilon c^2} > 0. \tag{284}$$

Thus replacing φ/τ by a bounded principal-argument representative does not make the known fast-gradient gate arbitrarily cheap. The proposition does not exclude a different diffractive finite-frame gate which does not approximate q_τ.

Proof. Choose a coordinate patch U on which $\mathrm{d}\varphi \neq 0$. By the coarea formula, the values of φ/τ become equidistributed modulo 2π on a smaller regular patch. In particular,

$$\liminf_{\tau\downarrow 0} \|q_\tau - 1\|_{L^2(U)} > 0.$$

Since $|e^{it} - 1| \leq |t|$, the triangle inequality and (281) give

$$\|w_\tau\|_2 \geq \|m_\tau - 1\|_2 \geq \|q_\tau - 1\|_{L^2(U)} - o(1),$$

which proves the first assertion in (282).

For the derivative bound, choose a smooth vector field X on U with $X\varphi = 1$, and a smaller patch $U_0 \Subset U$. If Φ_h is its local flow and $h = \pi\tau$, then

$$q_\tau \circ \Phi_h = -q_\tau$$

on U_0 for all sufficiently small τ. Pullback by Φ_h is uniformly bounded on L^2, so (281) implies

$$\|m_\tau \circ \Phi_h - m_\tau\|_{L^2(U_0)} \geq 2|U_0|^{1/2} - o(1).$$

The fundamental theorem of calculus along the flow gives the reverse bound

$$\|m_\tau \circ \Phi_h - m_\tau\|_{L^2(U_0)} \leq C_U h \|\nabla m_\tau\|_{L^2(U)}.$$

This proves the second assertion in (282).

Because $|m_\tau| = 1$, one has $\|m_\tau\|_2^2 = 4\pi$, and hence $D_\tau^2 \geq b_\varphi^2/(4\pi\tau^2) + o(\tau^{-2})$. The first condition in (283) gives

$$T_\tau \leq \frac{4\pi\varepsilon R_\tau^2 \tau^2}{b_\varphi^2 + o(1)}.$$

If the aggregate phase is subdivided as $w_\tau = \sum_j w_{\tau,j}$ over lengths $T_{\tau,j}$, Cauchy–Schwarz yields

$$\mathrm{Cost}_{X_3} = \sum_j \frac{\|w_{\tau,j}\|_2^2}{T_{\tau,j}} \geq \frac{\|w_\tau\|_2^2}{T_\tau}.$$

Now use (282) and $\tau R_\tau \leq c$ to obtain (284). □

11 Conditional completion and the single remaining theorem

For an inner Jost matrix J and the transmission/reflection coefficients T, R of one outer shell, exact radial composition is

$$J^+ = J\mathsf{T} + \overline{J}\,\mathsf{R} = J(\mathsf{T} + H\mathsf{R}), \qquad H = J^{-1}\overline{J} \in \mathrm{U}(\mathcal{H}). \tag{285}$$

Problem 10.8 is the single missing strict global contraction required by Theorem 7.1. If it is proved, every finite physical generation is closed by the scale order

$$D \ll R \ll J \ll L, \qquad 2M_{W+v,K}^2 \frac{\|W+v\|_\infty^2}{(J+1)(J+2)/R^2 - \|W+v\|_\infty - k_1^2} < \eta, \tag{286}$$

together with the Neumann smallness condition in Proposition 8.9.

Proposition 11.1 (Conditional completion of the counterexample)**.** *If Problem 10.8 holds at every finite stage, then there is a bounded real $V \in X_3$ and a nonempty compact $K \Subset (0,\infty)$ such that*

$$P_{\mathrm{ac}}(-\Delta+V)\mathbf{1}_K(-\Delta+V)=0. \tag{287}$$

Under the same hypothesis, Simon's multidimensional L^2 conjecture is false.

Proof. Choose a countable dense set of smooth compact probes. At stage j, apply the asserted global physical grouped Lemma B to the first j probes and include every previously declared witness and the first j continuous tests in its preservation list. Allocate the combined phase-placement, residual, outgoing-buffer, and X_3 error at most 2^{-2j}. The stage contracts each of the first j fractional moments by $3/4+o(1)$ and preserves all declared spectral tests. Repeating the finite grouped generation three times gives $(3/4)^3<1/2$ and hence the factor in (28), after the allocated error is included. Put all cells on successive disjoint radial shells. Their amplitudes have one uniform bound and

$$\|V\|_{X_3}^2 \le \sum_{j\ge 1} 2^{-2j} < \infty. \tag{288}$$

Theorem 7.1 and Corollary 7.2 give vanishing of the exterior absolutely continuous projection on K. Strong resolvent convergence identifies the weakly locked limits with the spectral measures of the summed potential. Changing the operator inside a fixed ball and recoupling the exterior Dirichlet boundary are compactly supported scattering perturbations, so completeness of the local wave operators transfers the conclusion to the whole-space operator. □

12 Conclusions and the remaining radialization problem

The supervisor's proposed one-cell full-block stopping test has a definite answer: Proposition 3.1 disproves it. Theorem 10.3 supplies the correct repair: a finite product of fixed-height scalar cells is one random effective Poisson barrier, and it contracts every compact bundle of singular reciprocal spectral measures by a prescribed factor:

one-cell dimension-free Lemma B is false;
fixed-height grouped singular-bundle Lemma B is true;
short-time radialization of its lens compiler remains open.

The order of quantifiers is decisive. First select the finite group length from the infinite-dimensional compact singular bundle. Only afterwards take one finite smooth safety enlargement for the already fixed recursion. The root-Zeno and Möbius-orbit examples show why the former local-window argument could not close the global step. Theorem 10.69 supplies the required abstract fixed-band angular gate, but Proposition 10.70 prevents its present adiabatic realization from being inserted automatically on remote radial shells. Problem 10.78 is therefore the only remaining physical input to Problem 10.8 and Proposition 11.1.

Proposition 10.11 rules out the shortcut of obtaining a broadband inverse from free-flight recurrence. Propositions 10.12, 10.15, and 10.16 instead reduce the local problem to high-gap drift-decorated population swaps. Positive echo times absorb the terminal drift, and the internal two-level phase cancels. The adiabatic and direct single-profile constructions fail for the reasons proved in Propositions 10.24 and 10.25; adjacent-shell carriers fail on a fixed band because of the growing resonant cluster in Proposition 10.28.

The new step is to combine a saturated nonadjacent transition with a shrinking window. Proposition 10.36 proves arithmetically that

$$L=\ell+2^k, \qquad J=\ell+L$$

gives exactly one coupled positive gap $N = \lambda_L - \lambda_\ell$. Proposition 10.37 proves that both carrier quadratures of the degree-J profiles generate $\mathfrak{su}(\mathcal{H}_\ell \oplus \mathcal{H}_L)$. Theorem 10.39 then realizes any gate in that group on

$$S_\lambda = [c - \lambda R/N, c + \lambda R/N]$$

with full-sphere operator-norm leakage $o(1)$, peak control $O(\lambda)$, fixed L^1 budget, and quadratic control cost $O(\lambda)$. Corollary 10.40 therefore proves the exact drift-decorated swaps needed by the echo on a nonempty window about every centre. This is the previously missing nonlinear local lift. Proposition 10.35 remains correct but no longer blocks the local route: it excludes delayed first-order filters, whereas the carrier theorem compresses the complete Lie word before taking the window scale to zero.

Proposition 10.42 and Corollary 10.43 remove the last fixed-band algebraic ambiguity. Degree-one carrier quadratures generate $\mathfrak{su}(4a+4)$ on every adjacent pair and separate any finite family of resonant lines, so arbitrary line-by-line profiles are reachable on each fixed detuning compactum. Proposition 10.44 removes one possible quantitative obstruction: the resonant $O(\lambda)$ normal-form term preserves every spherical shell, while the first nonadjacent resonance is the three-photon $O(\lambda^2)$ path. Proposition 10.45 identifies the complete first defect, and Proposition 10.46 cancels it to $o(\lambda)$ on every fixed safety space and detuning compactum. Proposition 10.48 then decides the naive global patch negatively: every nonzero dipole pulse has a sharp $1/\nu$ Stark tail, so the individual operator-norm tails cannot be summed over the carrier lattice. Proposition 10.50 rules out importing that word from unconstrained matrix Zakharov–Shabat inverse scattering: its first-Born map has matrix directions absent from every physical higher dipole line, and independent line potentials violate the common-control constraint. Proposition 10.52 nevertheless removes the harmonic-tail obstruction for any elementary family with a common Stark residue: after only a growing local Chebyshev cluster is retained, the whole remote product is uniformly $O(M^{-1} + M/N)$. Corollary 10.55 isolates the corresponding conditional analytic criterion: Proposition 10.54 equalizes every fixed finite collection by locally invisible symmetric sidebands, while Corollary 10.53 identifies the sufficient growth target $K_N = o(\sqrt{N})$ together with vanishing joint synthesis error on the $O(\sqrt{N})$ local cluster. Proposition 10.20 and Corollary 10.21 now show that the selected adjacent-block implementation cannot meet that target with arbitrarily prescribed X_3 cost: the determinant trace identity imposes a positive bandwidth–energy floor independent of the passband partition. Thus the Chebyshev construction repairs the remote operator-norm tail but does not rescue the resonant multiband carrier route; that route is closed structurally, not merely missing an estimate.

The lens audit identifies the full-sphere mechanism outside the selected carrier model. Proposition 10.59 proves that projective positivity is compatible with inverse drift, while Proposition 10.60 rules out the direct conically-connected shortcut at the ground harmonic. Propositions 10.63 and 10.64, Theorem 10.65, and Proposition 10.66 prove the required uniform, spectator-refocused, full-sphere population permutations and remove their dynamical and real-holonomy phases. Theorem 10.67 consequently proves the prescribed-band inverse drift. Proposition 10.68 adds a natural tunnel generator and a signed floor generator in the compatible scale window $\gamma \ll \delta \ll G_{\mathrm{ext}}$; this avoids the invalid shortcut of normalizing a vanishing off-diagonal multiplication overlap. Theorem 10.69 gives the resulting full-sphere ensemble word. Proposition 10.70 then supplies the final stopping test: the word must be compressed to total positive A_3-time $o(R^{-1})$ before it can enter the remote-shell Riesz–Talbot ledger. This precise missing assertion is Problem 10.78; until it is proved, the global contraction and counterexample are not claimed. Proposition 10.71 shows that the outward budget is not by itself a no-go. In the compatible window $\sqrt{N} \ll R_N \ll N$, a continuous slab realizes $A_3/N + M_{Q_N}$ with angular time $O(N^{-1})$, forward error $O(R_N/N)$, and charged cost $O(N/R_N^2)$; Proposition 10.72 also makes its counter-propagating block $o(1)$. Proposition 10.73 supplies the decisive stopping test: the same compatibility conditions freeze all multiplication observables, while the inverse drift changes one explicitly displayed two-harmonic density. Hence the single-scale direct slab is false as a lens compiler. Proposition 10.74 telescopes that commutator estimate and excludes every bounded-depth multiscale cascade as well. A survivor within the frozen-radius slab model must have unbounded depth and keep the product of its total frozen-radius error with its total reciprocal-radius budget away from zero. Proposition 10.75 treats the exact logarithmic scaling $V(r) = r^{-1}W(\log(r/R))$: its cost is $O(R^{-1})$, but a uniformly smooth W converges to a phase multiplier. Thus this escape requires angular frequency at least $\sqrt{R}$. Proposition 10.76 supplies the density-level stopping test: under bounded rescaled action, every degree-$o(R)$ grating still freezes multiplication observables. A logarithmic-shell survivor must therefore reach degree comparable to R, where exact radial dispersion and the whole growing resonant cluster must be retained. Proposition 10.77 then excludes an isolated Born-level carrier on the fixed coupling interval by a positive physical cost floor. Proposition 10.80 shows that rotation-averaged area-preserving twists would give an exact abstract kinetic amplifier. Proposition 10.81 supplies its physical stopping test: the standard fast-gradient realization requires $1/\tau$ phase kicks, whose weak-slab quadratic cost has a positive floor (and under the frozen-radius ledger actually diverges) once $\tau \lesssim R^{-1}$. Proposition 10.82 closes the immediate modulo-2π escape: a bounded wrapped representative still has nonvanishing L^2 size, while the multiplier itself has intrinsic angular frequency $D_\tau \gtrsim \tau^{-1}$. The angular-motion ledger therefore restores a positive charged-cost floor. Thus the available small-time phase/diffeomorphism theorem does not repair the radial gap. Any successful compiler must use either an unbounded-depth frozen-radius cascade that violates the product criterion or a non-paraxial logarithmic-scale full-sphere excursion of angular degree comparable to R; neither the standard fast-gradient construction, a single scaled slab, any bounded-depth cascade, nor any bounded-action degree-$o(R)$ logarithmic shell can serve as that solution. In the logarithmic case the remaining candidate is a nonperturbative collective gate controlling the entire growing resonant cluster; an isolated Born-level ballistic carrier is also excluded.

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References

[1] W. Zhang and J.-S. Li, *Ensemble control on Lie groups*, arXiv:2008.03243 (2020). https://arxiv.org/abs/2008.03243

[2] T. Chambrion, *A sufficient condition for partial ensemble controllability of bilinear Schrödinger equations with bounded coupling terms*, 52nd IEEE Conference on Decision and Control (2013), 3708–3713. https://arxiv.org/abs/1303.0298

[3] U. Boscain, M. Caponigro, and M. Sigalotti, *Controllability of the bilinear Schrödinger equation with several controls and application to a 3D molecule*, arXiv:1204.6017 (2012). https://arxiv.org/abs/1204.6017

[4] U. Boscain, M. Caponigro, and M. Sigalotti, *Multi-input Schrödinger equation: controllability, tracking, and application to the quantum angular momentum*, arXiv:1302.4173 (2013). https://arxiv.org/abs/1302.4173

[5] N. Augier, U. Boscain, and M. Sigalotti, *Adiabatic ensemble control of a continuum of quantum systems*, SIAM J. Control Optim. **56** (2018), 4045–4068. https://doi.org/10.1137/17M1140327

[6] U. Boscain, J.-P. Gauthier, F. Rossi, and M. Sigalotti, *Approximate controllability, exact controllability, and conical eigenvalue intersections for quantum mechanical systems*, Comm. Math. Phys. **333** (2015), 1225–1239. https://arxiv.org/abs/1309.1970

[7] P. Mason and M. Sigalotti, *Generic controllability properties for the bilinear Schrödinger equation*, Comm. Partial Differential Equations **35** (2010), 685–706. https://arxiv.org/abs/0911.5650

[8] A. Duca, R. Joly, and D. Turaev, *Permuting quantum eigenmodes by a quasi-adiabatic motion of a potential wall*, J. Math. Phys. **61** (2020), 101511. https://arxiv.org/abs/1912.07425

[9] R. Liang, U. Boscain, and M. Sigalotti, *Ensemble control of n-level quantum systems with a scalar control*, Automatica **185** (2026), 112733. https://arxiv.org/abs/2501.12357

[10] J.-S. Li and N. Khaneja, *Ensemble control of Bloch equations*, IEEE Trans. Automat. Control **54** (2009), 528–536. https://doi.org/10.1109/TAC.2009.2012983

[11] J. F. Magland, *Discrete inverse scattering theory for NMR pulse design*, arXiv:0903.4363 (2009). https://arxiv.org/abs/0903.4363

[12] C. van der Mee, *Matrix Zakharov–Shabat systems with zero diagonal entry*, arXiv:2511.15348 (2025). https://arxiv.org/abs/2511.15348

[13] B. Luy, K. Kobzar, T. E. Skinner, N. Khaneja, and S. J. Glaser, *Construction of universal rotations from point-to-point transformations*, J. Magn. Reson. **176** (2005), 179–186. https://doi.org/10.1016/j.jmr.2005.06.002

[14] T. Chambrion, *Periodic excitations of bilinear quantum systems*, Automatica **48** (2012), 2040–2046. https://arxiv.org/abs/1103.1130

[15] N. Boussaïd, M. Caponigro, and T. Chambrion, *Implementation of logical gates on infinite dimensional quantum oscillators*, Proc. 2012 American Control Conference, 5825–5830. https://arxiv.org/abs/1110.6634

[16] N. Boussaïd, M. Caponigro, and T. Chambrion, *Efficient finite dimensional approximations for the bilinear Schrödinger equation with bounded variation controls*, Proc. 21st International Symposium on Mathematical Theory of Networks and Systems (2014), 1889–1891. https://arxiv.org/abs/1406.2260

[17] T. Chambrion and E. Pozzoli, *Small-time bilinear control of Schrödinger equations with application to rotating linear molecules*, Automatica **153** (2023), 111028. https://arxiv.org/abs/2207.03818

[18] K. Beauchard and E. Pozzoli, *Small-time approximate controllability of bilinear Schrödinger equations and diffeomorphisms*, Ann. Inst. H. Poincaré C Anal. Non Linéaire (2025). https://arxiv.org/abs/2410.02383

[19] Y. Zhang and N. K. Fontaine, *Multi-plane light conversion: a practical tutorial*, arXiv:2304.11323 (2023). https://arxiv.org/abs/2304.11323